# THE REPRODUCTIVE SYSTEM

**ANATOMY AND PHYSIOLOGY (MC1)**
**COLLEGE OF NURSING**
**PRESIDENT RAMON MAGSAYSAY STATE UNIVERSITY**

# PRAYER BEFORE THE START OF THE CLASS

We praise you Lord and we thank you
for the opportunity to study
and prepare for a good life in the future.
Bless our teachers who inspire and guide us.
Bless our parents who work hard to support us.
Bless our classmates and all the people
who care for us.
Bless us, help us to be more attentive, patient and diligent to
understand the lessons that our teachers teach us.
This we pray in the name of Jesus Christ our Lord.
Amen.

# COPYRIGHT

PREPARED BY: REIKA ATSUMI, MD

MARIA NOVIEL DOMINGUEZ, RN, MD

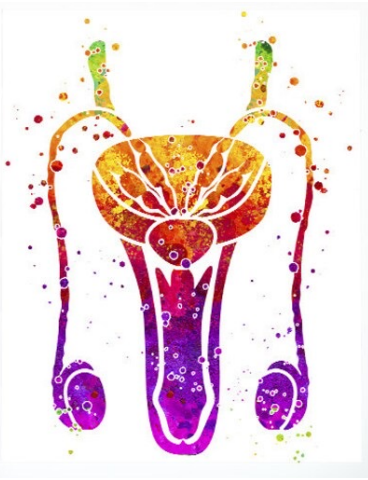
COLLEGE OF NURSING, PRESIDENT RAMON MAGSAYSAY STATE UNIVERSITY

ANATOMY AND PHYSIOLOGY

2020

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# OBJECTIVES



AT THE END OF THE SESSION, STUDENTS ARE EXPECTED TO:

1. ENUMERATE THE FUNCTIONS OF MALE AND FEMALE REPRODUCTIVE SYSTEM
2. ENUMERATE AND IDENTIFY THE PARTS OF MALE AND FEMALE REPRODUCTIVE ORGAN



# THE REPRODUCTIVE SYSTEM

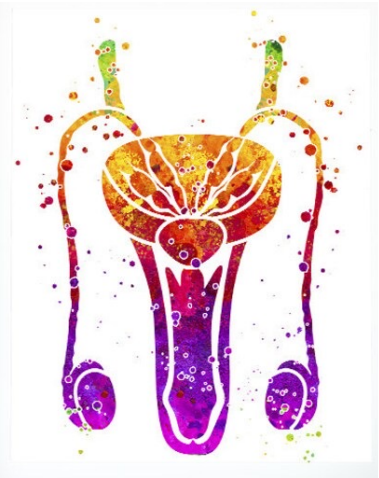


Gonads—primary sex organs

- Testes in males
- Ovaries in females

Gonads produce gametes (sex cells) and secrete hormones

- Sperm—male gametes
- Ova (eggs)—female gametes

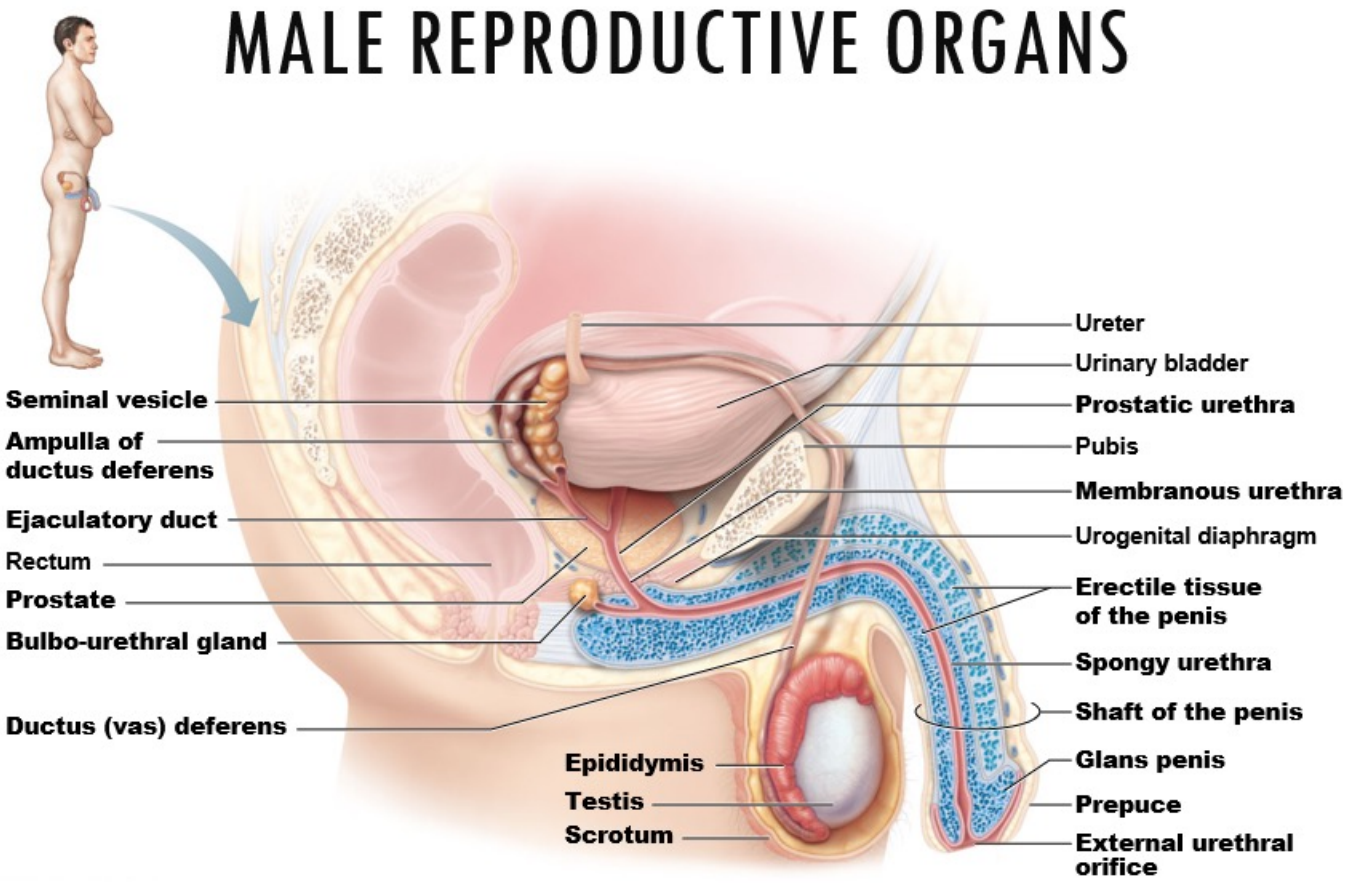


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# MALE REPRODUCTIVE SYSTEM

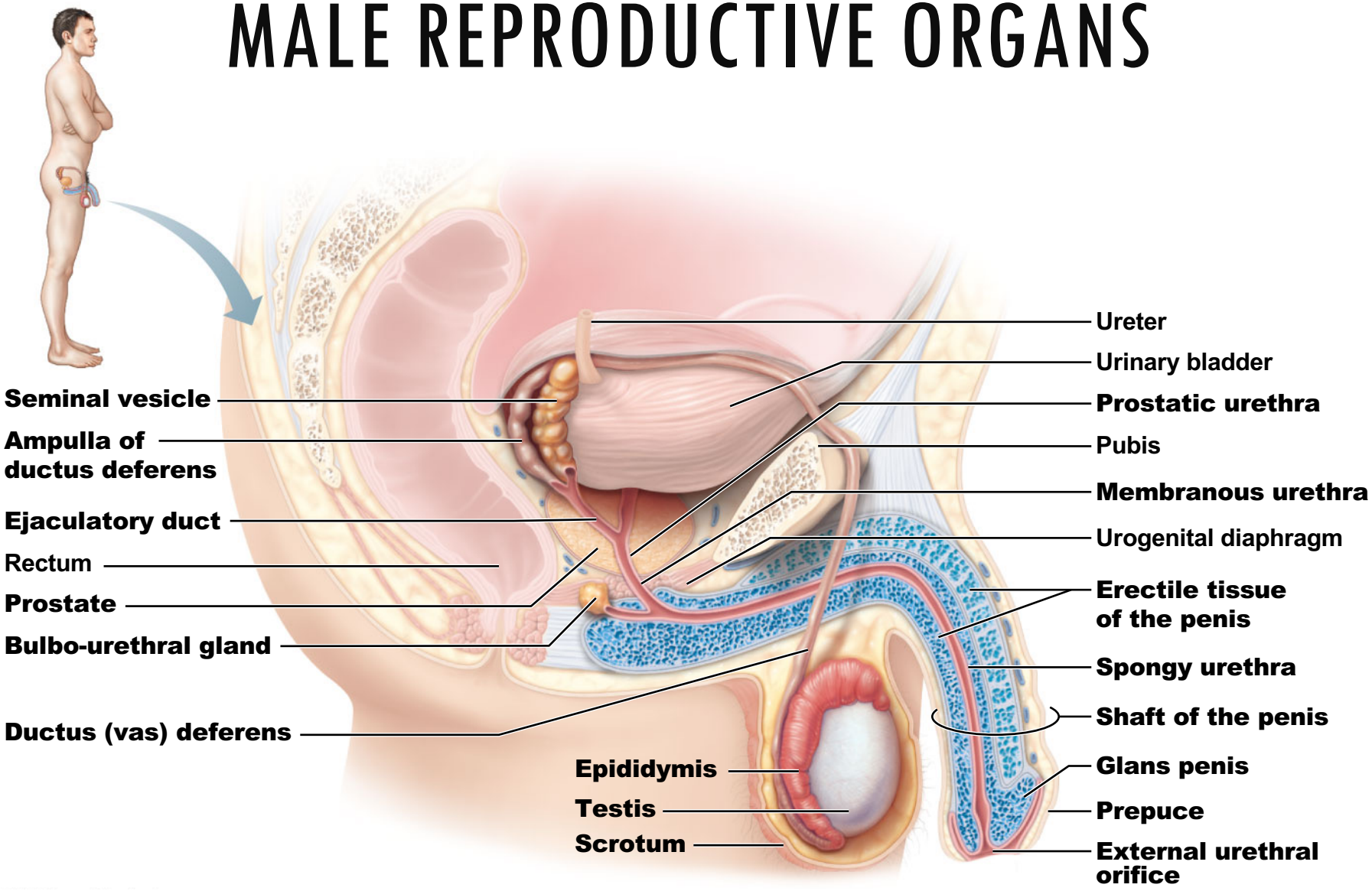
## ANATOMY

| | |
|---|---|
| **Testes** | **Accessory organs**<br>Seminal glands (vesicles)<br>Prostate<br>Bulbourethral glands |
| **Duct system**<br>Epididymis<br>Ductus (vas) deferens<br>Urethra | **External genitalia**<br>Penis<br>Scrotum |



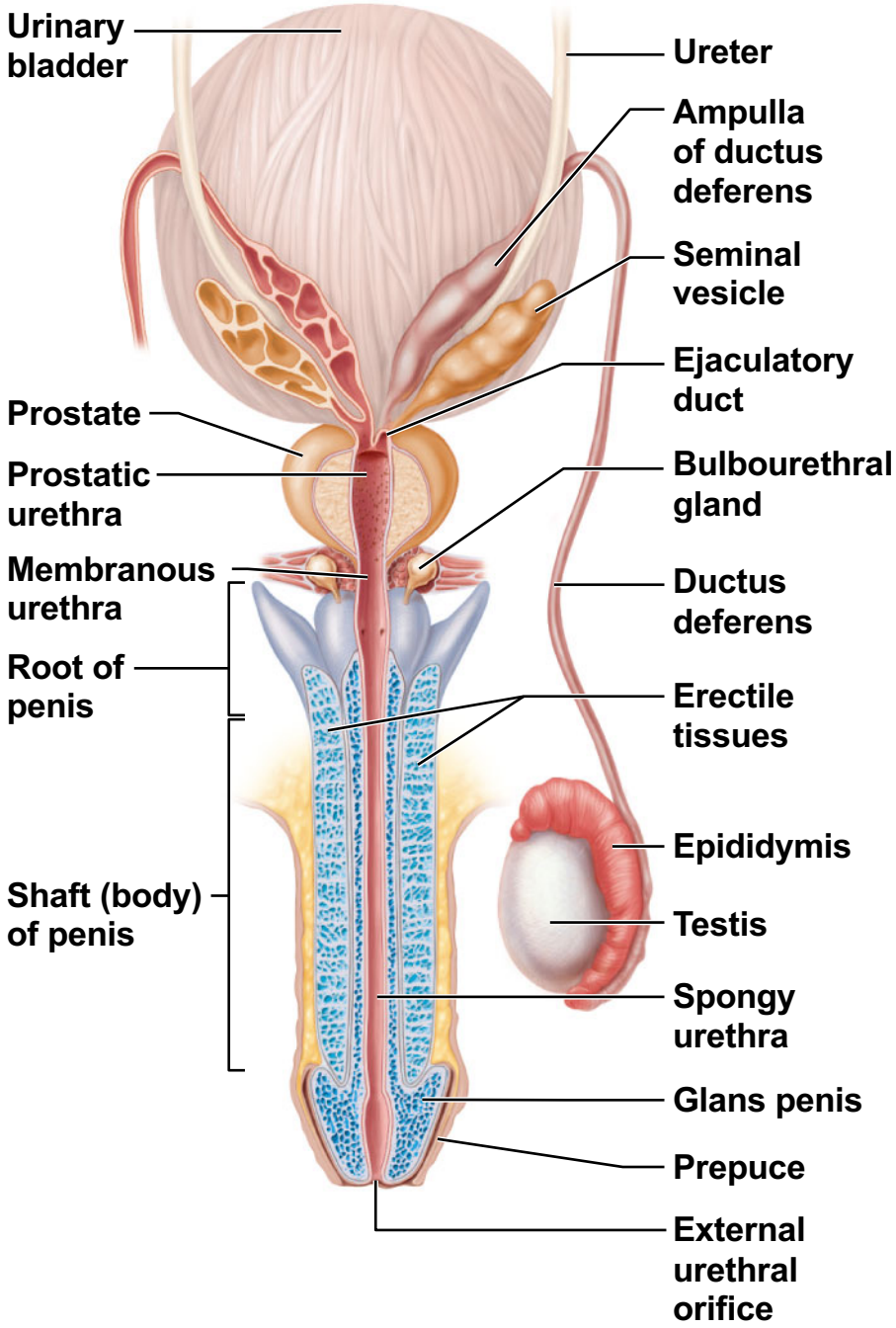

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# MALE REPRODUCTIVE ORGANS




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# MALE REPRODUCTIVE ORGANS

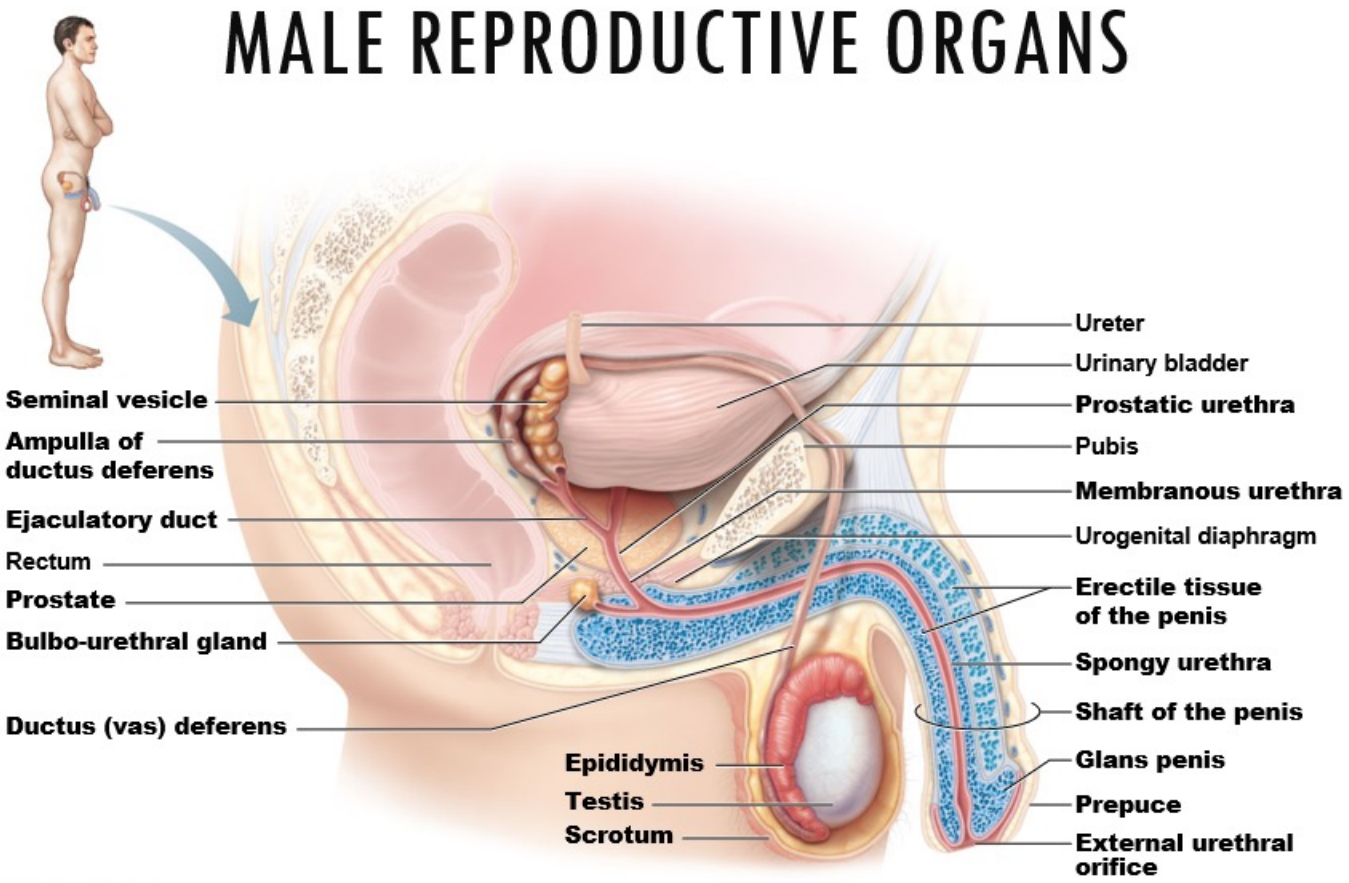



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# MALE REPRODUCTIVE SYSTEM

## ANATOMY

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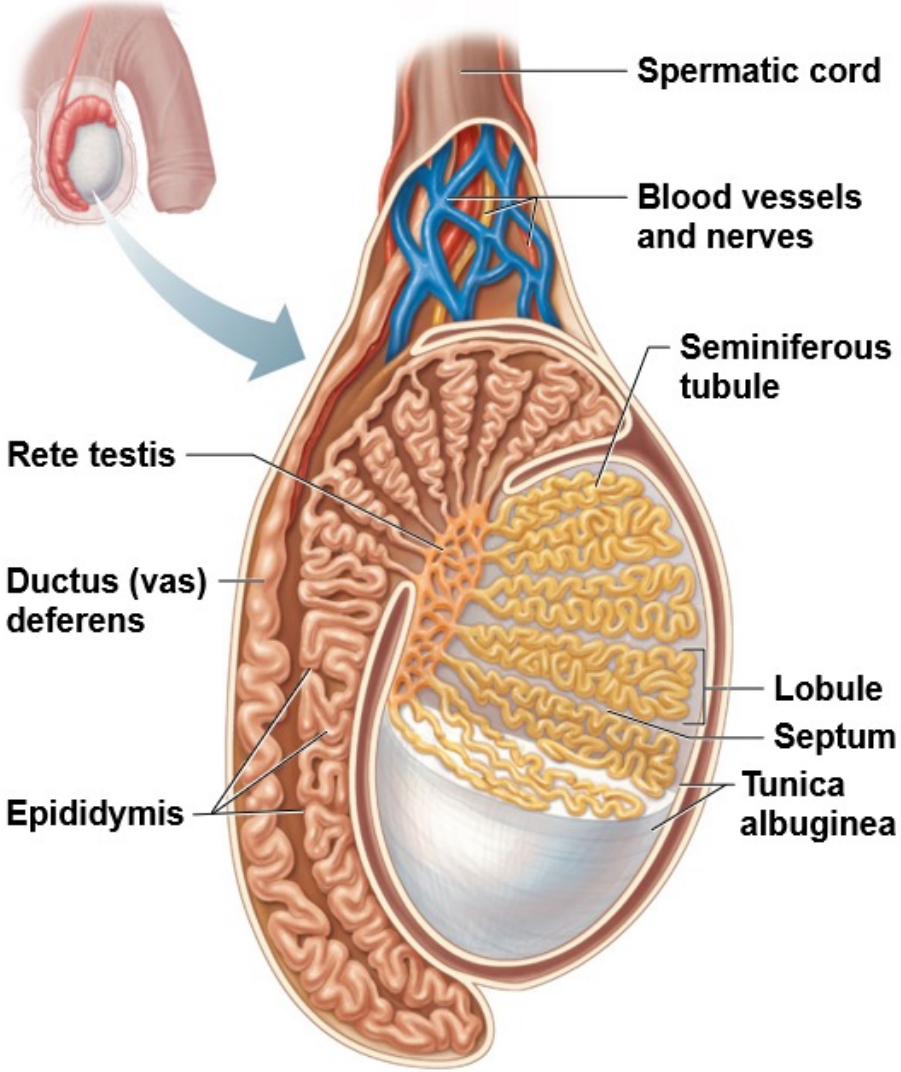
# ANATOMY OF MALE REPRODUCTIVE SYSTEM
## TESTES

Each testis is connected to the trunk via the spermatic cord, which houses:

- Blood vessels
- Nerves
- Ductus deferens

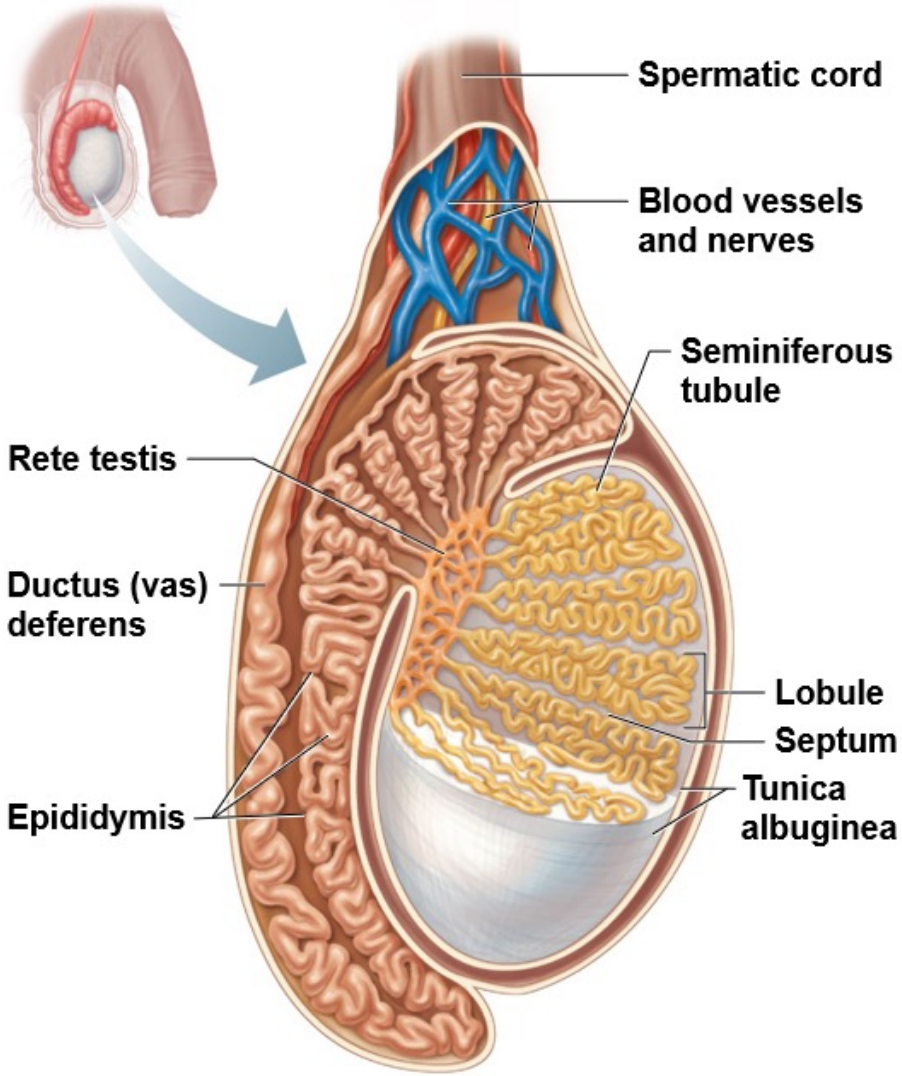
### Coverings of the testes

- Tunica albuginea—capsule that surrounds each testis
- Septa—extensions of the capsule that extend into the testis and divide it into lobules



# ANATOMY OF MALE REPRODUCTIVE SYSTEM
## TESTES

- Each lobule contains one to four seminiferous tubules
  - Tightly coiled structures
  - Function as sperm-forming factories
  - Empty sperm into the rete testis
- Sperm travels from the rete testis to the epididymis
- Interstitial cells in the seminiferous tubules produce androgens such as testosterone

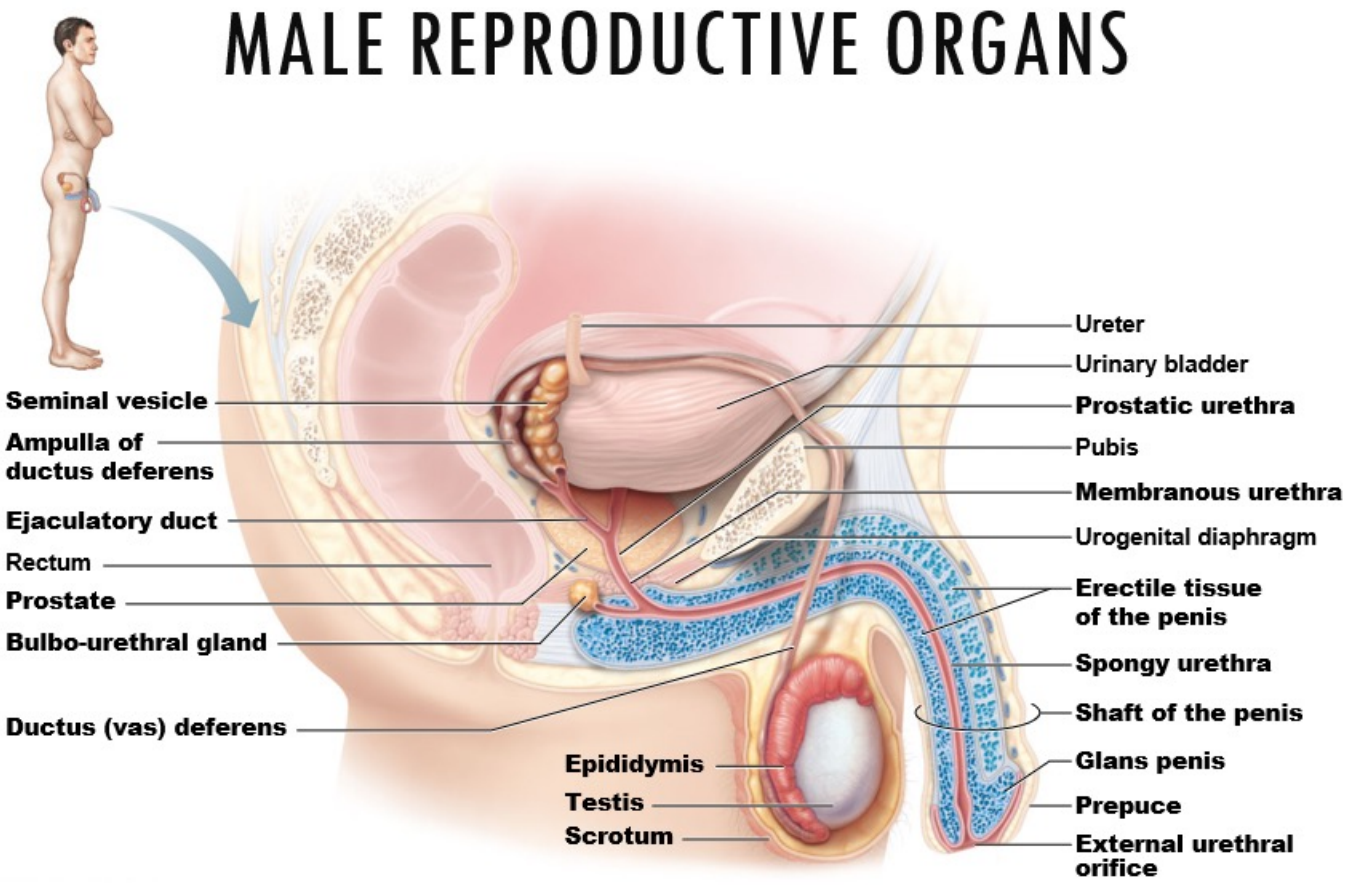



SAGITTAL SECTION OF THE TESTIS AND ASSOCIATED EPIDIDYMIS.

# MALE REPRODUCTIVE SYSTEM

## ANATOMY

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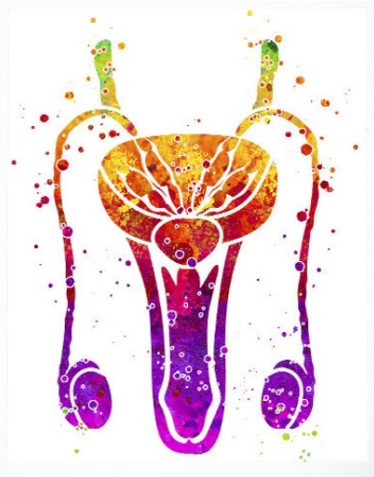



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# ANATOMY OF MALE REPRODUCTIVE SYSTEM
## DUCT SYSTEM

The duct system transports sperm from the body and includes:

- Epididymis
- Ductus deferens
- Urethra

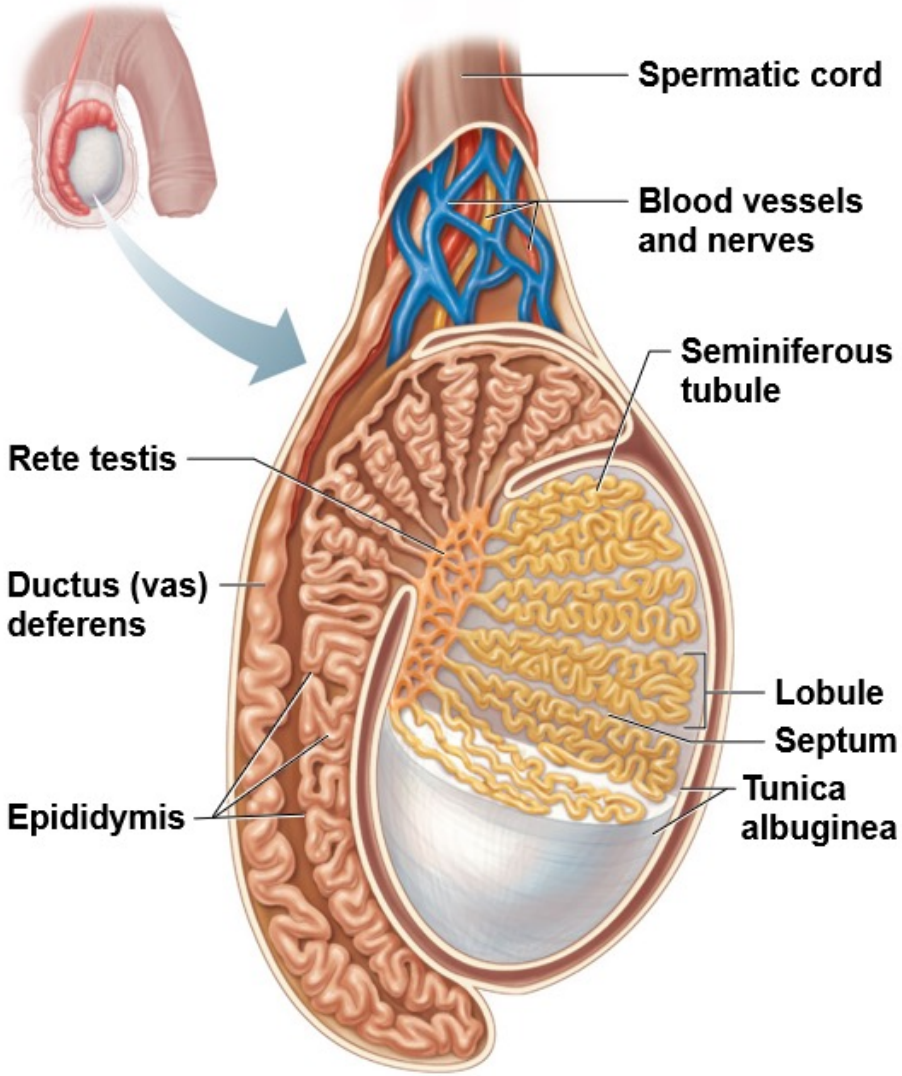


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# ANATOMY OF MALE REPRODUCTIVE SYSTEM
## DUCT SYSTEM

### Epididymis

- Highly convoluted tube 6 m (20 ft) long
- Found along the posterior lateral side of the testis
- First part of the male duct system
- Temporary storage site for immature sperm
- Sperm mature as they journey through the epididymis
- During ejaculation, sperm are propelled to the ductus deferens



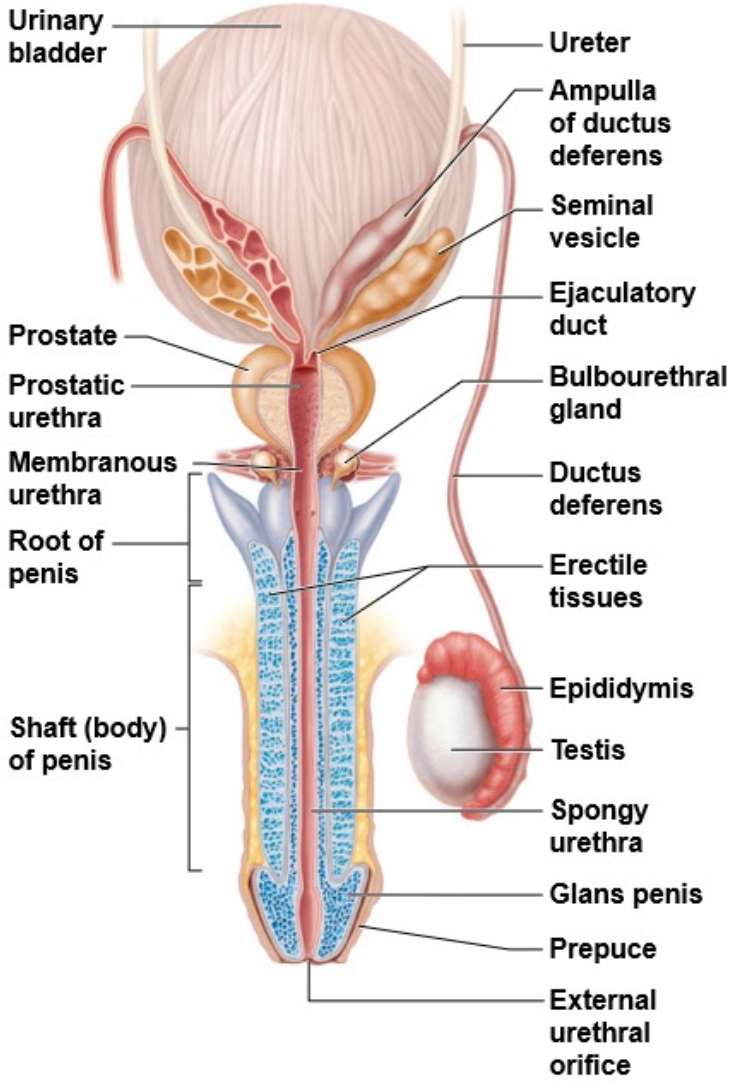

SAGITTAL SECTION OF THE TESTIS AND ASSOCIATED EPIDIDYMIS.

# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## DUCT SYSTEM

### Ductus (vas) deferens

- Runs from the epididymis via the spermatic cord through the inguinal canal and arches over the urinary bladder
  - Ampulla—end of the ductus deferens, which empties into the ejaculatory duct
  - Ejaculatory duct—passes through the prostate to merge with the urethra
- Moves sperm by peristalsis into the urethra

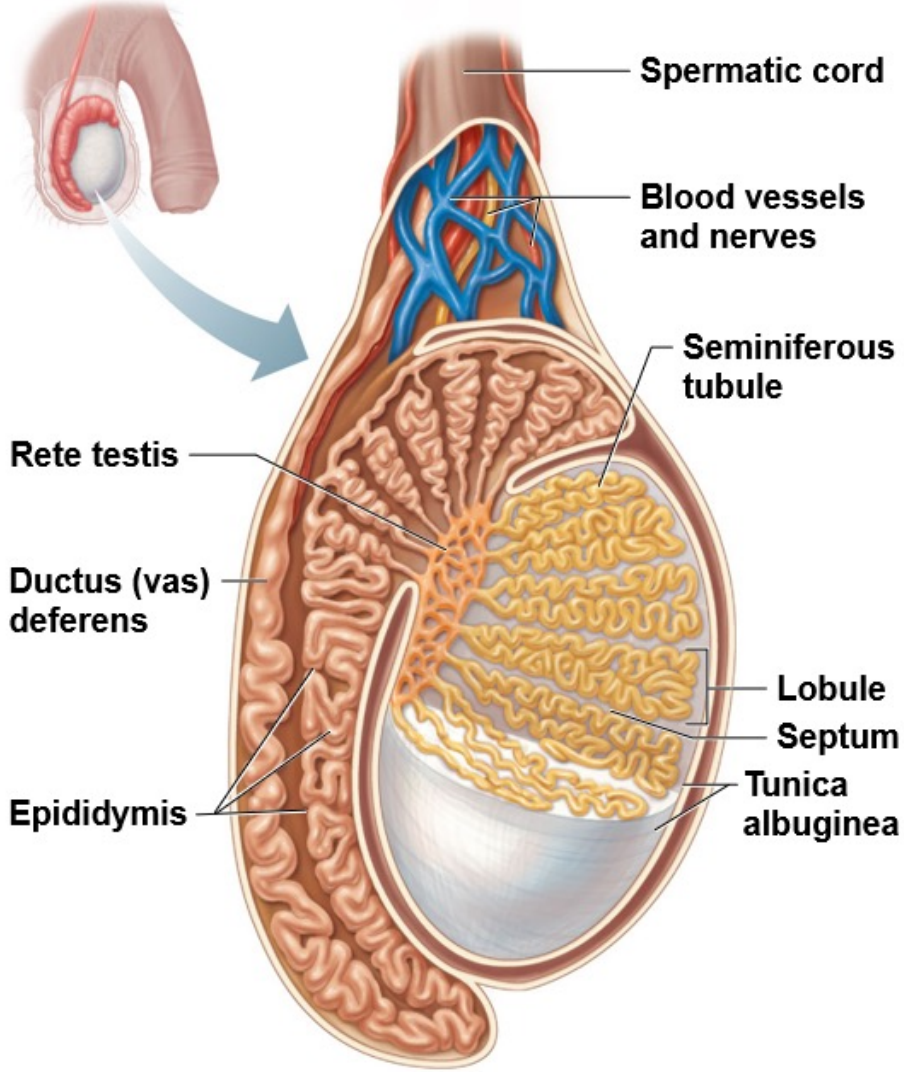


# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## DUCT SYSTEM

### Ductus (vas) deferens

- Ejaculation—smooth muscle in the walls of the ductus deferens create peristaltic waves to squeeze sperm forward
- Vasectomy—cutting of the ductus deferens at the level of the testes prevents transportation of sperm (form of birth control)



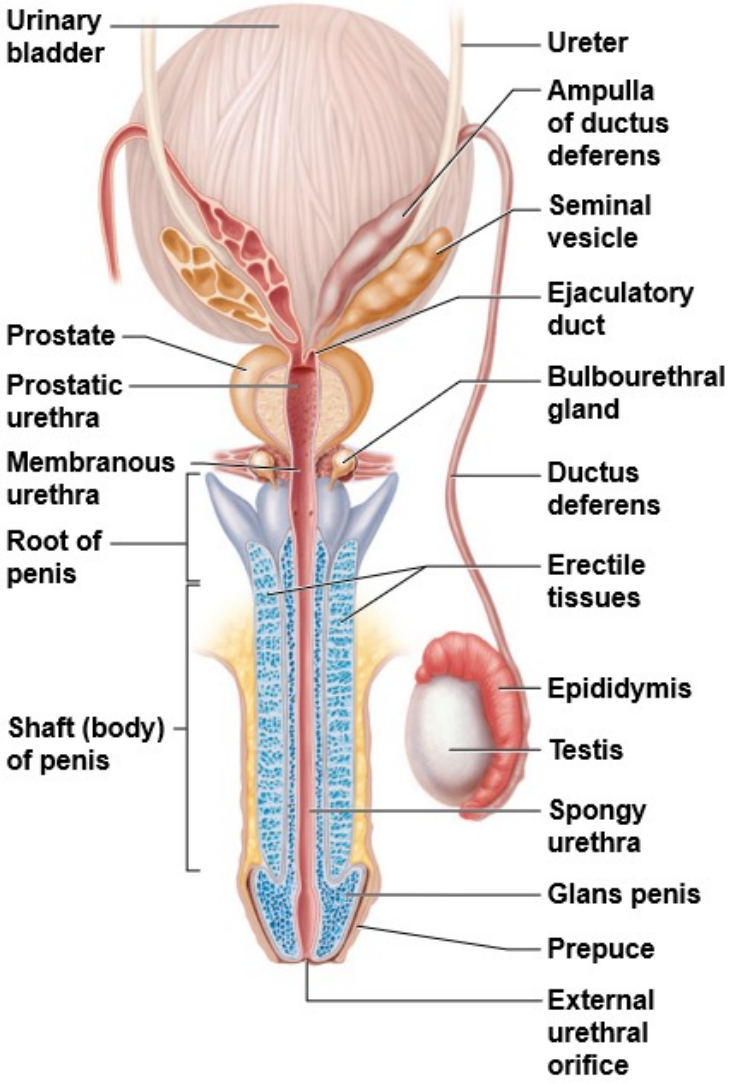

SAGITTAL SECTION OF THE TESTIS AND ASSOCIATED EPIDIDYMIS.

# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## DUCT SYSTEM

### Urethra

- Extends from the base of the urinary bladder to the tip of the penis
- Carries both urine and sperm
- Sperm enters from the ejaculatory duct



# ANATOMY OF MALE REPRODUCTIVE SYSTEM

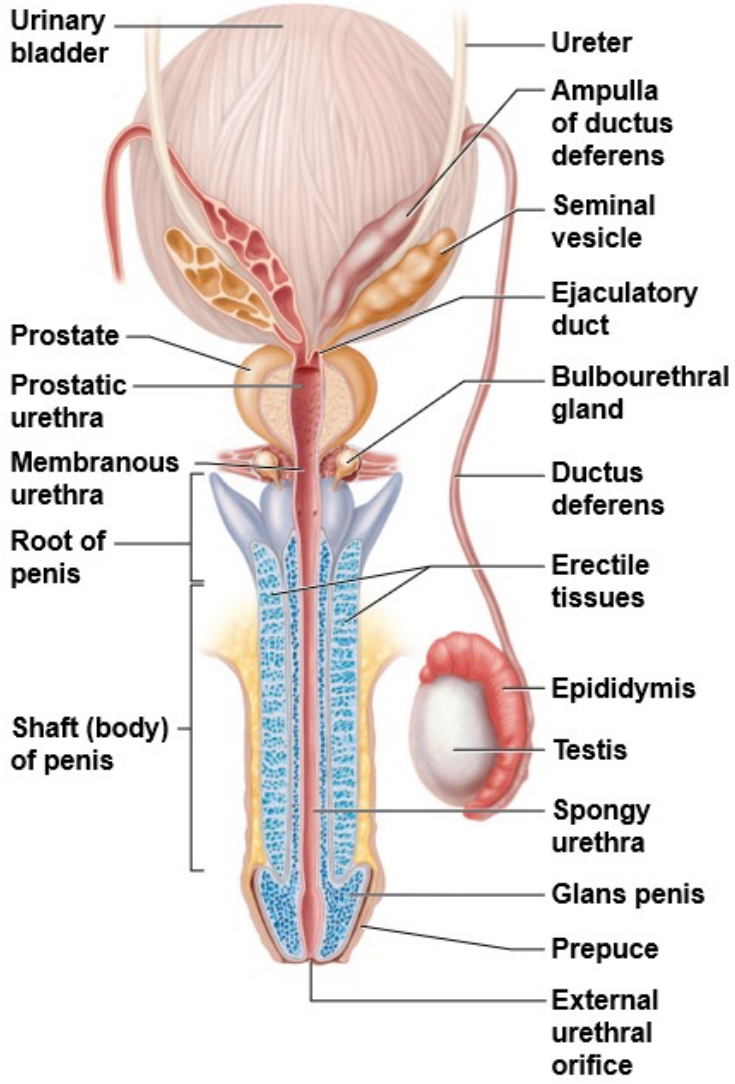
## DUCT SYSTEM

Urethra regions

1. Prostatic urethra—surrounded by prostate gland
2. Membranous urethra—prostatic urethra to penis
3. Spongy (penile) urethra—runs the length of the penis to the external urethral orifice

Ejaculation causes the internal urethra sphincter to close

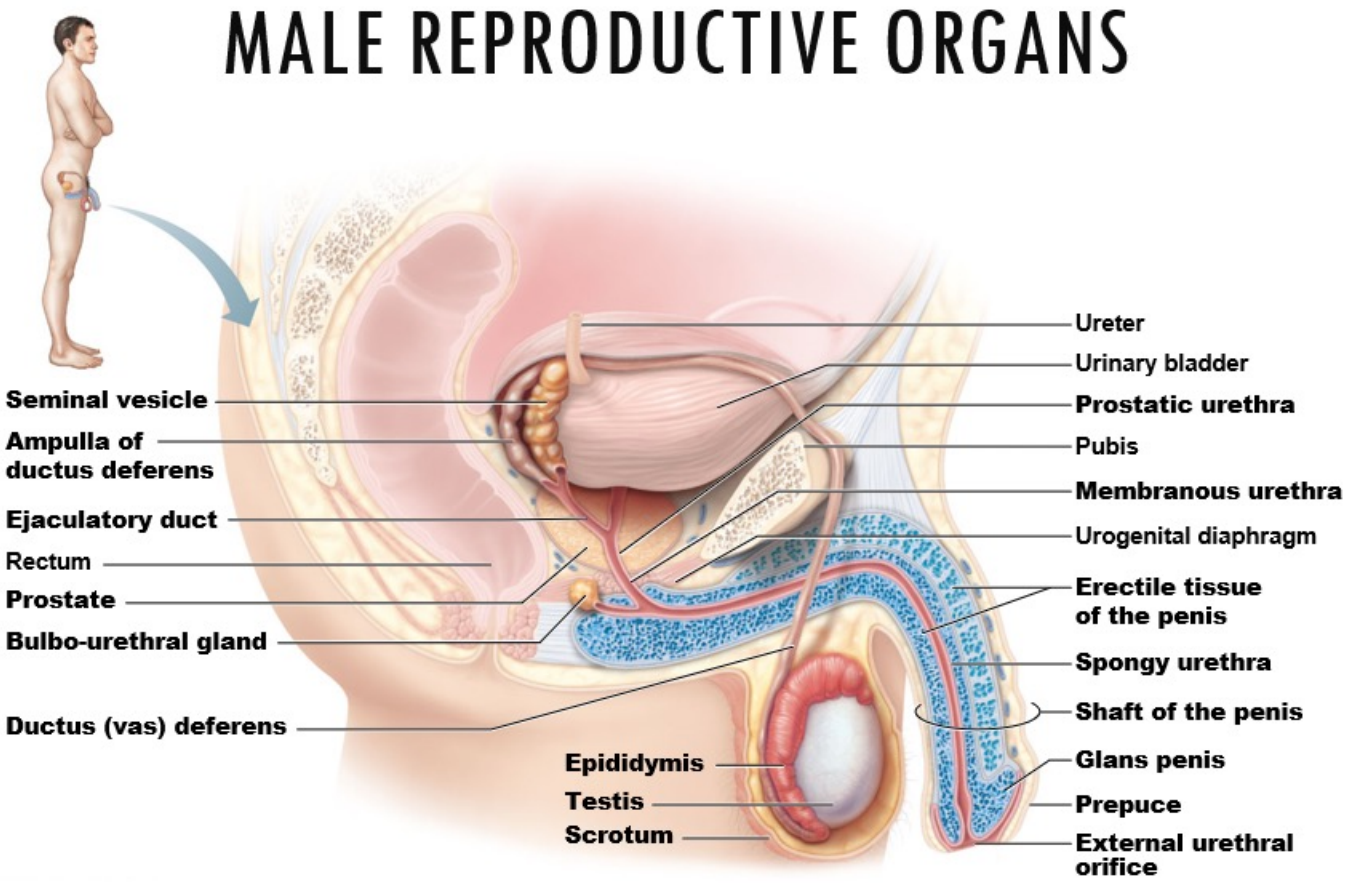
- Prevents urine from passing into the urethra
- Prevents sperm from entering the urinary bladder



# MALE REPRODUCTIVE SYSTEM

## ANATOMY

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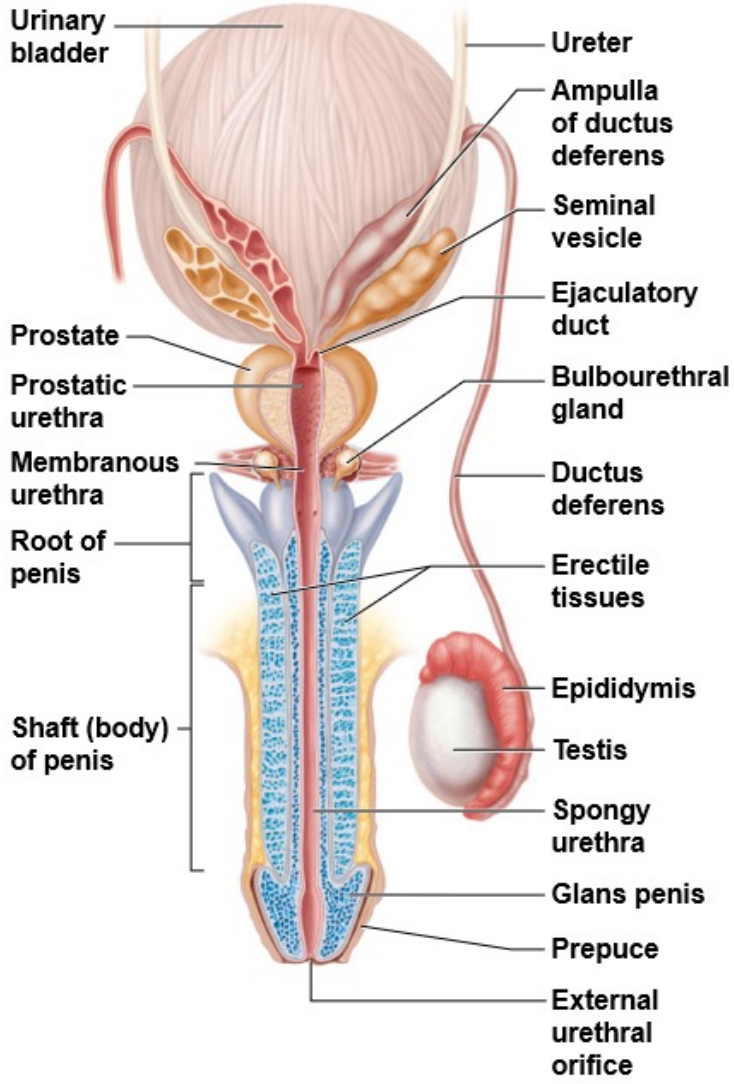

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# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## ACCESSORY GLANDS AND SEMEN

### Seminal vesicles

- Located at the base of the bladder
- Produce a thick, yellowish secretion (60% of semen) that contains:
  - Fructose (sugar)
  - Vitamin C
  - Prostaglandins
  - Other substances that nourish and activate sperm
- Duct of each seminal vesicle joins that of the ductus deferens on each side to form the ejaculatory duct

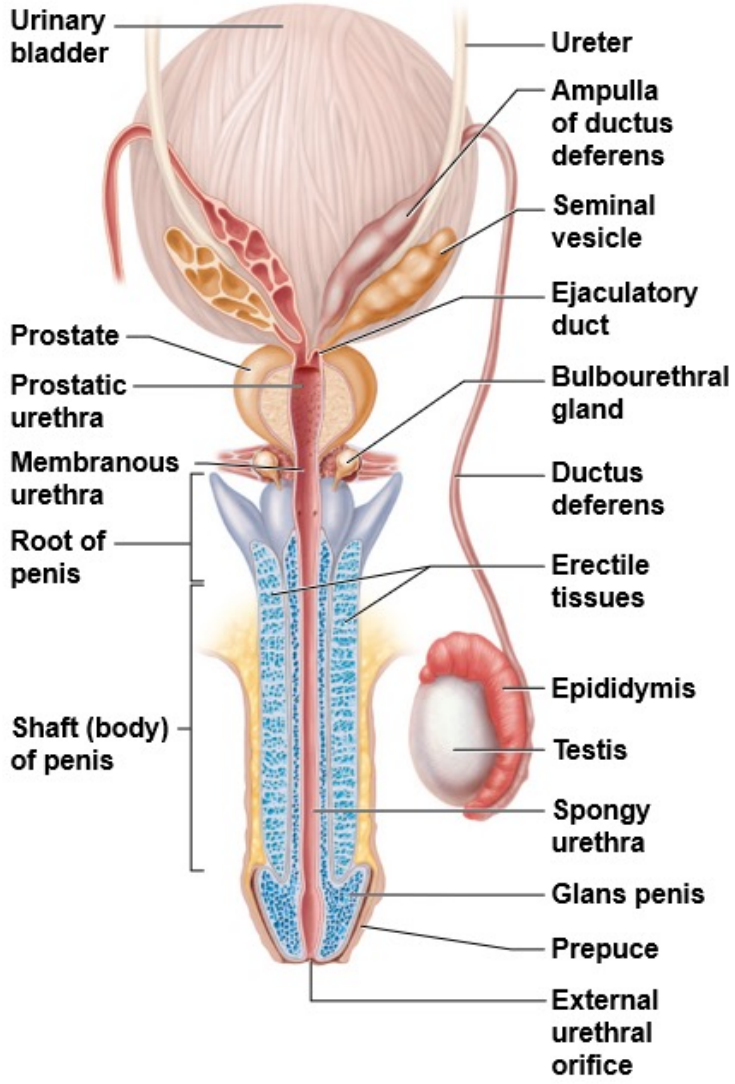


# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## ACCESSORY GLANDS AND SEMEN

### Prostate

- Encircles the upper (prostatic) part of the urethra
- Secretes a milky fluid
  - Helps to activate sperm
  - Fluid enters the urethra through several small ducts

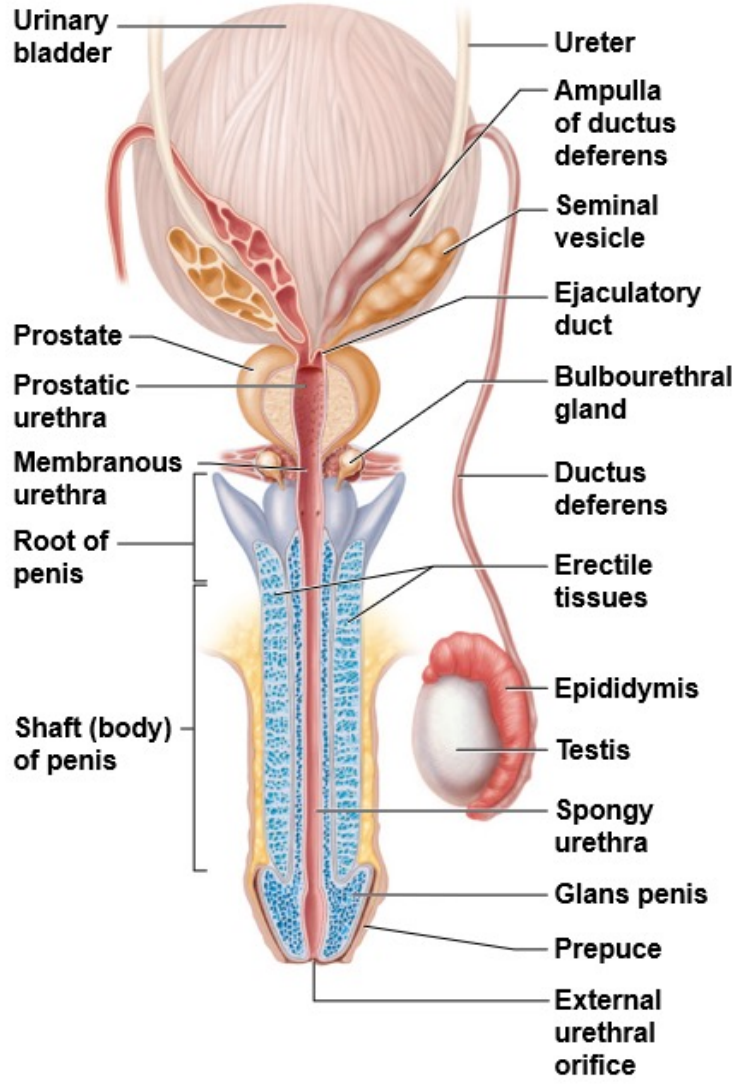


# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## ACCESSORY GLANDS AND SEMEN

### Bulbourethral glands

- Pea-sized glands inferior to the prostate
- Produce a thick, clear mucus
  - Mucus cleanses the spongy (penile) urethra of acidic urine prior to ejaculation
  - Mucus serves as a lubricant during sexual intercourse

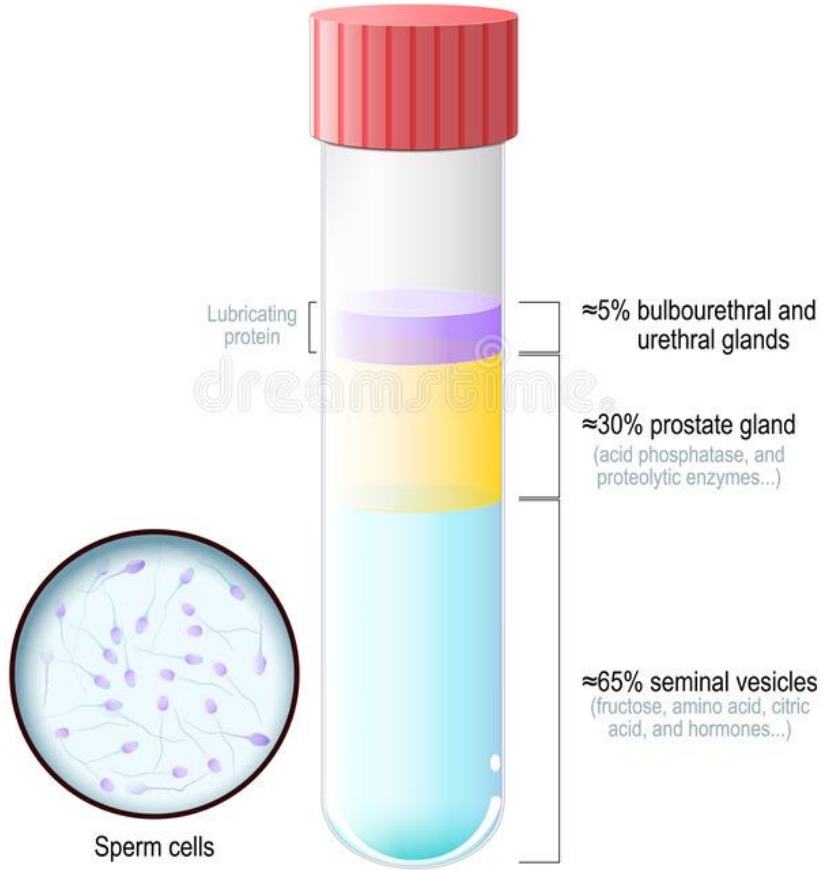


# ANATOMY OF MALE REPRODUCTIVE SYSTEM
## ACCESSORY GLANDS AND SEMEN



### Semen

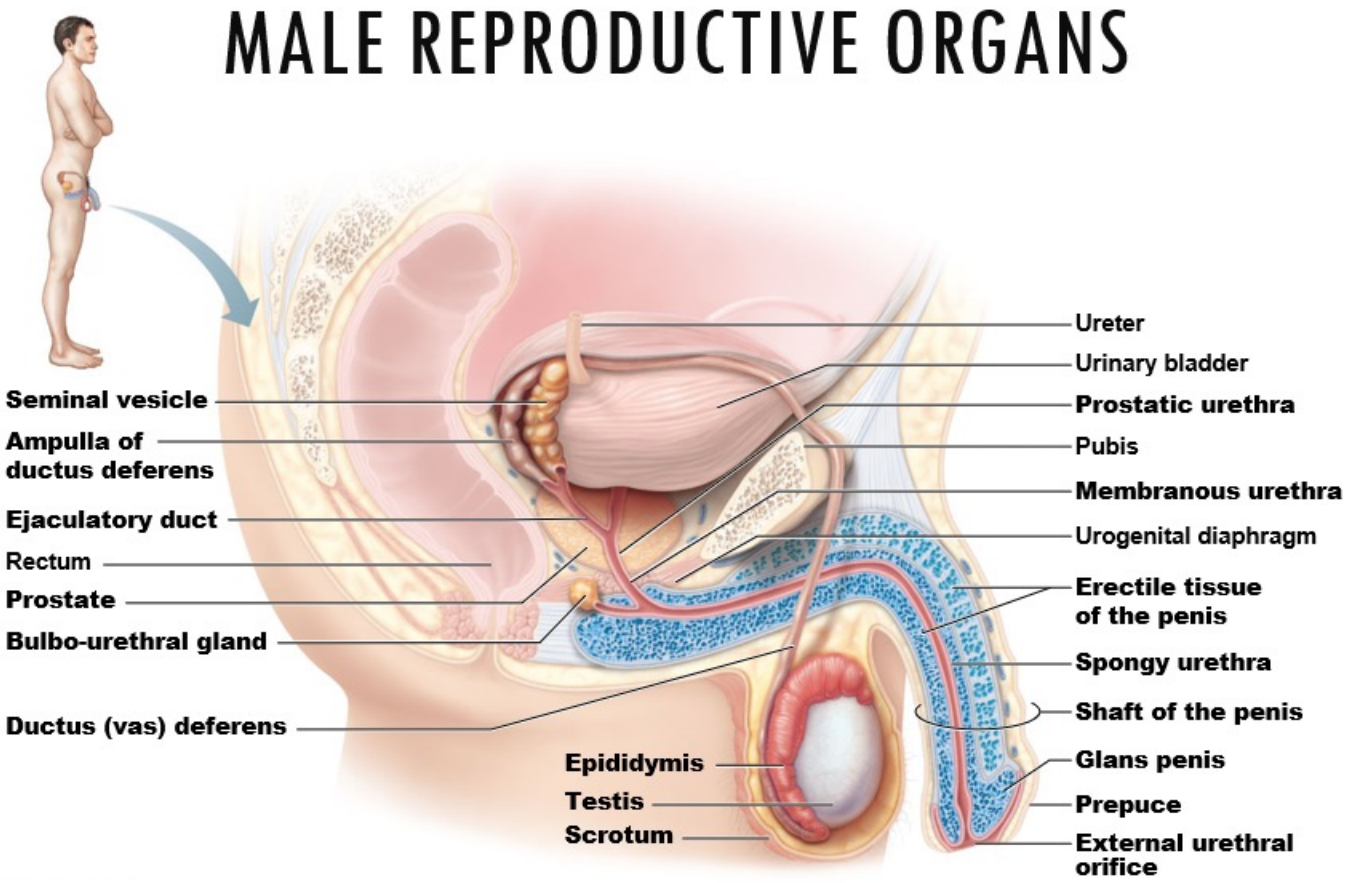
- Milky white mixture of sperm and accessory gland secretions
  - Liquid portion acts as a transport medium to dilute sperm
  - Sperm are streamlined cellular "tadpoles"
  - Fructose provides energy for sperm cells
  - Alkalinity of semen helps neutralize the acidic environment of vagina
- Semen inhibits bacteria



# MALE REPRODUCTIVE SYSTEM

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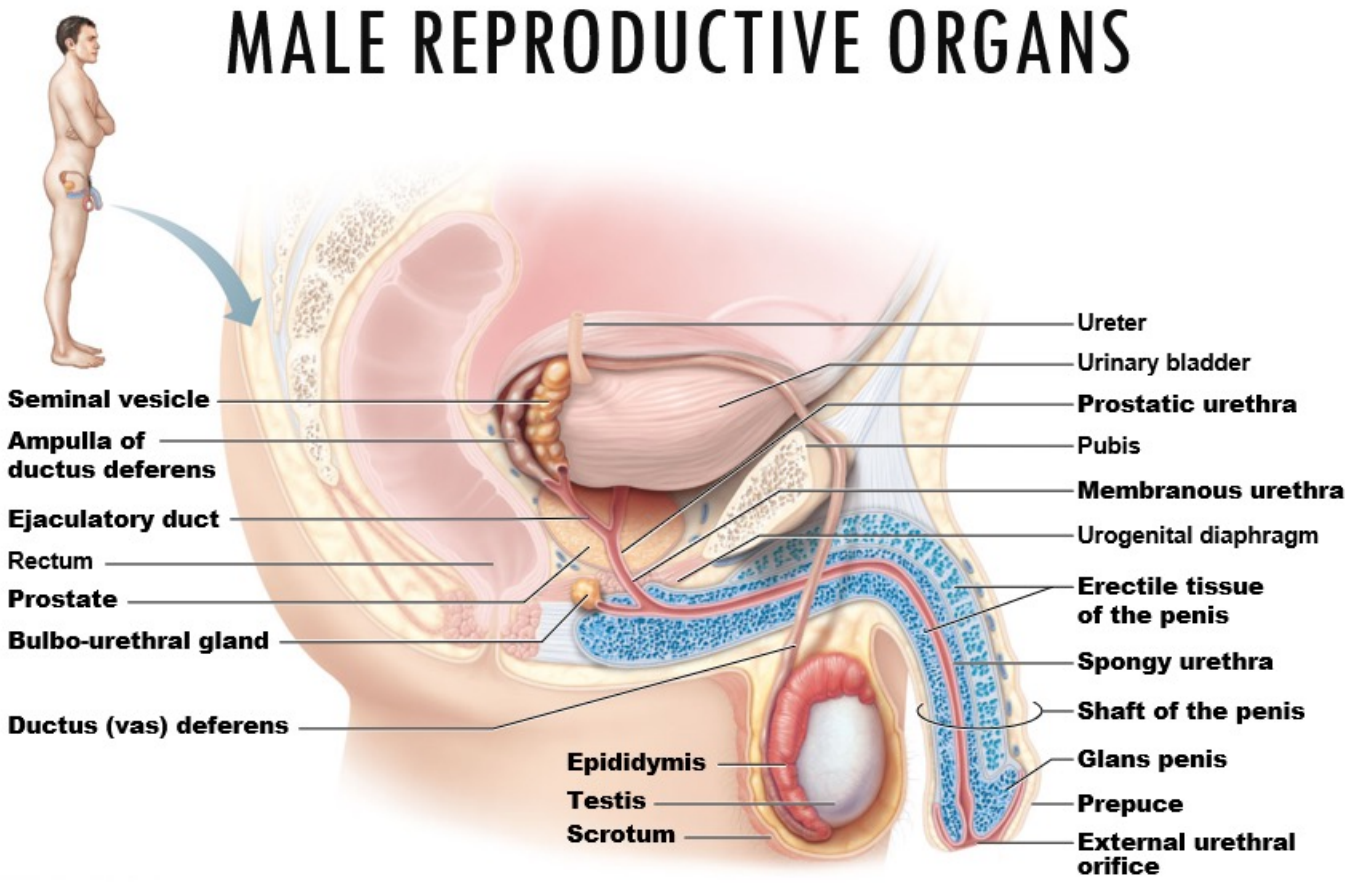

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# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## EXTERNAL GENITALIA

### Scrotum

- Divided sac of skin outside the abdomen that houses the testes
- Viable sperm cannot be produced at normal body temperature
- Maintains testes at 3°C lower than normal body temperature



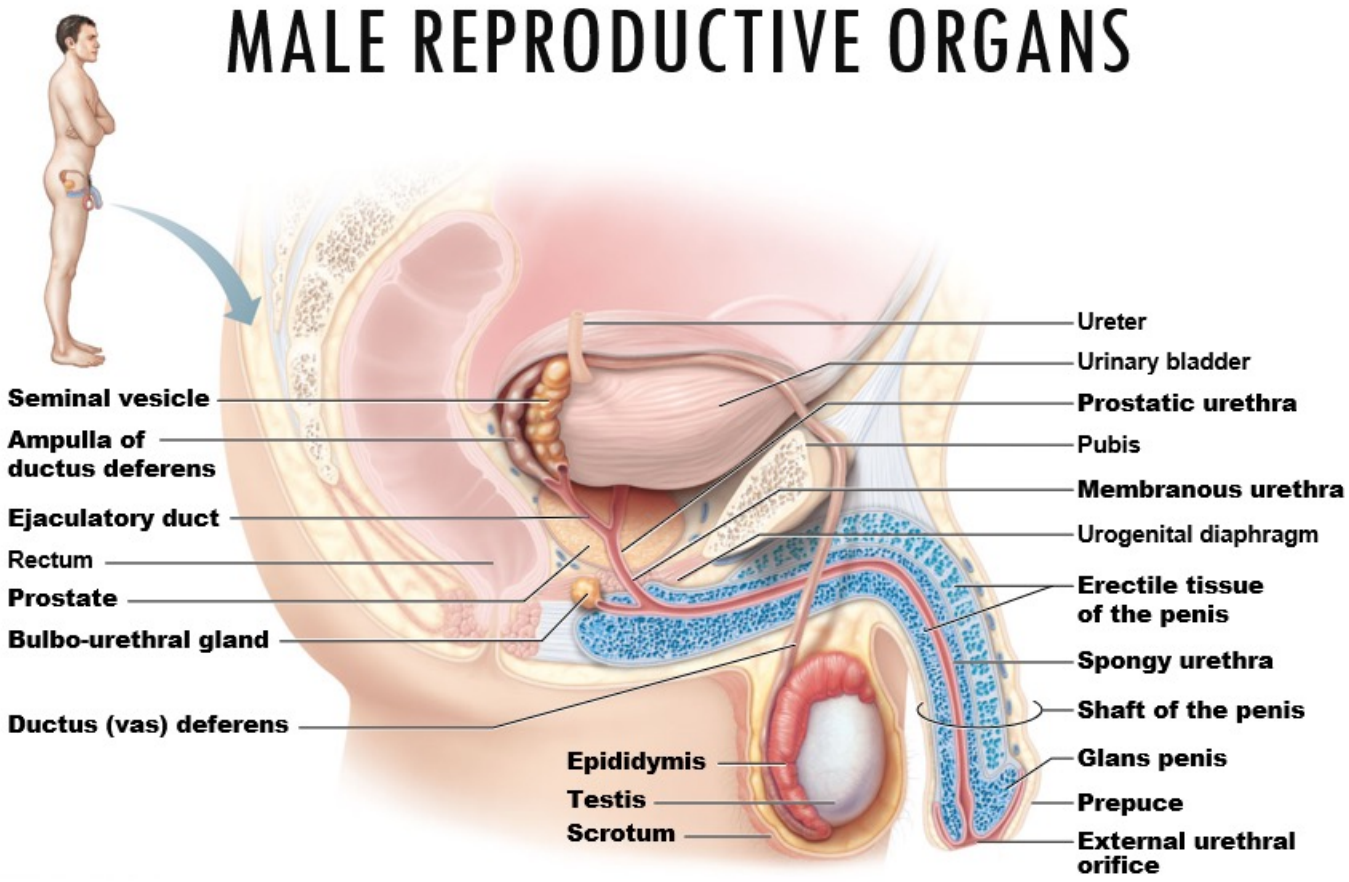

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# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## EXTERNAL GENITALIA

Penis

- Male organ of copulation that delivers sperm into the female reproductive tract
- Regions of the penis
  - Shaft
  - Glans penis (enlarged tip)
  - Prepuce (foreskin)
    - Folded cuff of skin around proximal end
    - Often removed by circumcision



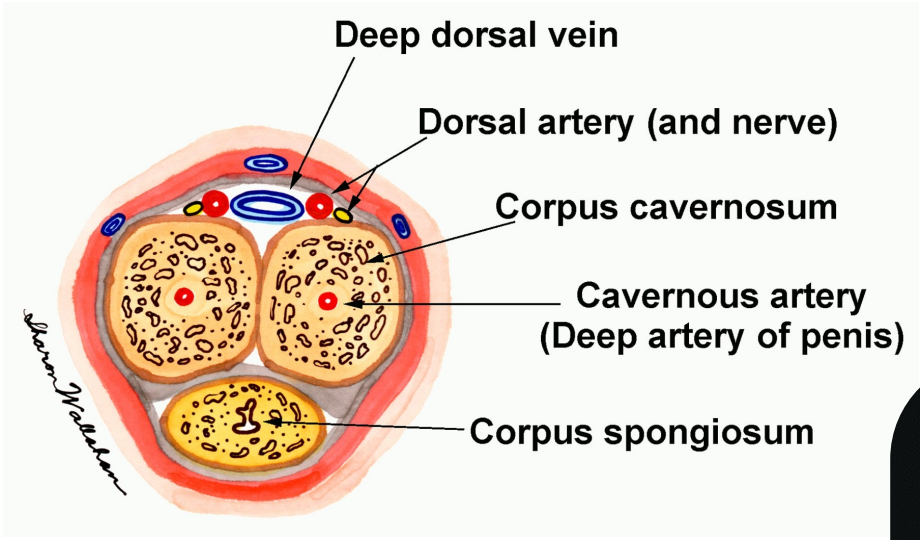

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# ANATOMY OF MALE REPRODUCTIVE SYSTEM

## EXTERNAL GENITALIA

### Penis

- Internally there are three areas of spongy erectile tissue around the urethra
- Erections occur when this erectile tissue fills with blood during sexual excitement

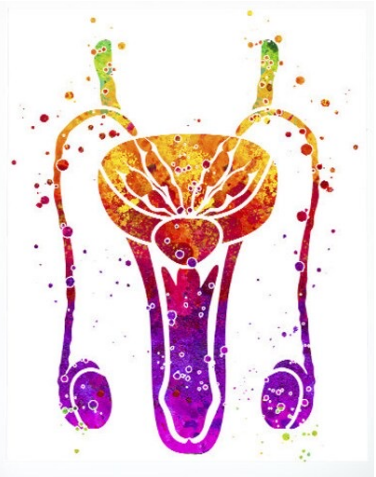





# MALE REPRODUCTIVE SYSTEM
## FUNCTIONS

Chief roles of the male in the reproductive process

- Produce sperm
- Produce a hormone, testosterone



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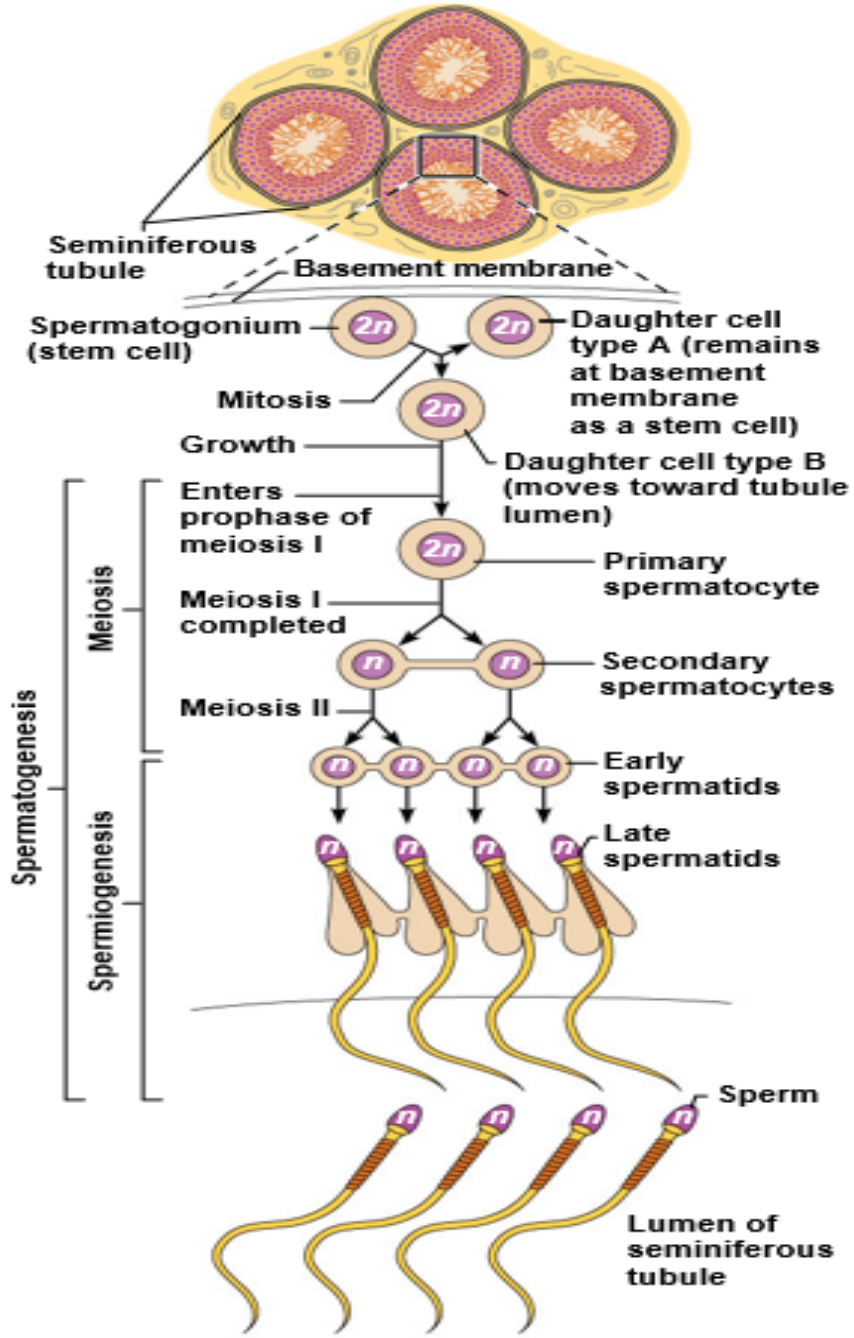
# MALE REPRODUCTIVE SYSTEM
## SPERMATOGENESIS

Sperm production

- Begins at puberty and continues throughout life
- Millions of sperm are made every day

Sperm are formed in the seminiferous tubules of the testis

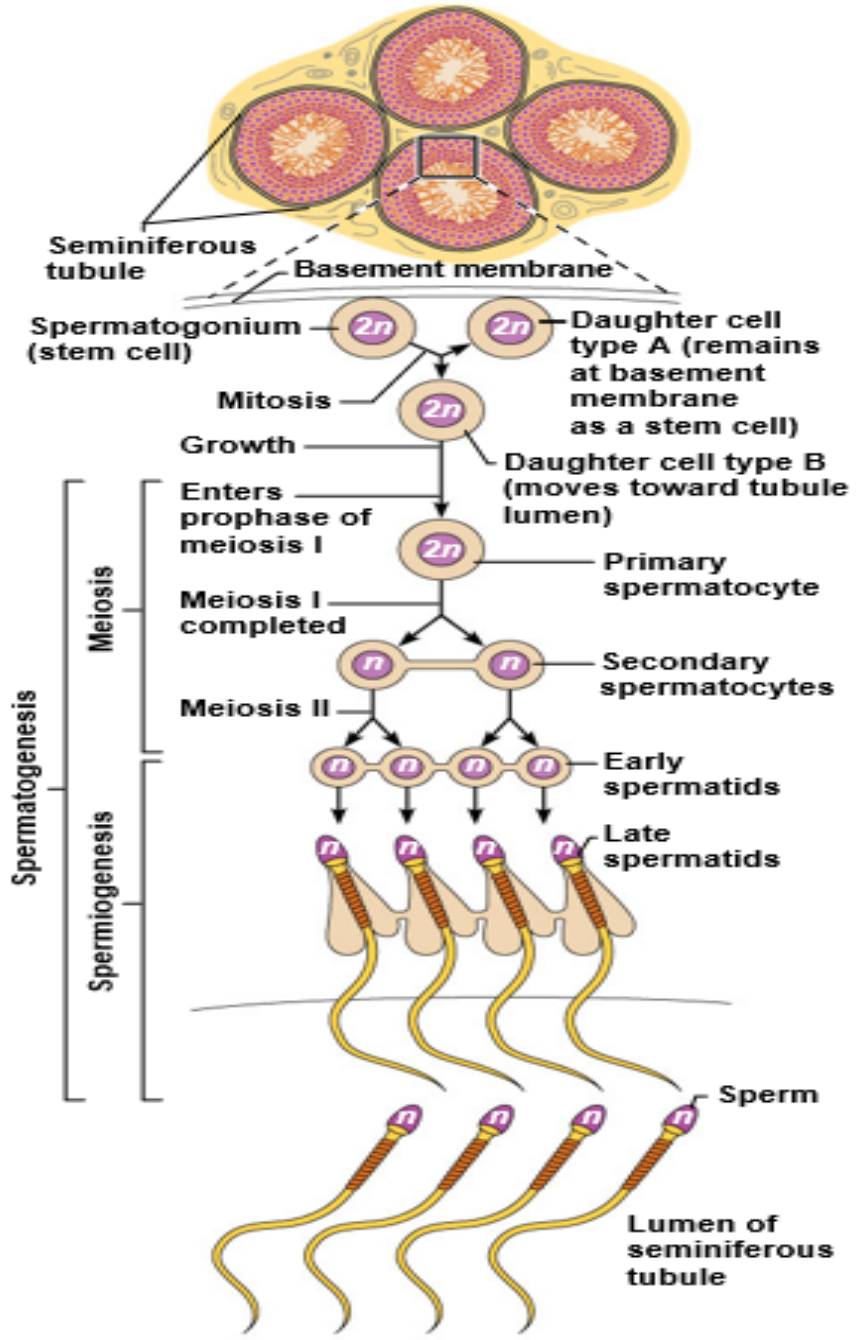
- Spermatogonia (primitive stem cells) begin the process by dividing rapidly
- During puberty, follicle-stimulating hormone (FSH) is secreted in increasing amounts



# MALE REPRODUCTIVE SYSTEM
## SPERMATOGENESIS

Each division of a spermatogonium stem cell produces:

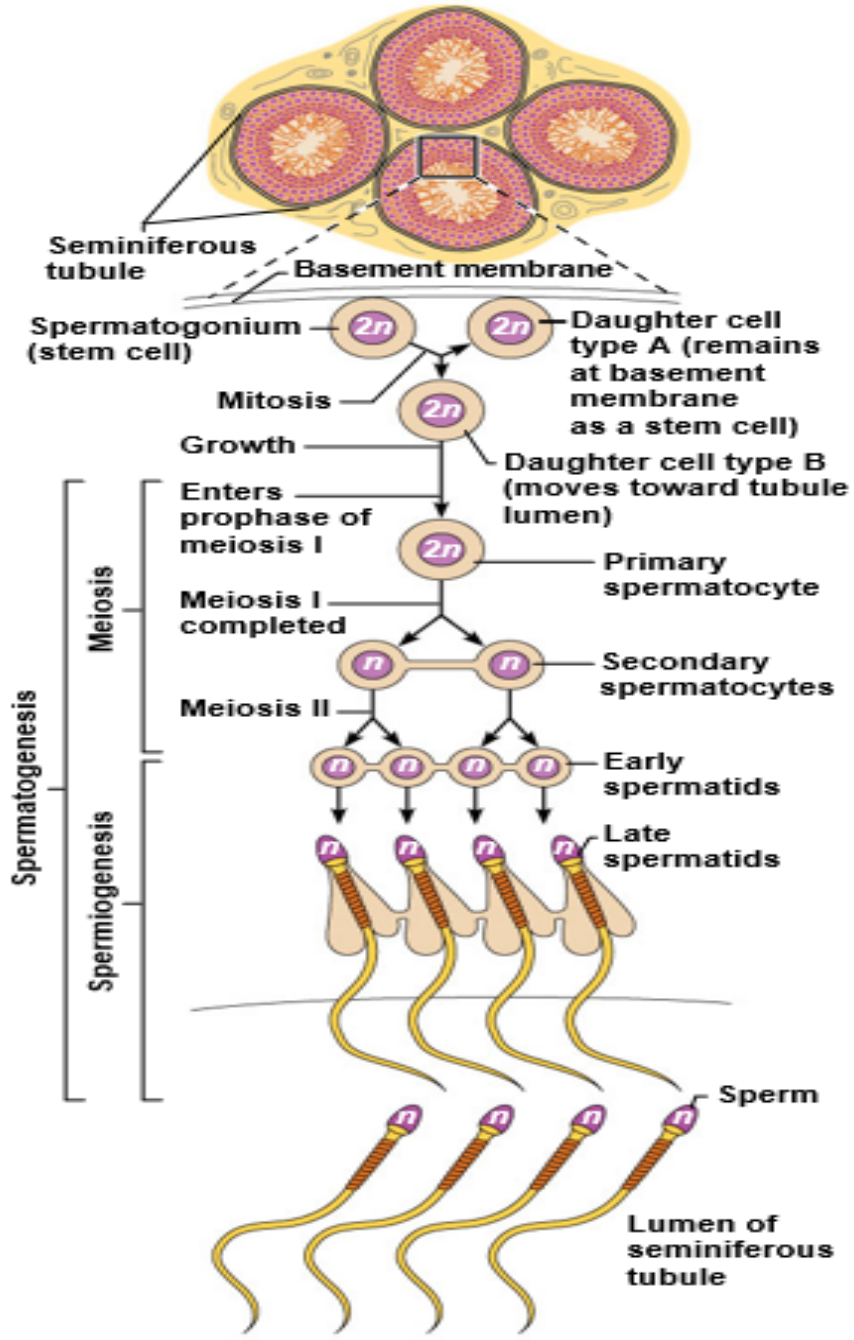
- *Type A daughter cell*, a stem cell, that continues the stem cell population
- *Type B daughter cell*, which becomes a primary spermatocyte, destined to undergo meiosis and spermiogenesis



# MALE REPRODUCTIVE SYSTEM
## SPERMATOGENESIS

### Meiosis

- Special type of nuclear division that differs from mitosis
- Occurs in the gonads
- Includes two successive divisions of the nucleus (meiosis I and II)
- Results in four daughter cells (gametes)

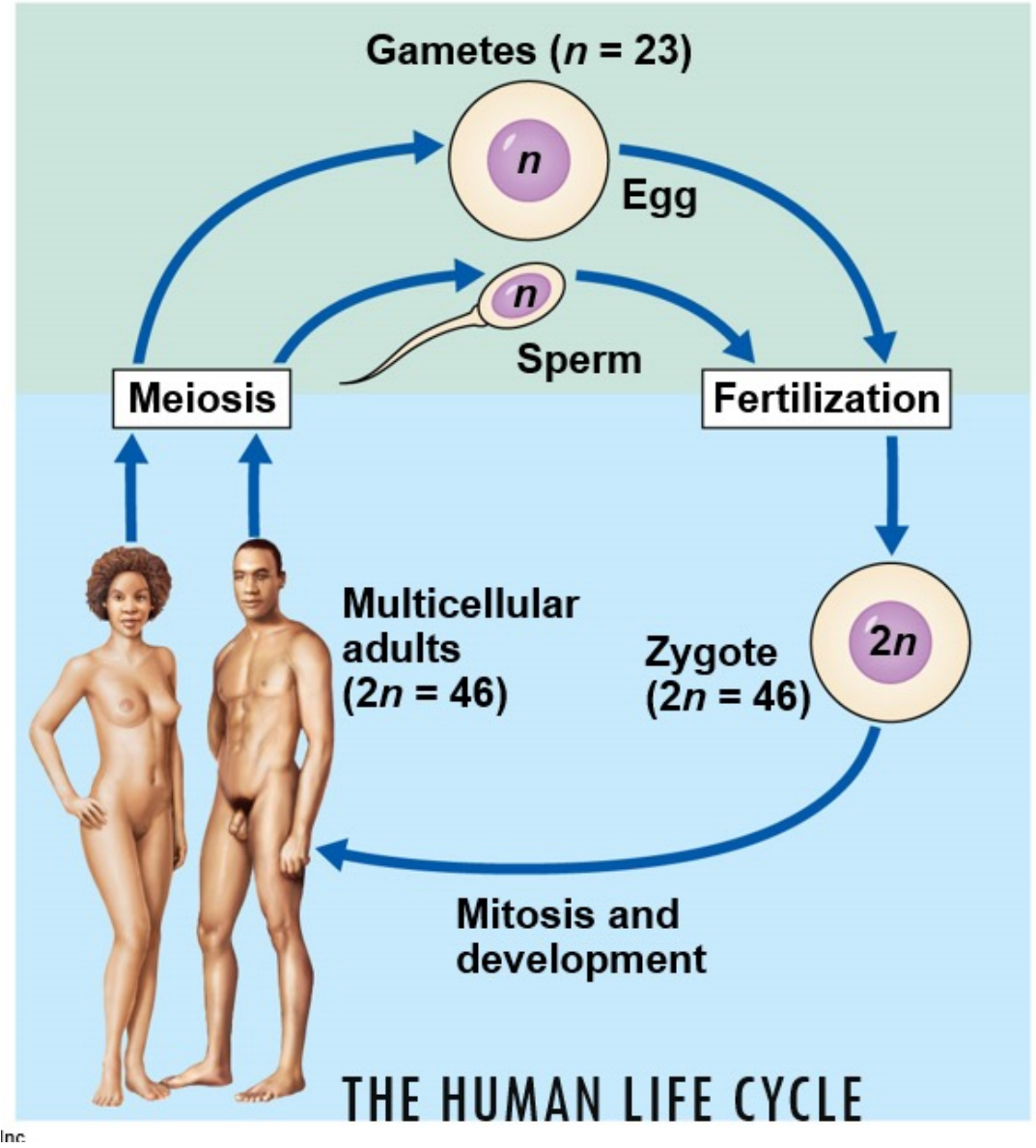


# HUMAN LIFE CYCLE

Gametes are spermatids with 23 chromosomes

- 23 chromosomes
  - are half the usual 46 found in other body cells
  - known as the haploid number (*n*)—half the genetic material as other body cells

Union of a sperm (23 chromosomes, *n*) with an egg (23 chromosomes, *n*) creates a zygote (2*n*, or 46 chromosomes)

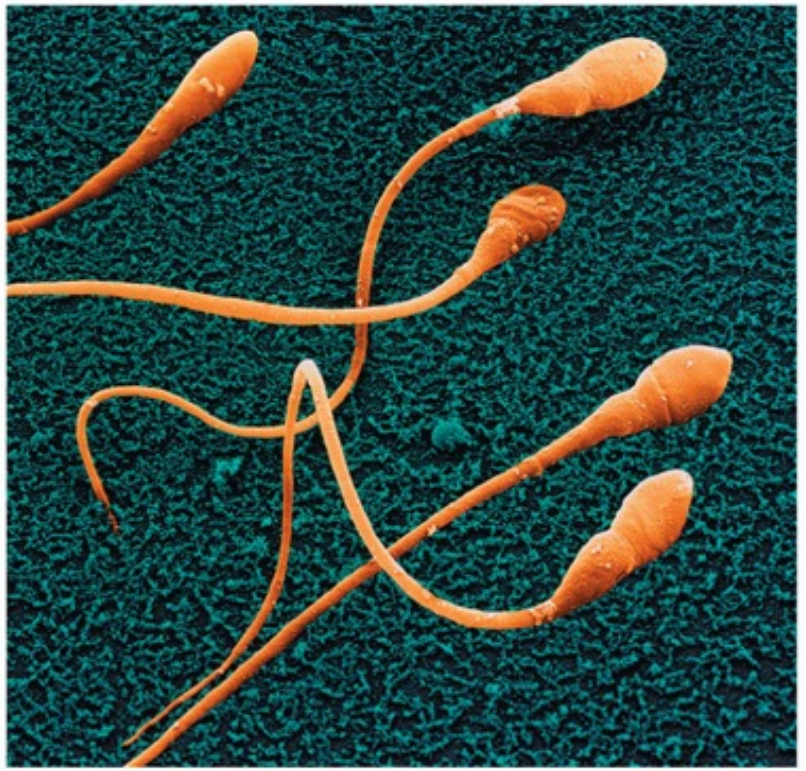



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# MALE REPRODUCTIVE SYSTEM
## SPERMIOGENESIS

Spermiogenesis

- Spermatids are non-motile and not functional as sperm
- A streamlining process is needed to strip excess cytoplasm from a spermatid and modify it into a sperm
- Creation of mature sperm

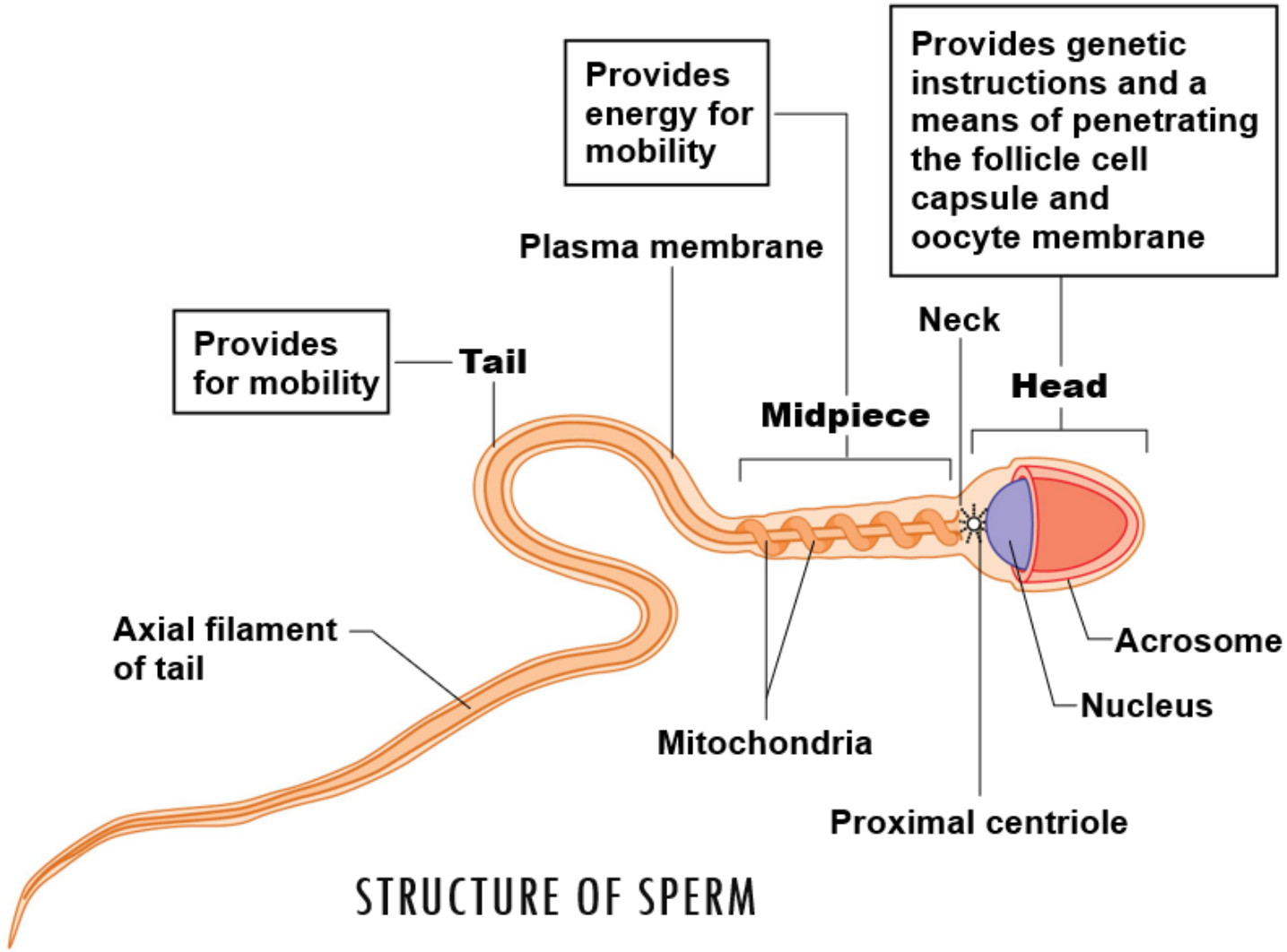


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# MALE REPRODUCTIVE SYSTEM
## SPERMIOGENESIS

### Spermiogenesis

- A sperm has three regions: head, midpiece, tail
- Acrosome sits anterior to the sperm head (nucleus)




STRUCTURE OF SPERM

# MALE HORMONES

During puberty:

- Follicle-stimulating hormone (FSH) begins prodding seminiferous tubules to produce sperm
- Luteinizing hormone (LH) begins activating the interstitial cells to produce testosterone

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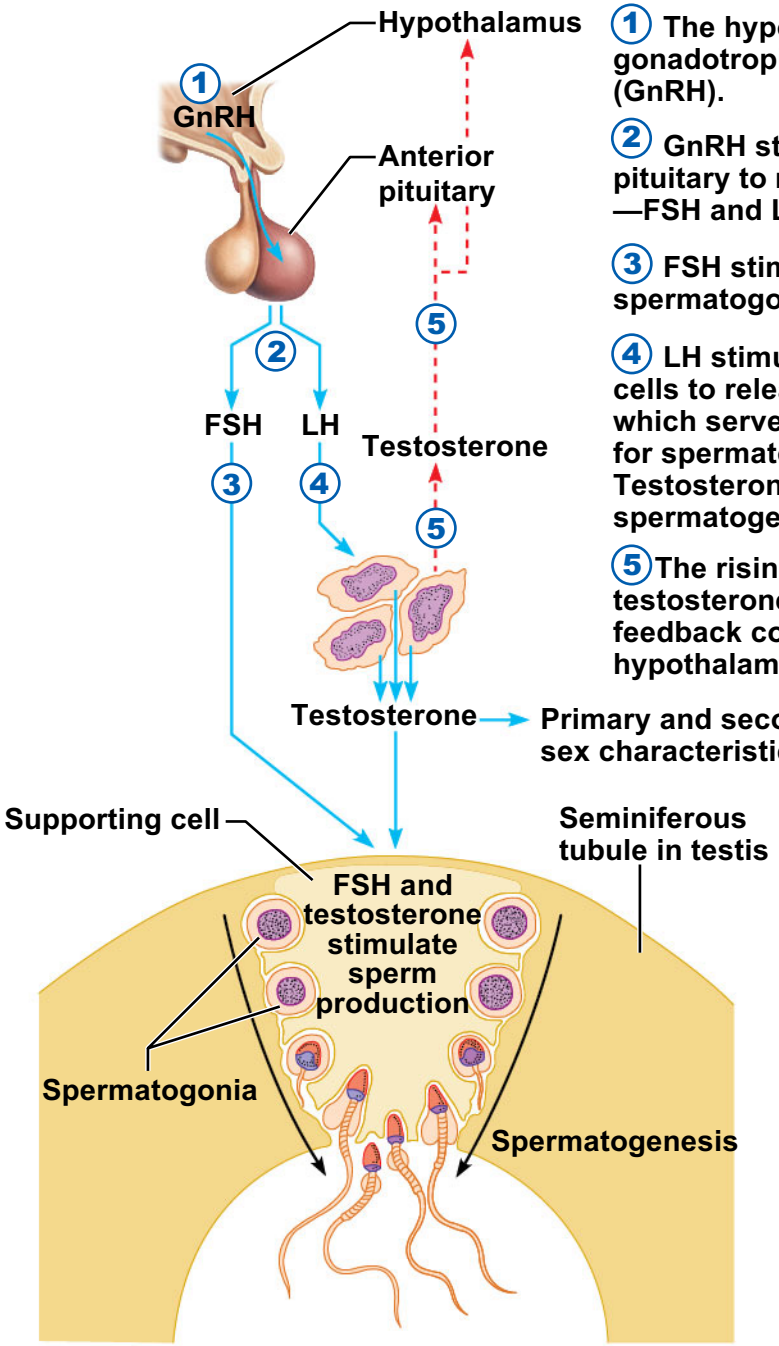
# MALE HORMONES

Testosterone

- Most important hormonal product of the testes
- Stimulates reproductive organ development
- Underlies sex drive
- Causes secondary sex characteristics
  - Deepening of voice
  - Increased hair growth
  - Enlargement of skeletal muscles
  - Increased bone growth and density

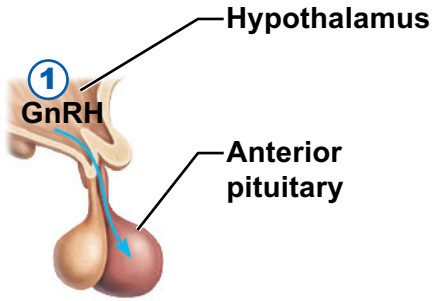
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# HORMONAL CONTROL OF TESTOSTERONE RELEASE AND SPERM PRODUCTION

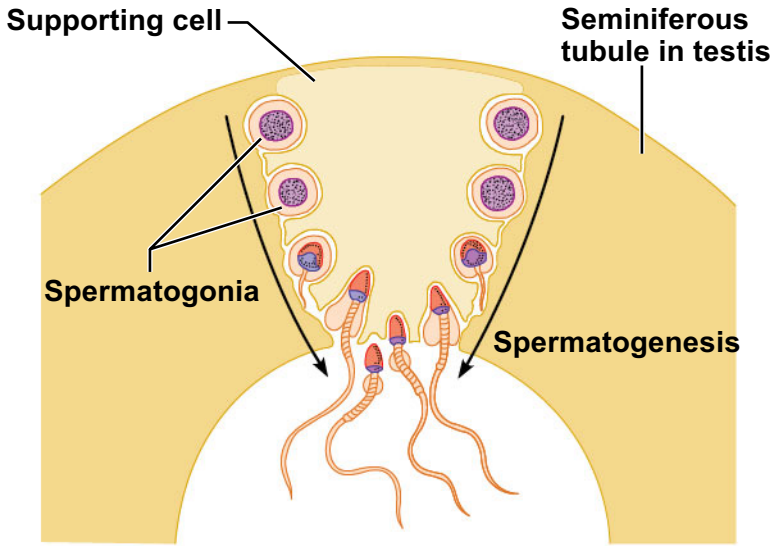



1. The hypothalamus releases gonadotropin-releasing hormone (GnRH).
2. GnRH stimulates the anterior pituitary to release gonadotropins—FSH and LH.
3. FSH stimulates spermatogonia to produce sperm.
4. LH stimulates the interstitial cells to release testosterone, which serves as the final trigger for spermatogenesis. Testosterone then enhances spermatogenesis.
5. The rising level of testosterone exerts negative feedback control on the hypothalamus and pituitary.

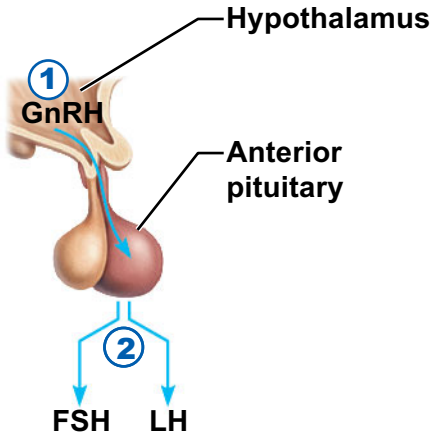
**KEY:**
- → Stimulates
- --→ Inhibits




1 The hypothalamus releases gonadotropin-releasing hormone (GnRH).

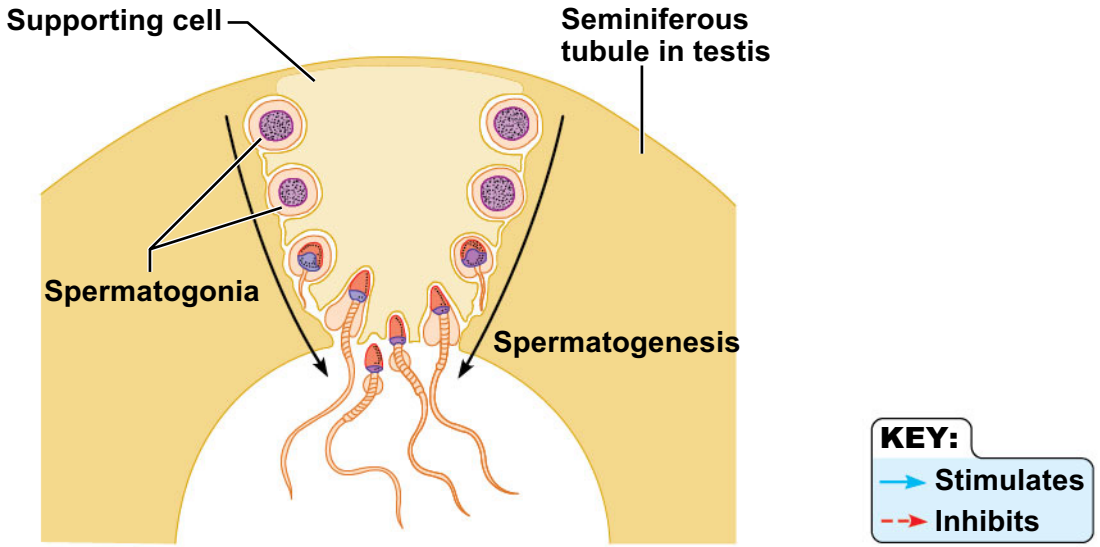


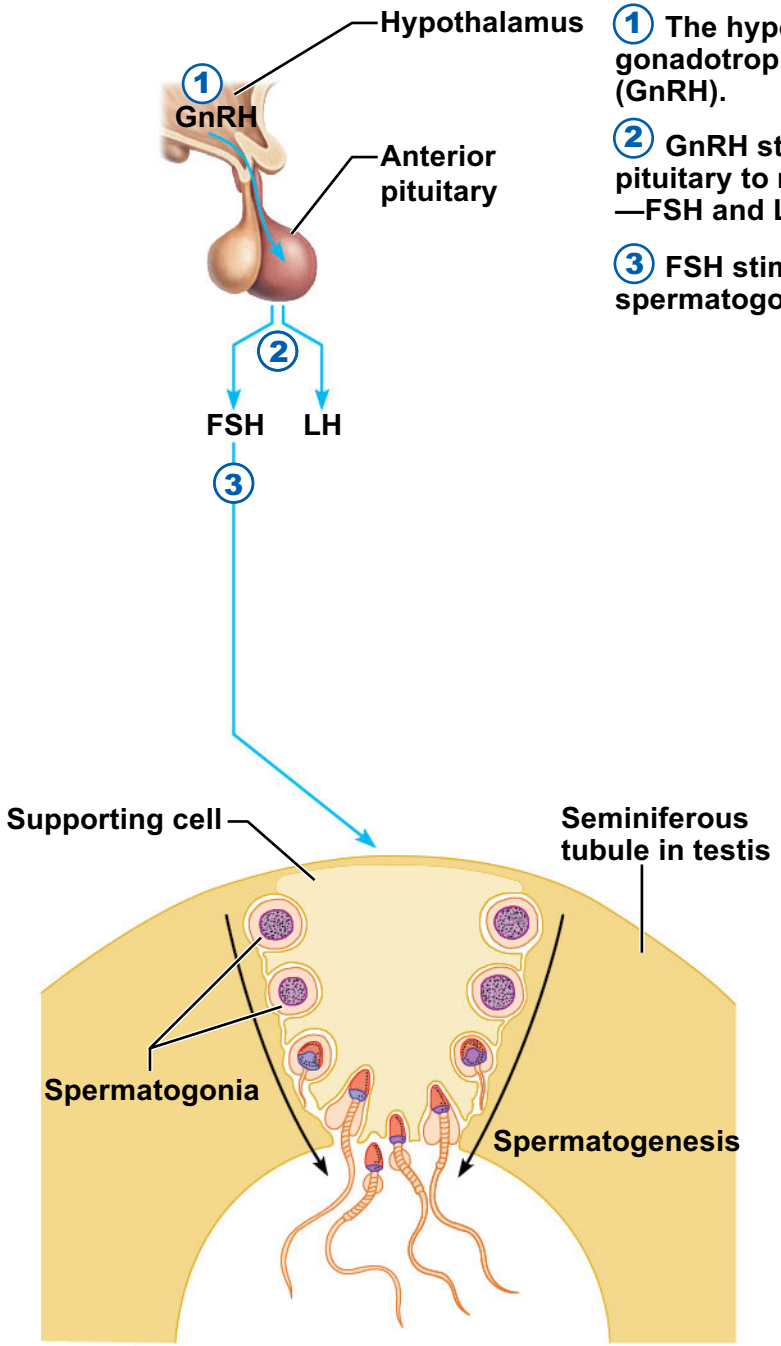

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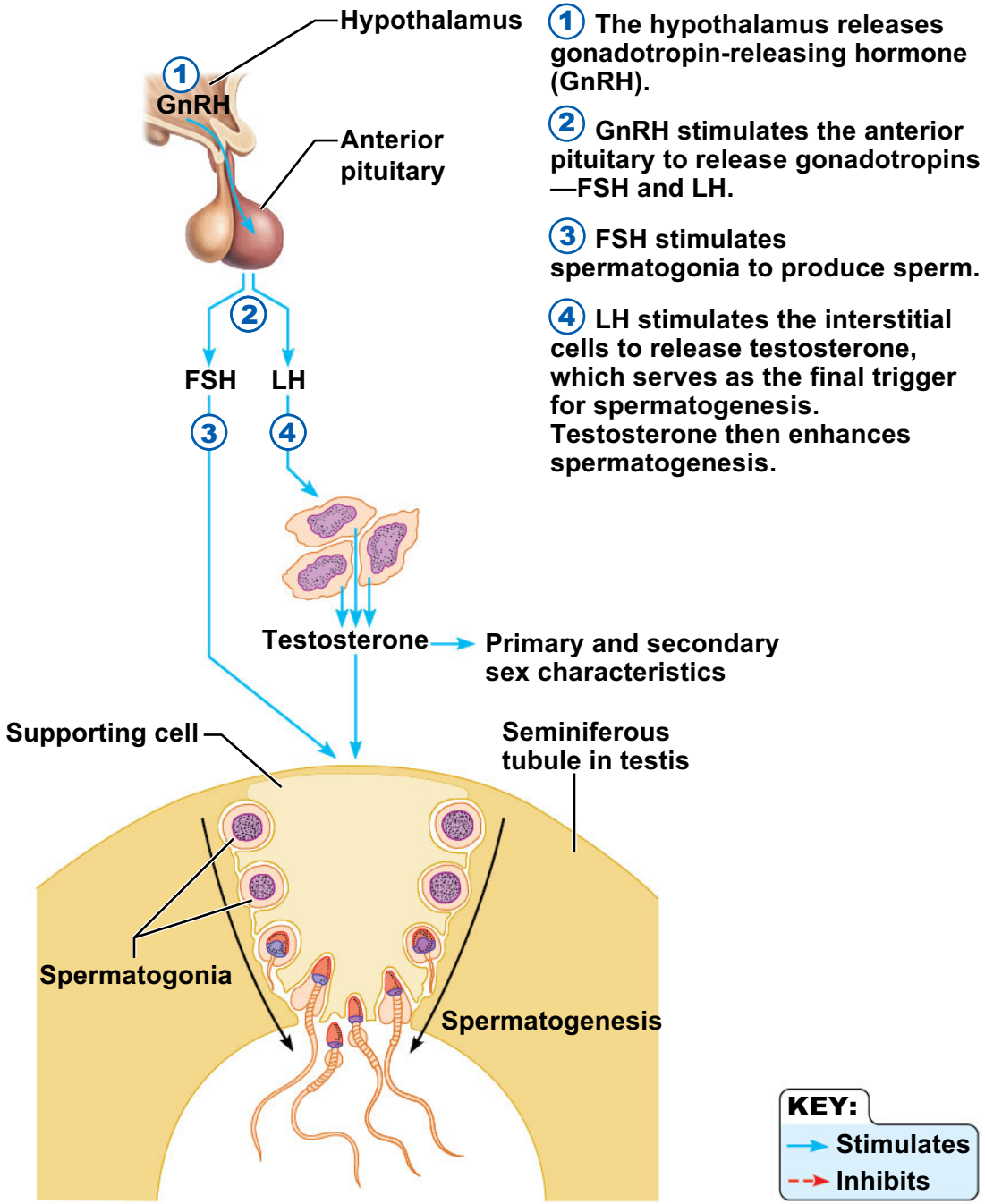




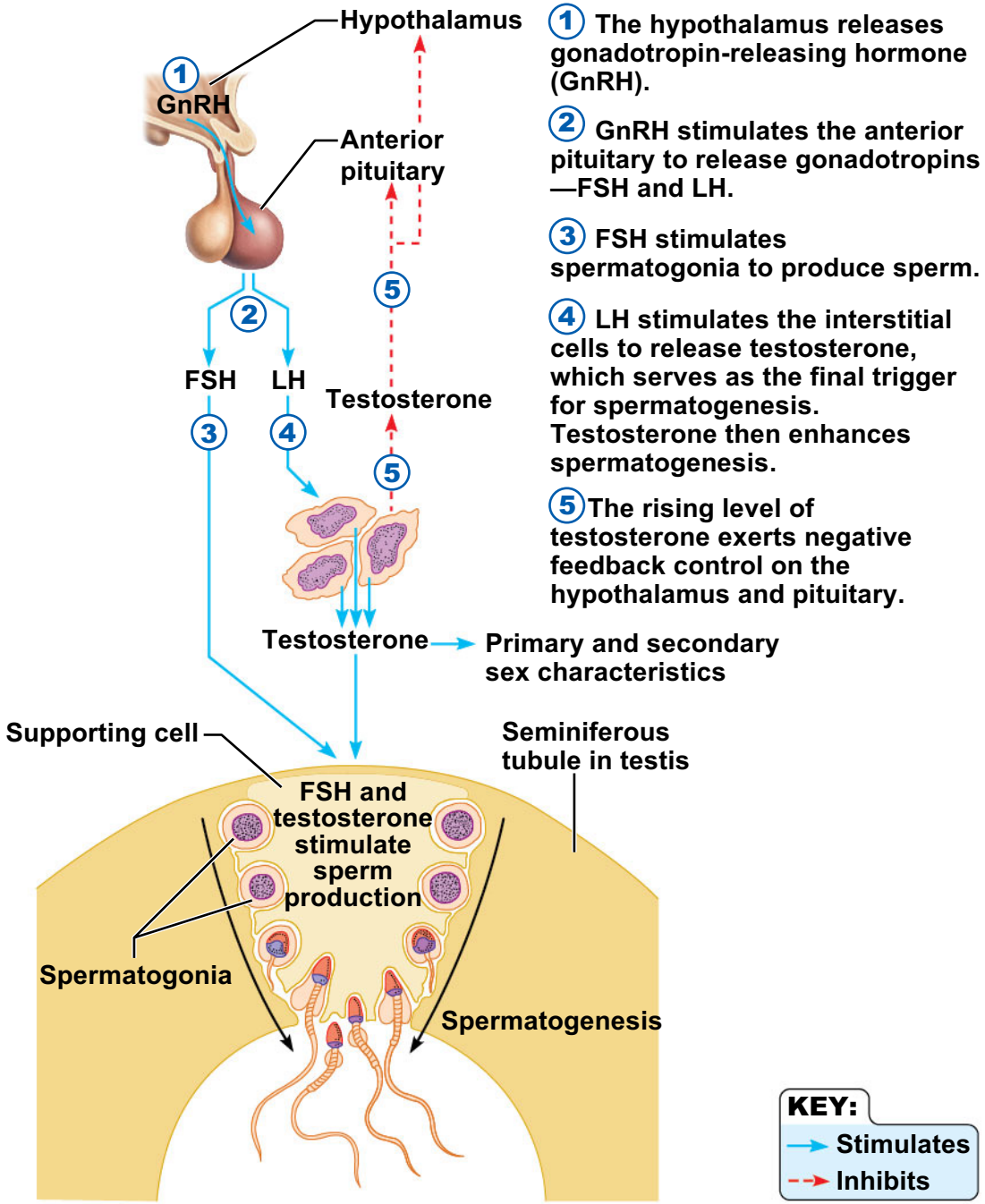

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# FEMALE REPRODUCTIVE SYSTEM
## ANATOMY

- Ovaries
- Duct system
  - Uterine (fallopian) tubes
  - Uterus
  - Vagina
- External genitalia

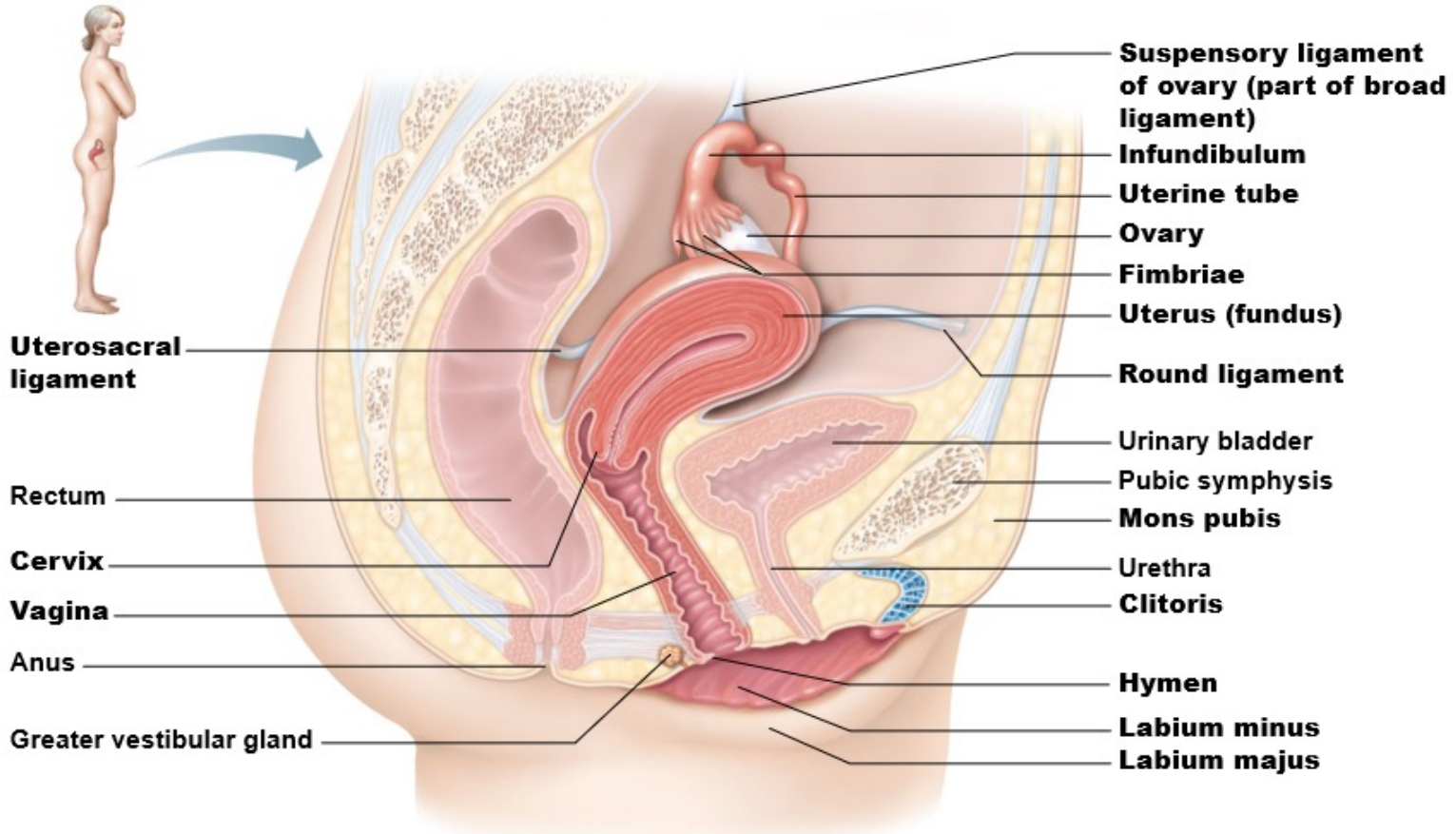


# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY



- Ovaries
- Duct system
  - Uterine (fallopian) tubes
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  - Vagina
- External genitalia




THE HUMAN FEMALE REPRODUCTIVE ORGANS

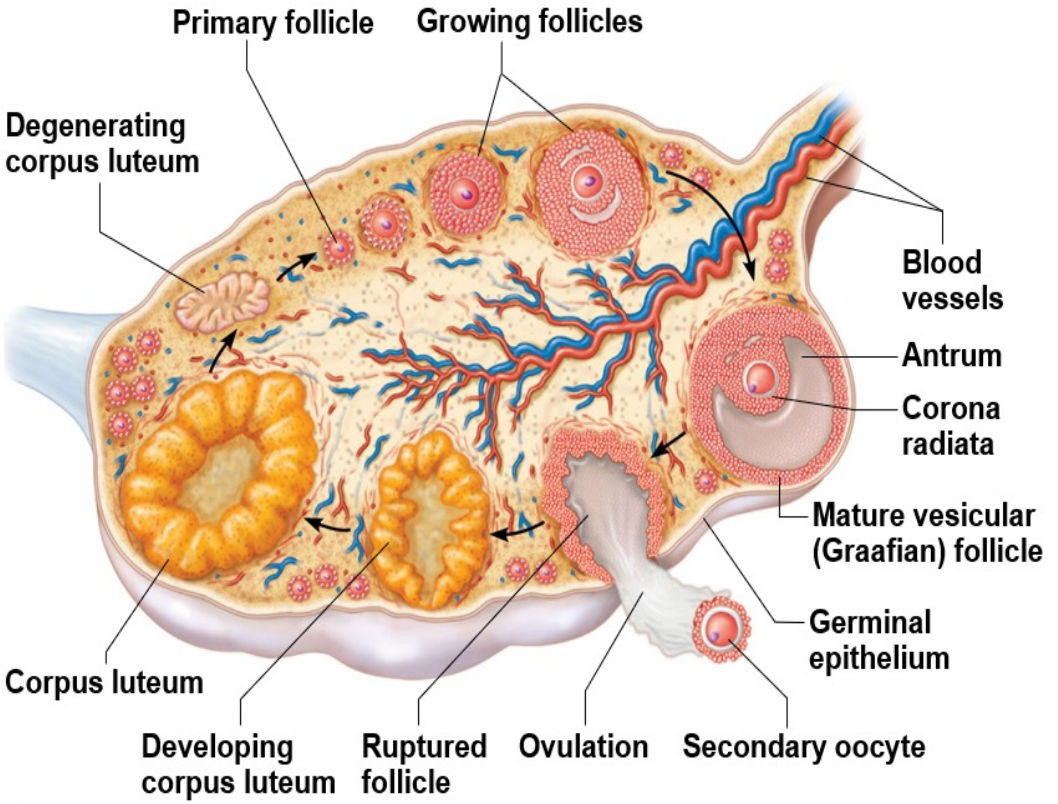
# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: OVARY

Ovaries

- Produce eggs (ova)
- Produce hormones (estrogen and progesterone)
- Each ovary houses ovarian follicles consisting of:
  - Oocyte (immature egg)
  - Follicle cells—layers of different cells that surround the oocyte

SAGITTAL SECTION OF A HUMAN OVARY SHOWING THE DEVELOPMENTAL STAGES OF AN OVARIAN FOLLICLE




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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: OVARY

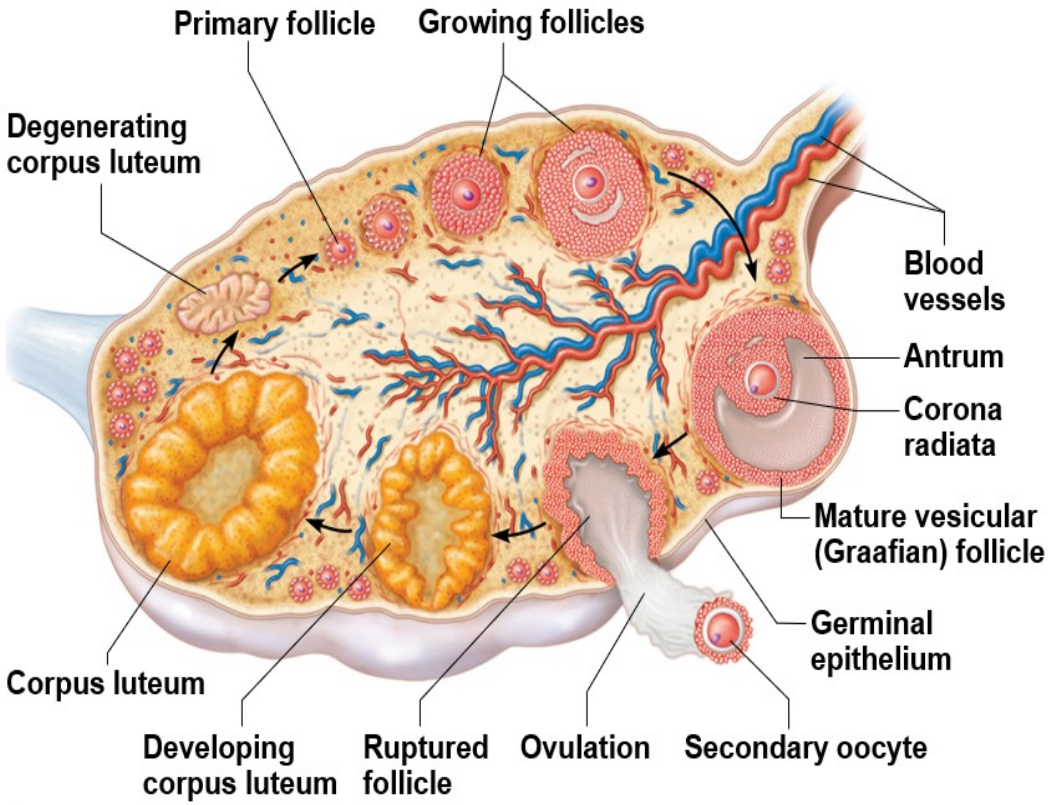
### Ovarian follicles

- Primary follicle—contains an immature oocyte
- Vesicular (Graafian) follicle—growing follicle with a maturing oocyte

### Ovulation

- the follicle ruptures when the egg is mature and ready to be ejected from the ovary
- occurs about every 28 days
- the ruptured follicle is transformed into a corpus luteum

SAGITTAL SECTION OF A HUMAN OVARY SHOWING THE DEVELOPMENTAL STAGES OF AN OVARIAN FOLLICLE



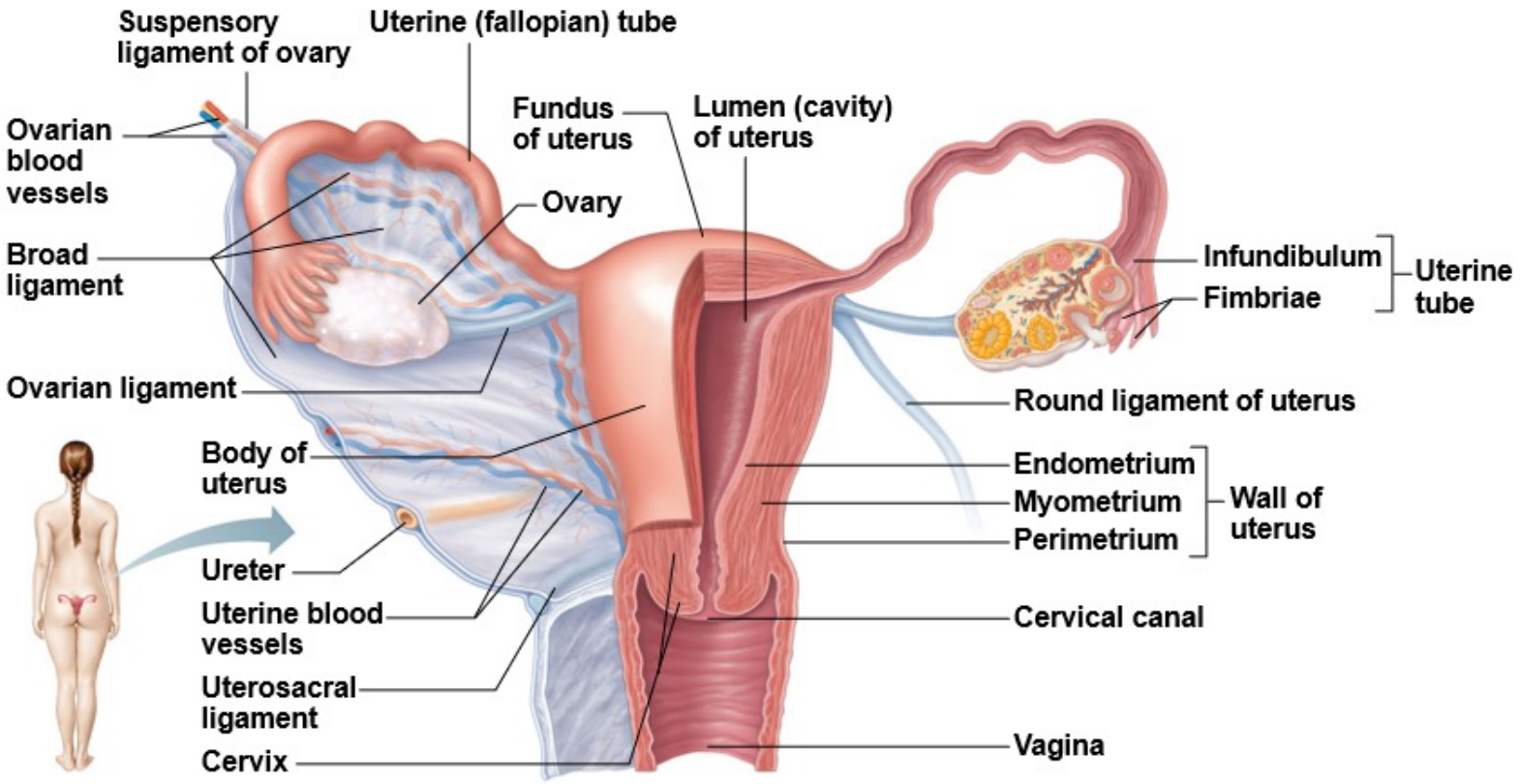

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: OVARY

### Ovary support

- Suspensory ligaments secure the ovaries to the lateral walls of the pelvis
- Ovarian ligaments anchor ovaries to the uterus medially
- Broad ligaments, a fold of peritoneum, enclose and hold the ovaries in place




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# FEMALE REPRODUCTIVE SYSTEM

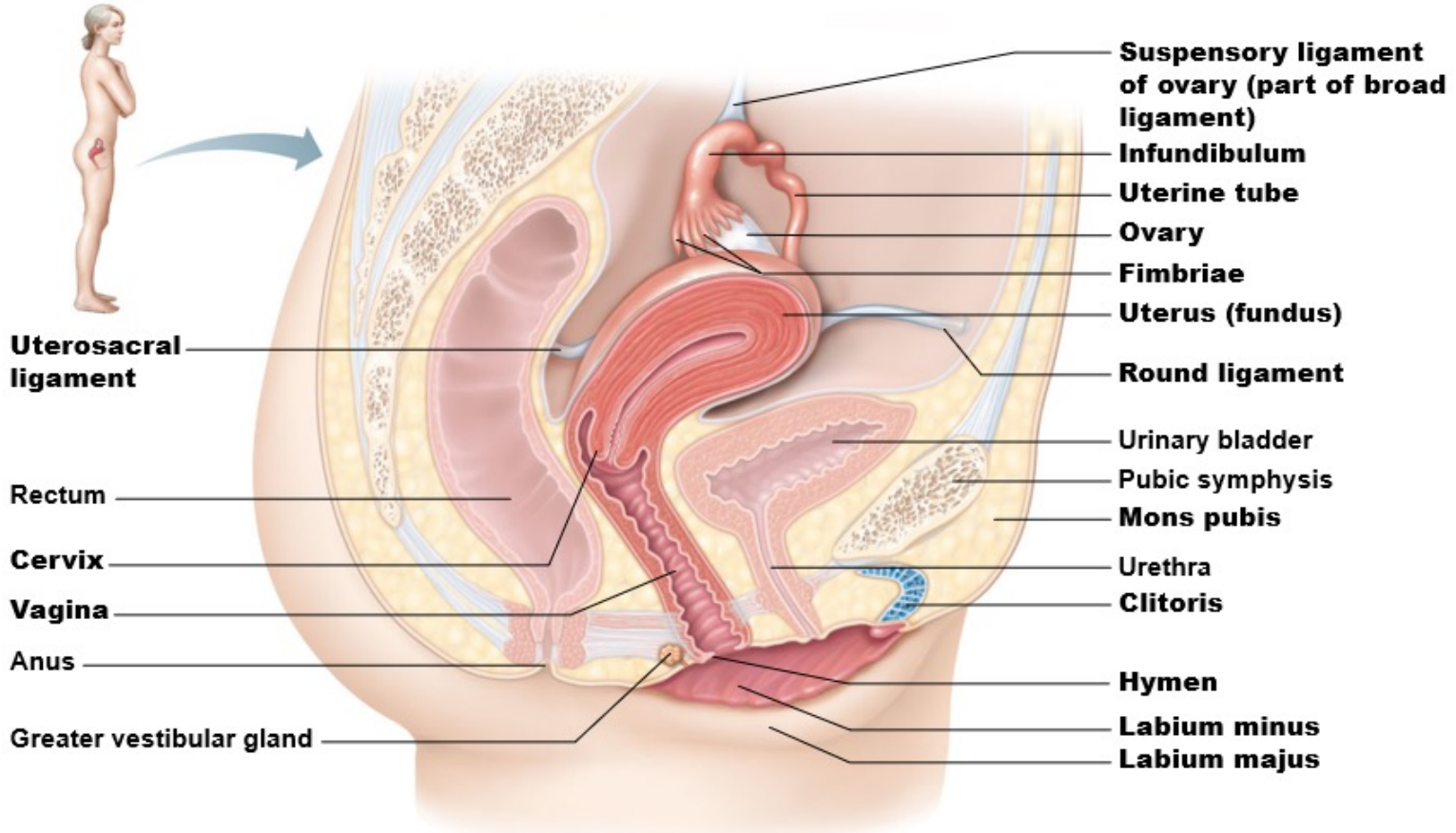
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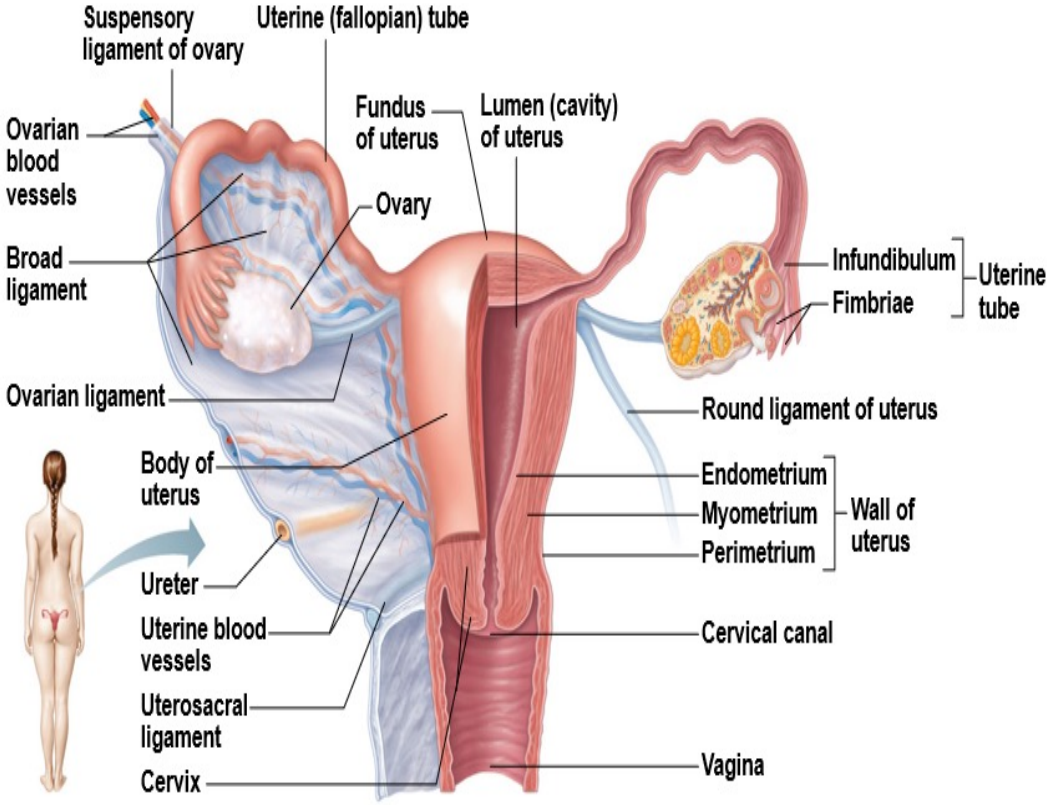

THE HUMAN FEMALE REPRODUCTIVE ORGANS

# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: DUCT SYSTEM

### Uterine / Fallopian tubes

- Form the initial part of the duct system
- Receive the ovulated oocyte from the ovaries
- Provide a site for fertilization
- Empty into the uterus
- Supported and enclosed by the broad ligament



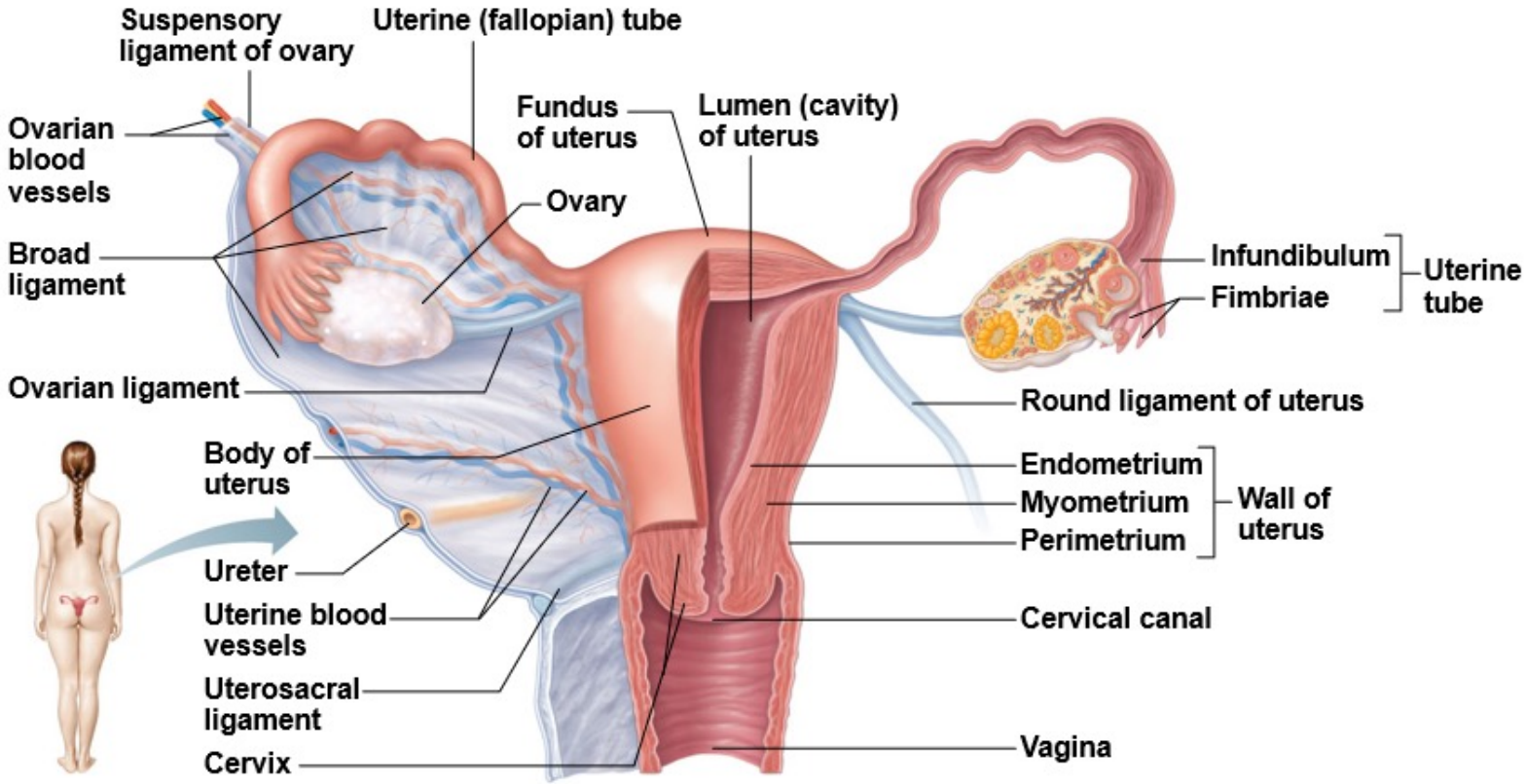

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: DUCT SYSTEM

Uterine /Fallopian tube

- Infundibulum
  - Distal, funnel-shaped end
- Fimbriae
  - Fingerlike projections of the infundibulum
  - Receive the oocyte from the ovary
  - Cilia located inside the uterine tube transport the oocyte



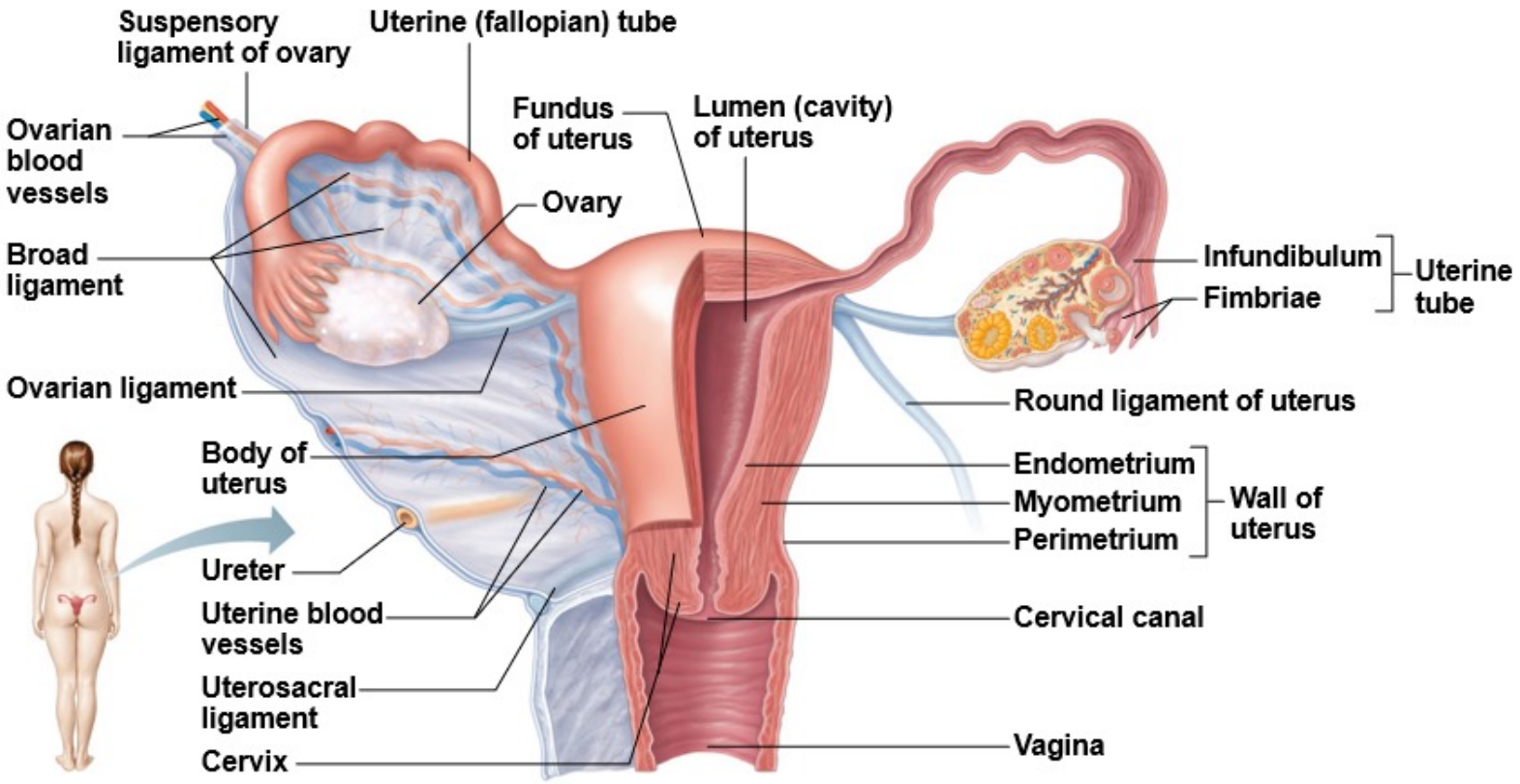

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: DUCT SYSTEM

### Uterus

- Situated between the urinary bladder and rectum
- Size and shape of a pear, in a woman who has never been pregnant
- Receives, retains, nourishes a fertilized egg



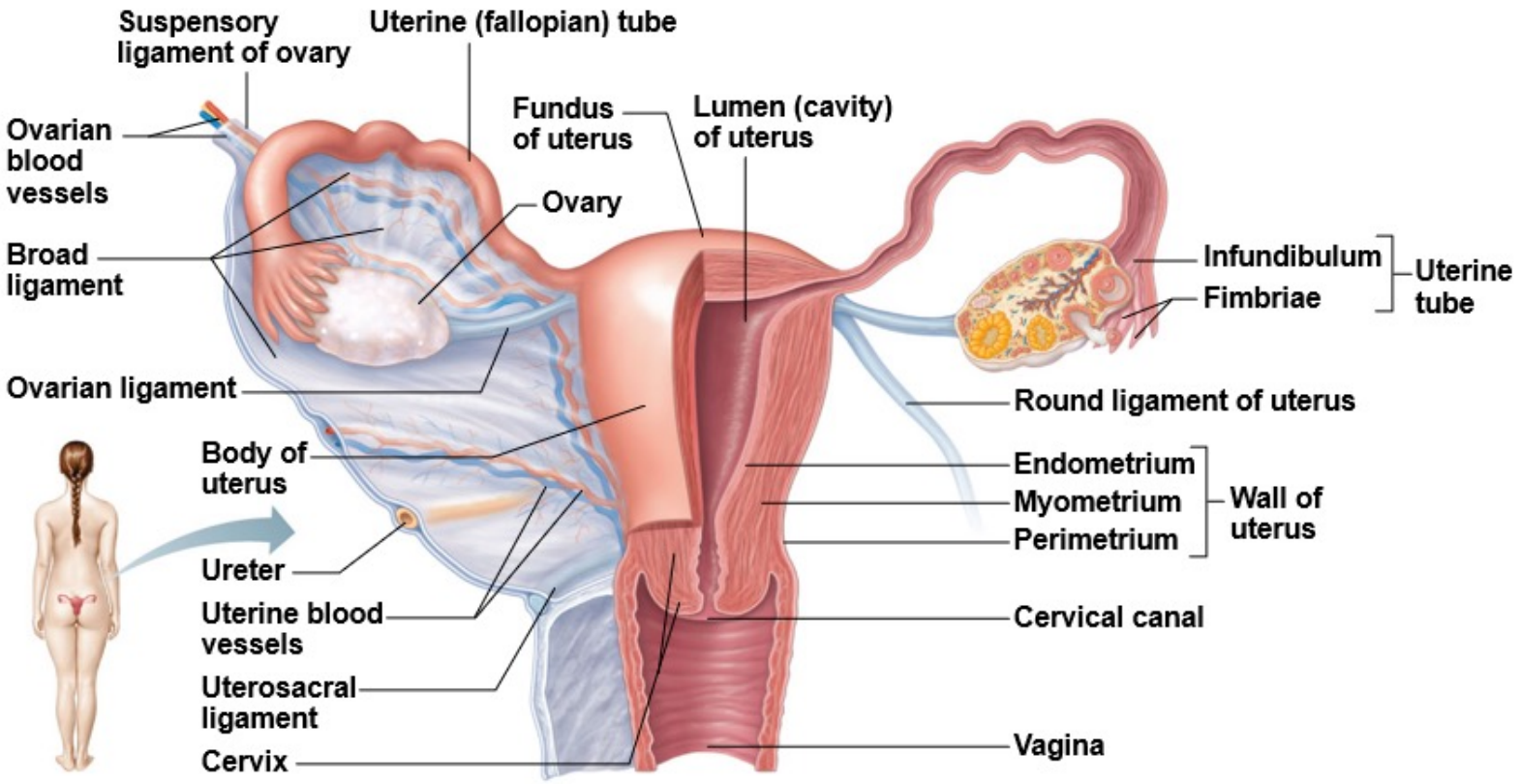

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: DUCT SYSTEM

### Uterine support

- Broad ligament suspends the uterus in the pelvis
- Round ligament anchors the uterus anteriorly
- Uterosacral ligament anchors the uterus posteriorly



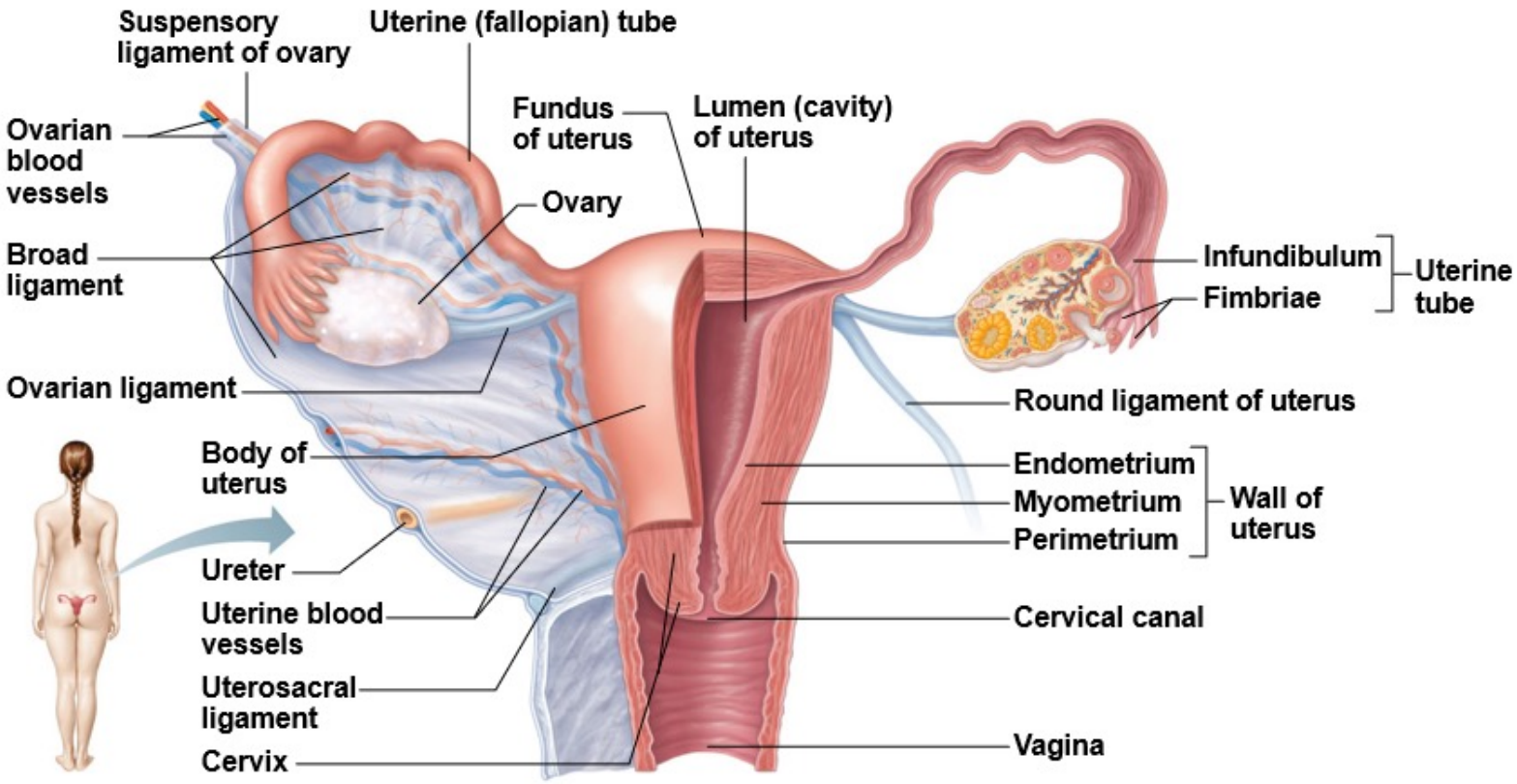

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: DUCT SYSTEM

Regions of the uterus

- Body—main portion
- Fundus—superior rounded region above where uterine tube enters
- Cervix—narrow outlet that protrudes into the vagina



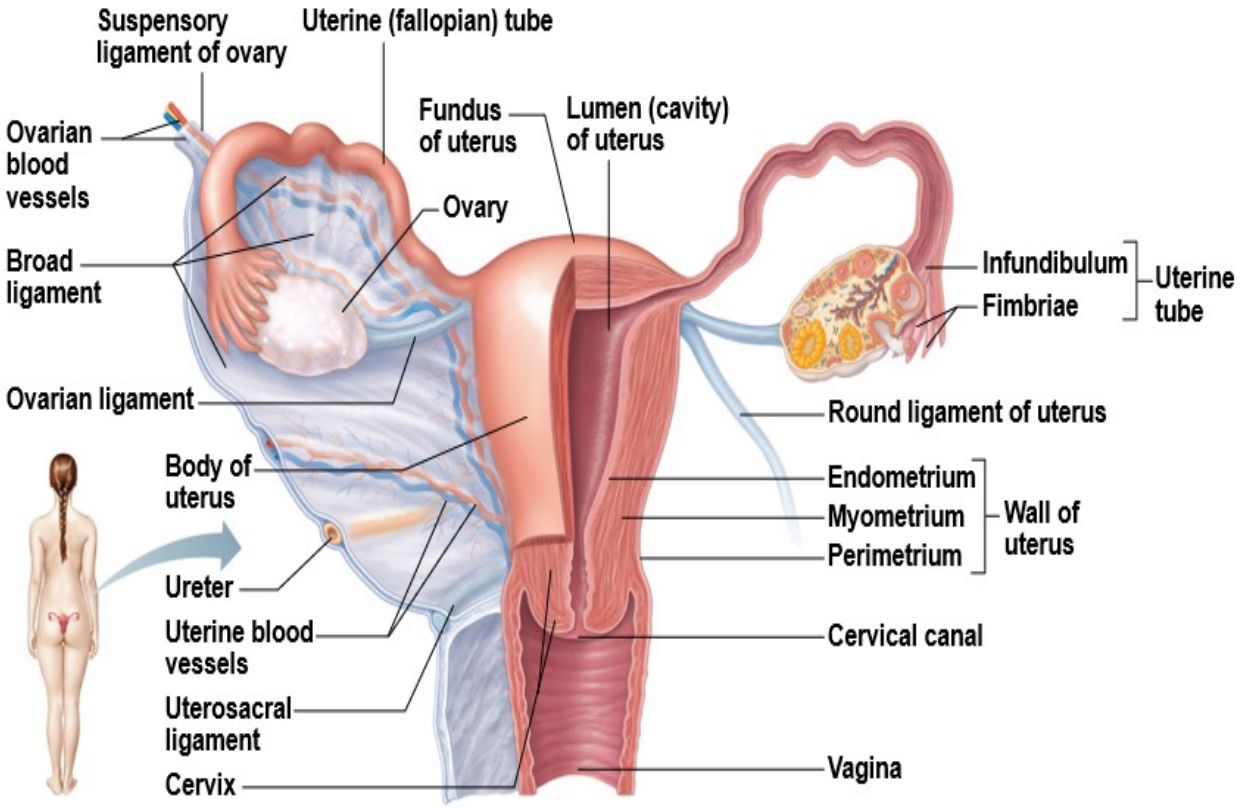

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: DUCT SYSTEM

Layers of the uterus

- Endometrium
  - Inner layer (mucosa)
  - Site of implantation of a fertilized egg
  - Sloughs off if no pregnancy occurs (menstruation or menses)
- Myometrium
  - the middle layer of smooth muscle that contracts during labor
- Perimetrium (visceral peritoneum)
  - the outermost serous layer of the uterus



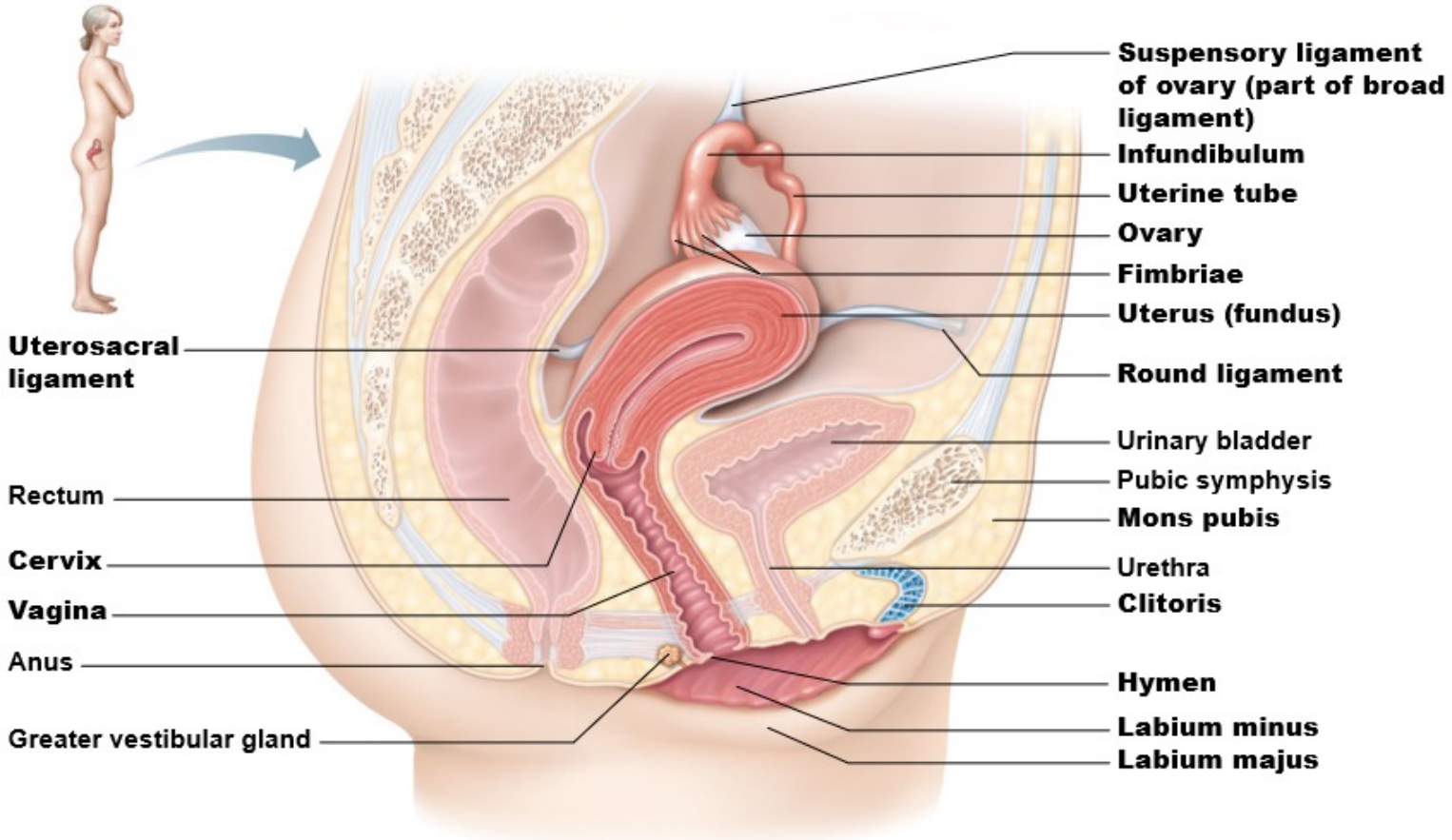

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: DUCT SYSTEM

### Vagina

- Passageway that extends from cervix to exterior of body and is located between urinary bladder and rectum
- Serves as the canal that allows a baby or menstrual flow to leave the body
- Female organ of copulation
- Receives the penis during sexual intercourse
- Hymen—partially closes the vagina until it is ruptured




THE HUMAN FEMALE REPRODUCTIVE ORGANS

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# FEMALE REPRODUCTIVE SYSTEM

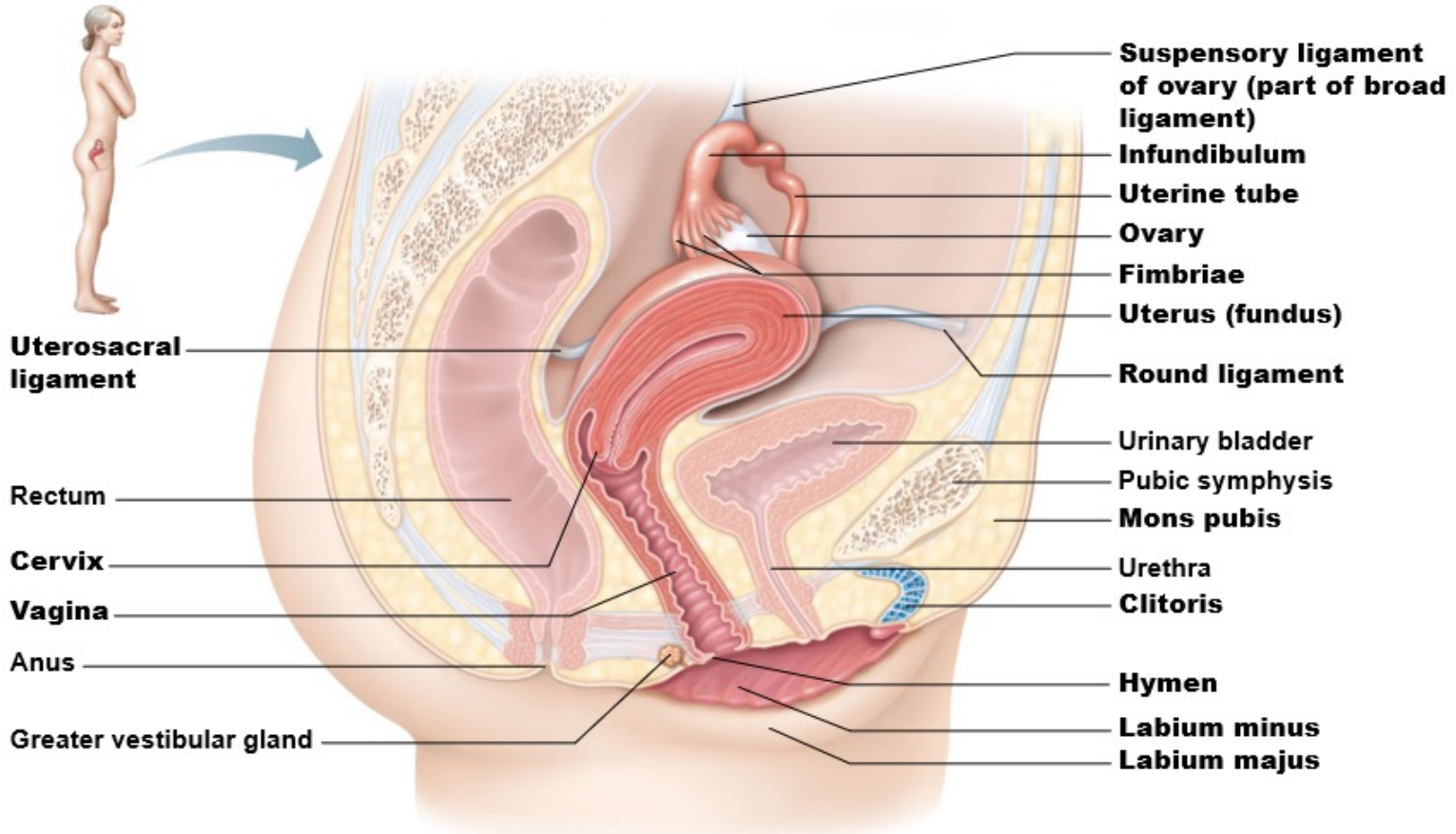
## ANATOMY

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**Duct system**

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External genitalia



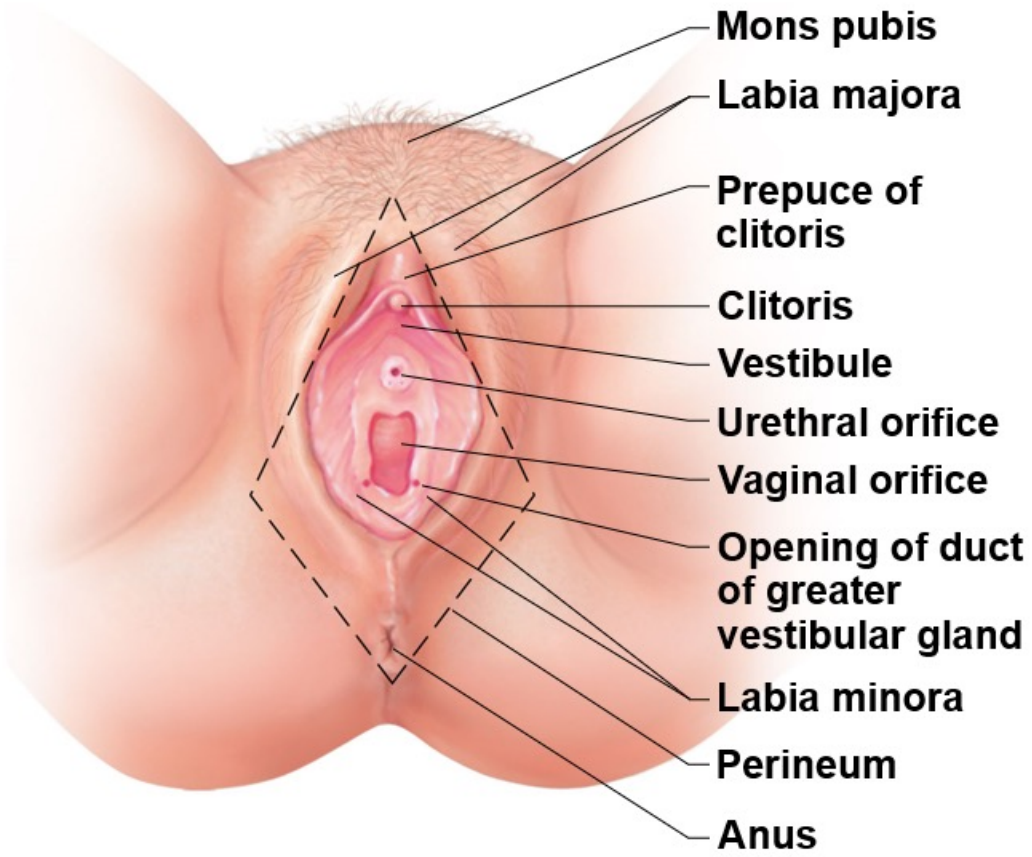

THE HUMAN FEMALE REPRODUCTIVE ORGANS

# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: EXTERNAL GENITALIA AND FEMALE PERINEUM

The female external genitalia, or vulva, includes:

- Mons pubis
- Labia
- Clitoris
- Greater vestibular glands
- Vestibule
  - Urethral orifice
  - Vaginal orifice



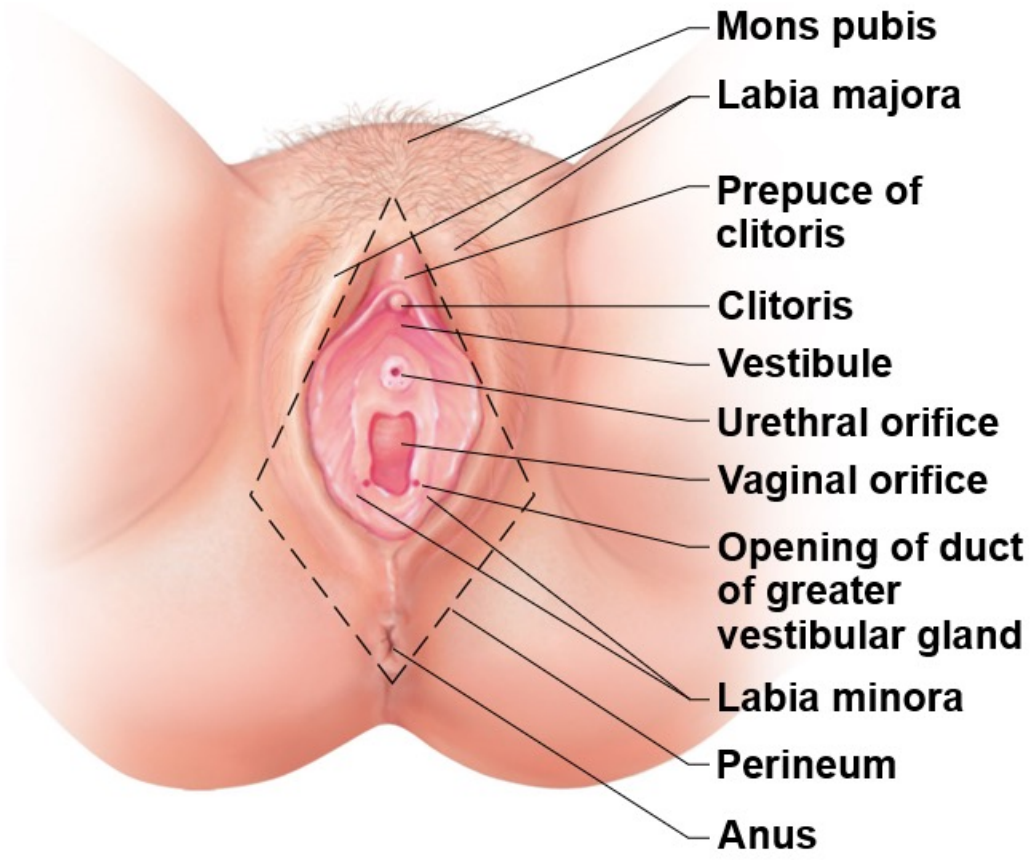

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: EXTERNAL GENITALIA AND FEMALE PERINEUM

### Mons pubis

- Fatty area overlying the pubic symphysis
- Covered with pubic hair after puberty



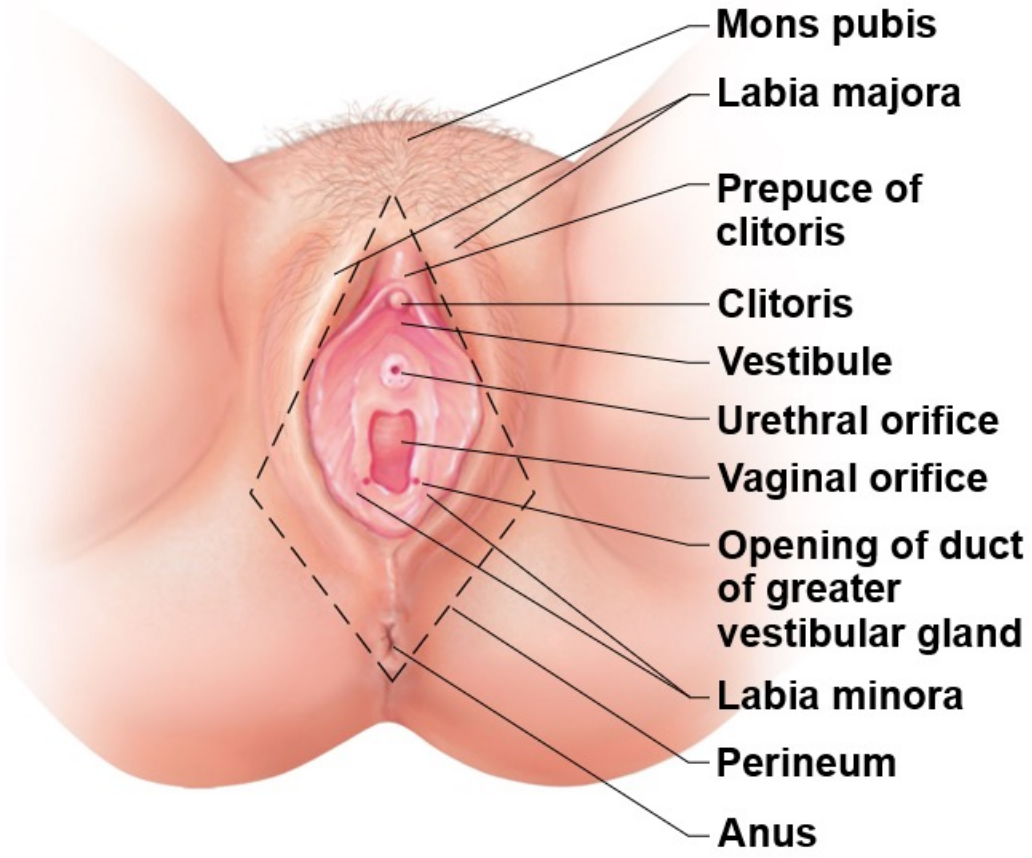

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: EXTERNAL GENITALIA AND FEMALE PERINEUM

Labia—skin folds

- Labia majora
  - Hair-covered skin folds
  - Enclose the labia minora
  - Also encloses the vestibule
- Labia minora—delicate, hair-free folds of skin

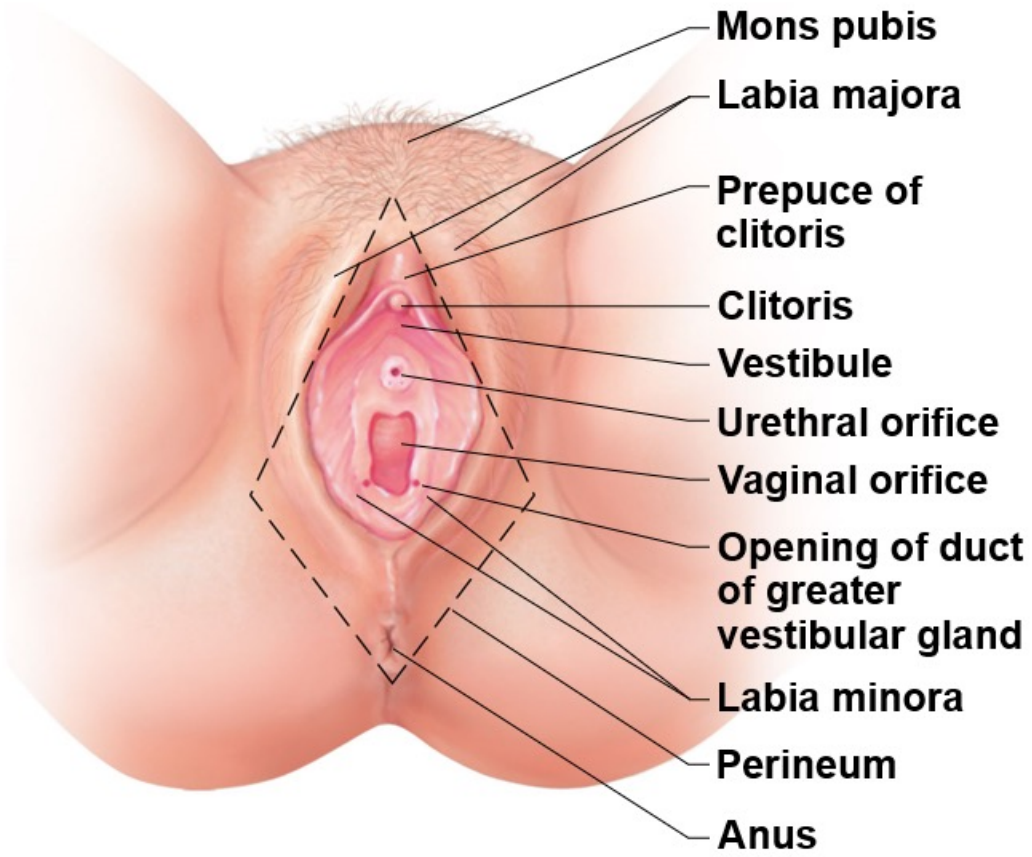



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# EXTERNAL GENITALIA AND FEMALE PERINEUM

Clitoris

- Contains erectile tissue
- Corresponds to the male penis
- The clitoris is similar to the penis in that it is:
  - Hooded by a prepuce
  - Composed of sensitive erectile tissue
  - Swollen with blood during sexual excitement
- The clitoris lacks a reproductive duct




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# FEMALE REPRODUCTIVE SYSTEM

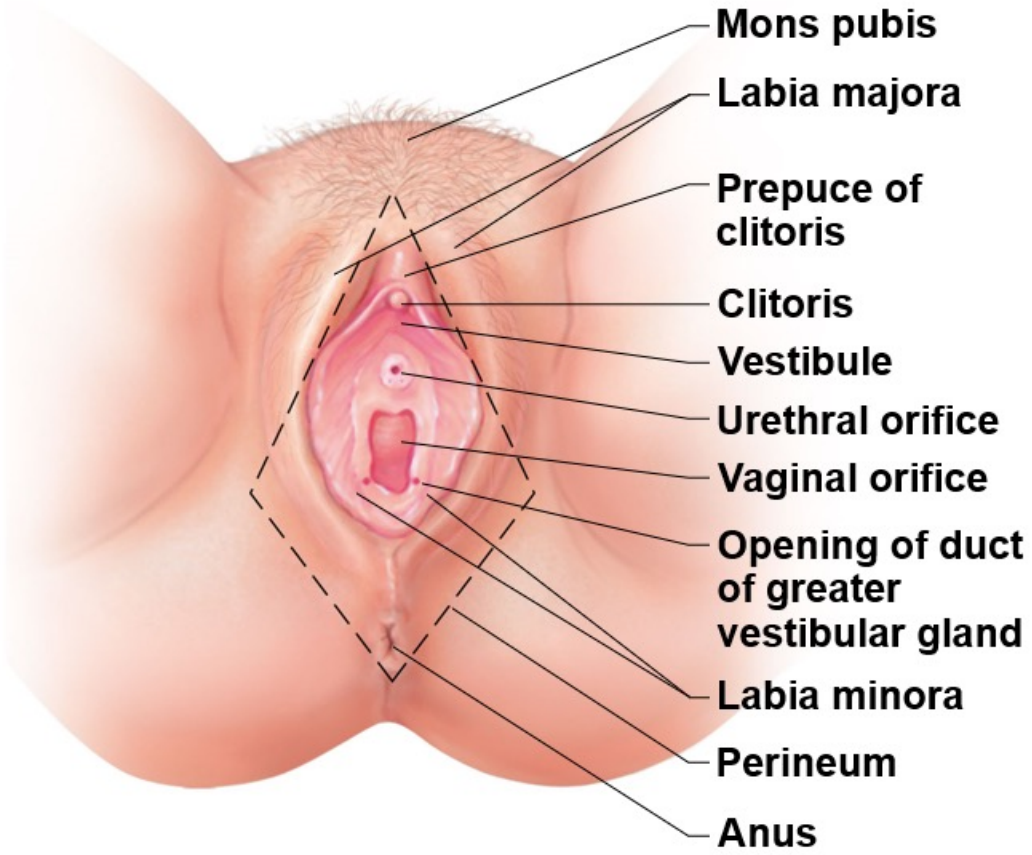
## ANATOMY: EXTERNAL GENITALIA AND FEMALE PERINEUM

### Vestibule

- Enclosed by labia majora
- Contains external openings of the urethra and vagina

### Greater vestibular glands

- One is found on each side of the vagina
- Secretions lubricate vagina during intercourse



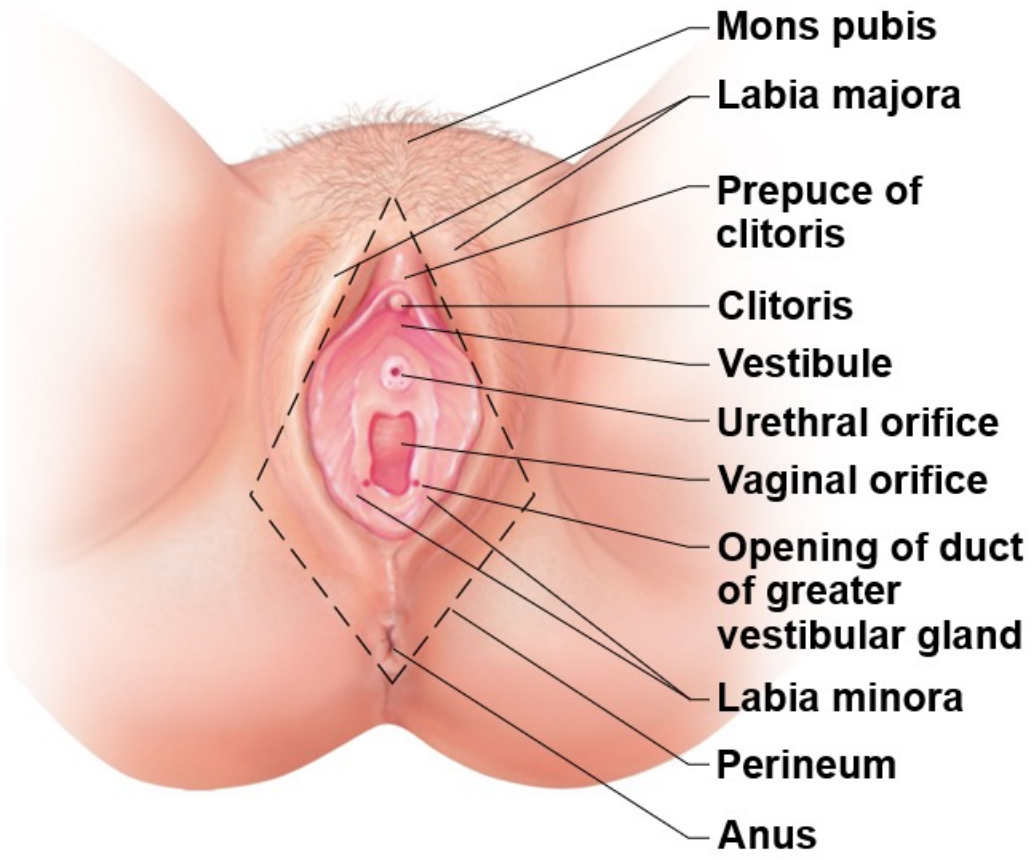

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# FEMALE REPRODUCTIVE SYSTEM

## ANATOMY: EXTERNAL GENITALIA AND FEMALE PERINEUM

### Perineum

- Diamond-shaped region between the anterior ends of the labial folds, anus posteriorly, and ischial tuberosities laterally




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# FEMALE REPRODUCTIVE SYSTEM

## FUNCTIONS AND CYCLES

The total supply of eggs is determined by the time a female is born

Ability to release eggs begins at puberty with the onset of the *menstrual cycle*

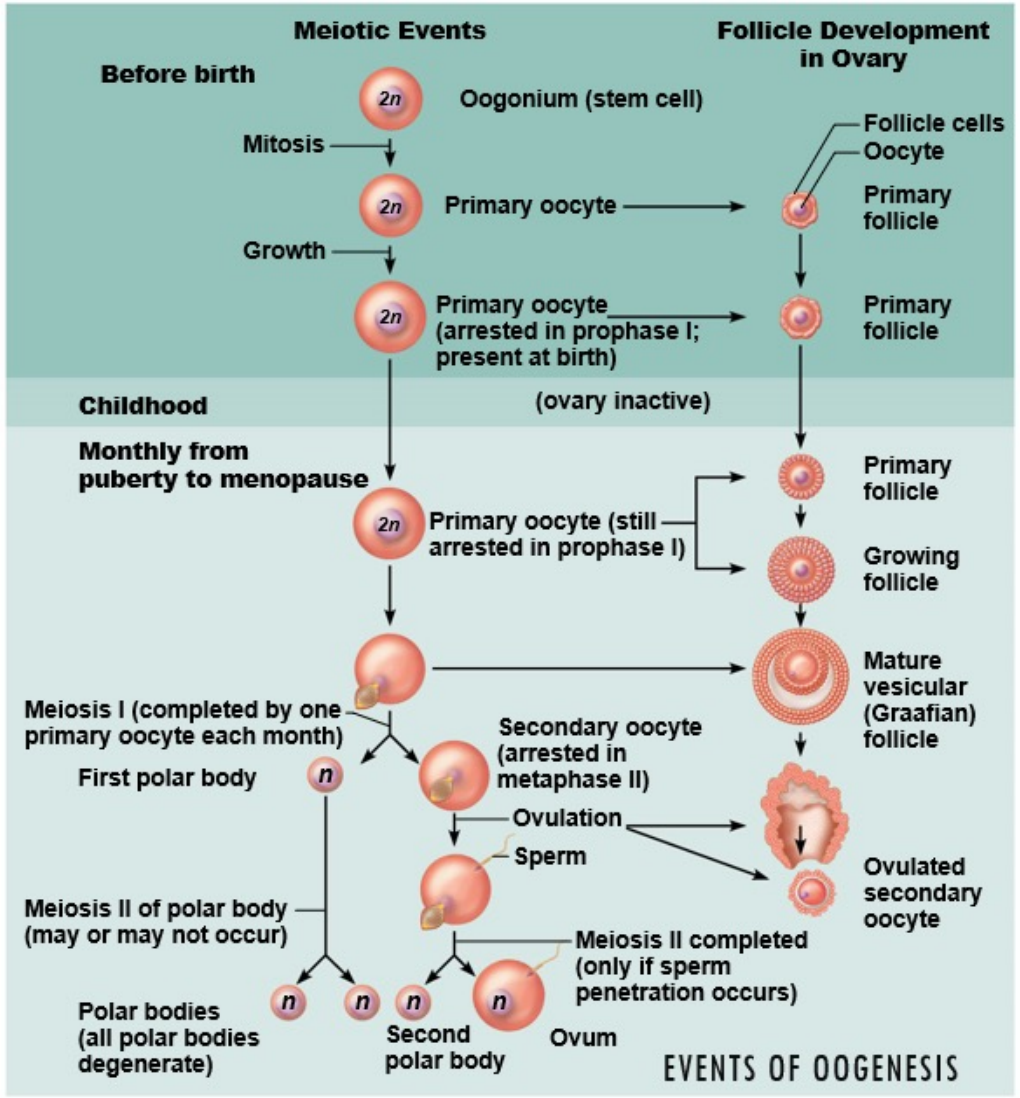
Reproductive ability ends at *menopause* (in female's fifties)

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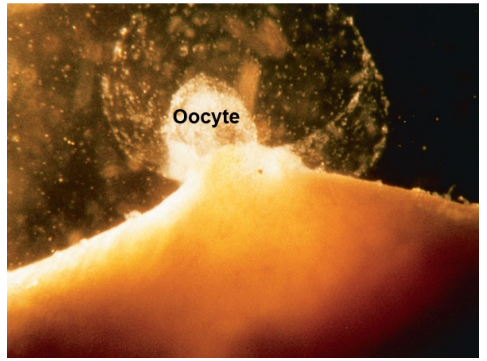
# OOGENESIS AND THE OVARIAN CYCLE

Oogenesis is the process of producing ova (eggs) in a female

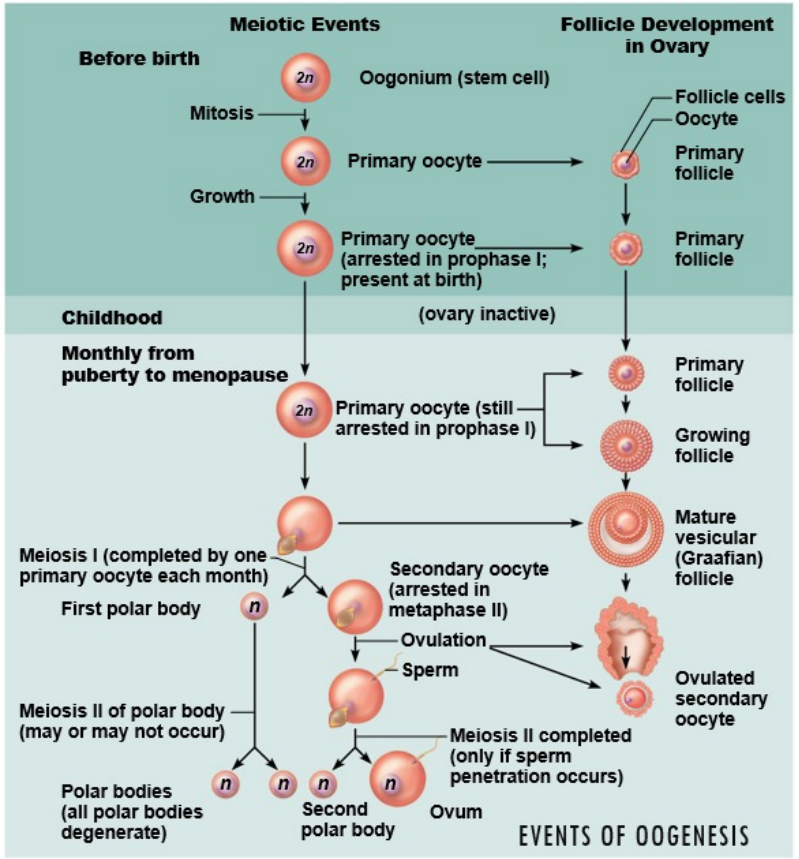
- Oogonia
  - are female stem cells found in a developing fetus
  - undergo mitosis to produce primary oocytes that are surrounded by cells that form primary follicles in the ovary



# OOGENESIS AND THE OVARIAN CYCLE




- Primary oocytes are inactive until puberty
- Follicle-stimulating hormone (FSH) causes some primary follicles to mature each month
- Cyclic monthly changes constitute the ovarian cycle




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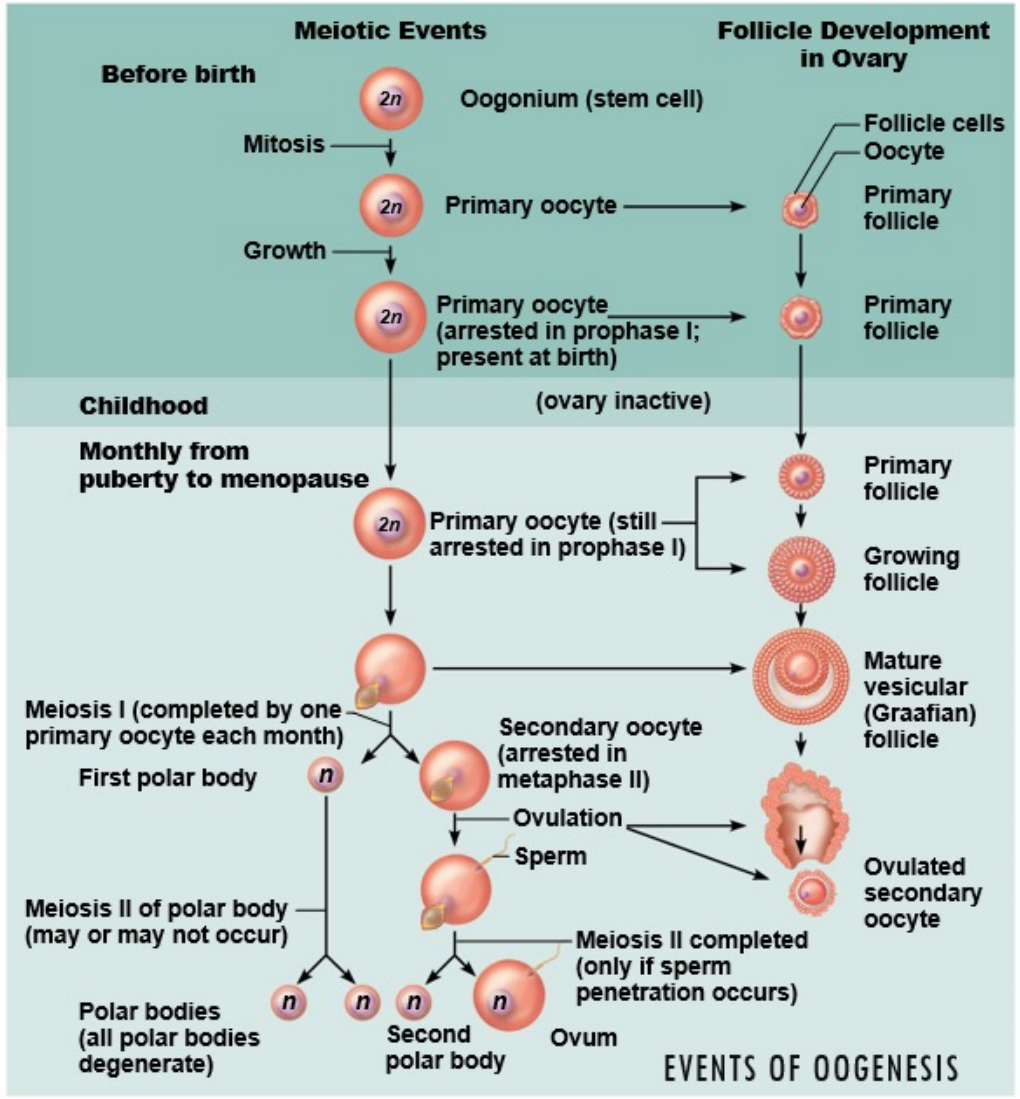
# OOGENESIS AND THE OVARIAN CYCLE

Meiosis starts inside maturing follicle

- First meiotic division produces a larger secondary oocyte and a smaller first polar body
- A vesicular follicle contains a secondary oocyte (maturation from a primary follicle takes about 14 days)

Ovulation of a secondary oocyte occurs with the release of luteinizing hormone (LH)

Secondary oocyte is released and surrounded by a corona radiata



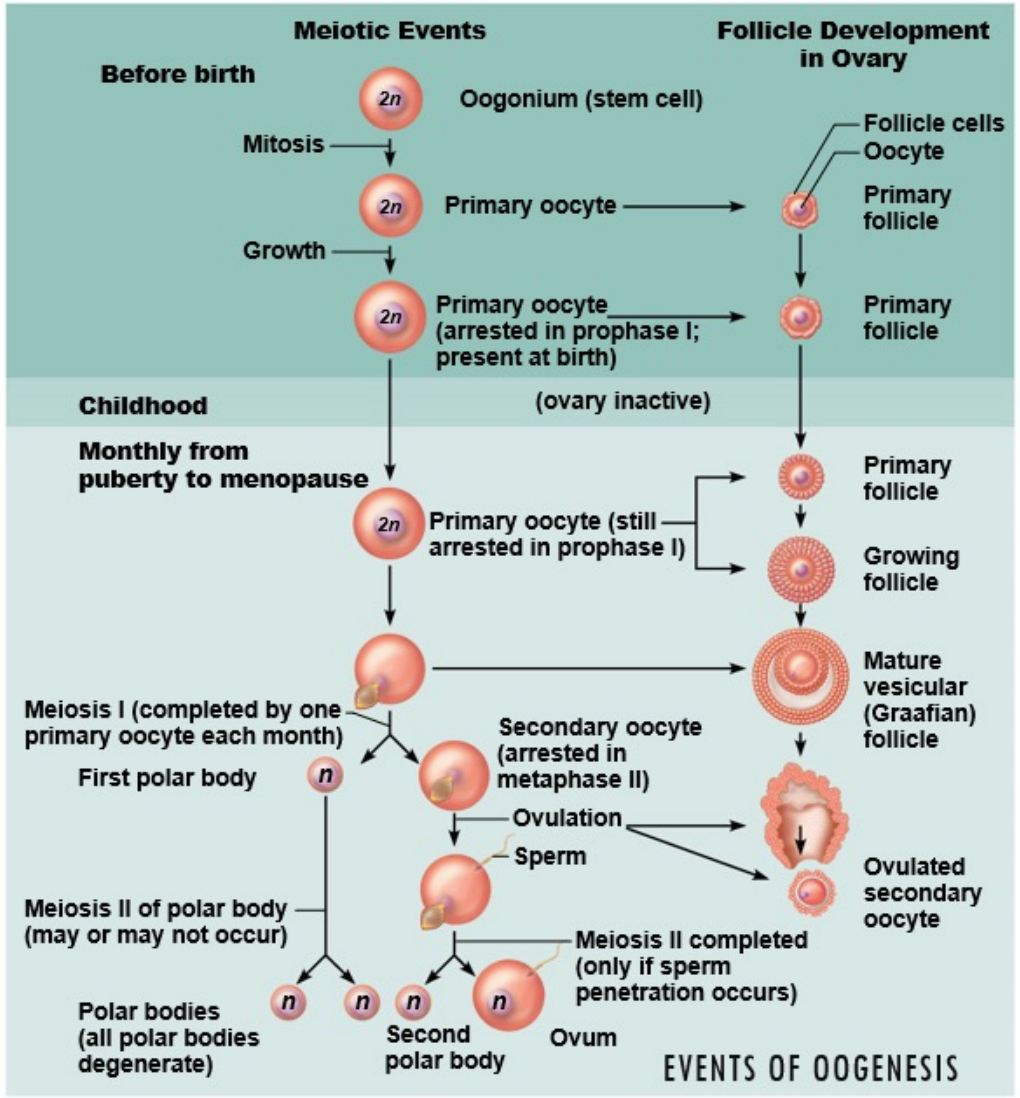
# OOGENESIS AND THE OVARIAN CYCLE

Meiosis is completed after ovulation *only if* sperm penetrates the oocyte

- Ovum is produced
- Two additional polar bodies are produced

Once ovum is formed, the 23 chromosomes can be combined with the 23 chromosomes of the sperm to form the fertilized egg (zygote)

If the secondary oocyte is *not* penetrated by a sperm, it dies and does not complete meiosis to form an ovum



# SUMMARY

## Meiosis

- Males—produces four functional sperm
- Females—produces one functional ovum and three tiny polar bodies

## Sex cell size and structure

- Sperm are tiny, motile, and equipped with nutrients in seminal fluid
- Egg is large, is nonmotile, and has nutrient reserves to nourish the embryo until implantation

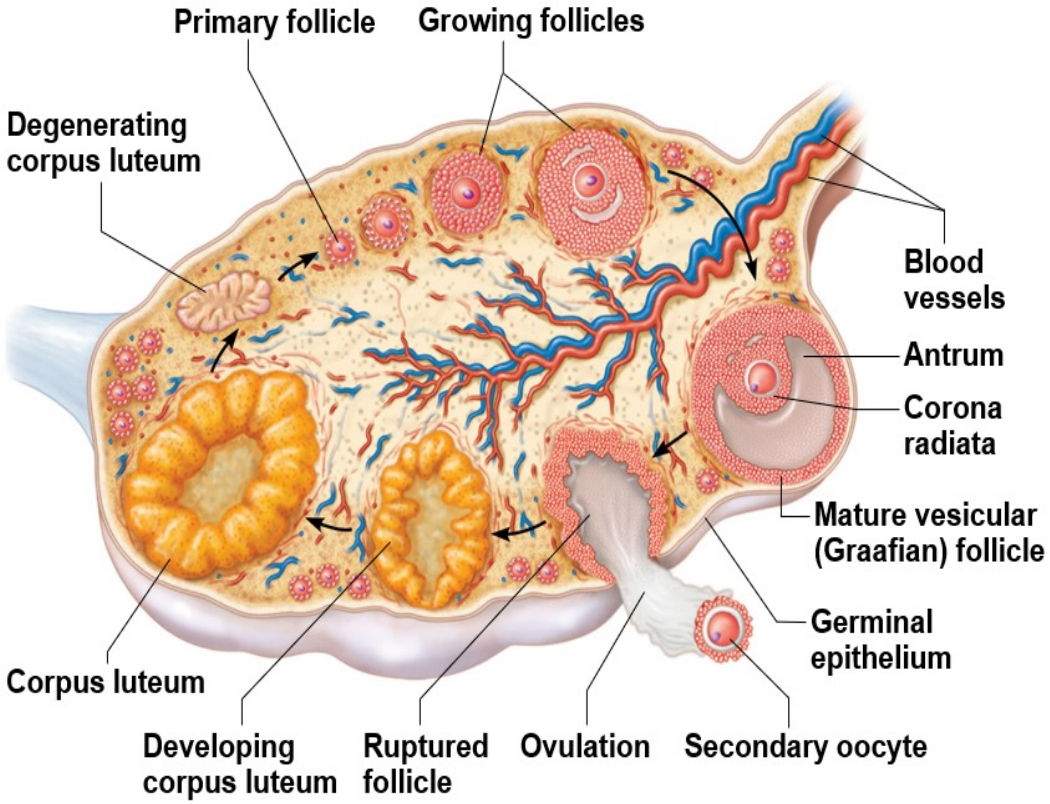
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# HORMONE PRODUCTION BY THE OVARIES

Estrogens are produced by follicle cells

- Cause secondary sex characteristics
  - Enlargement of accessory organs of the female reproductive system
  - Development of breasts
  - Appearance of axillary and pubic hair
  - Increase in fat beneath the skin, particularly in hips and breasts
  - Widening and lightening of the pelvis
  - Onset of menses (menstrual cycle)

SAGITTAL SECTION OF A HUMAN OVARY SHOWING THE DEVELOPMENTAL STAGES OF AN OVARIAN FOLLICLE



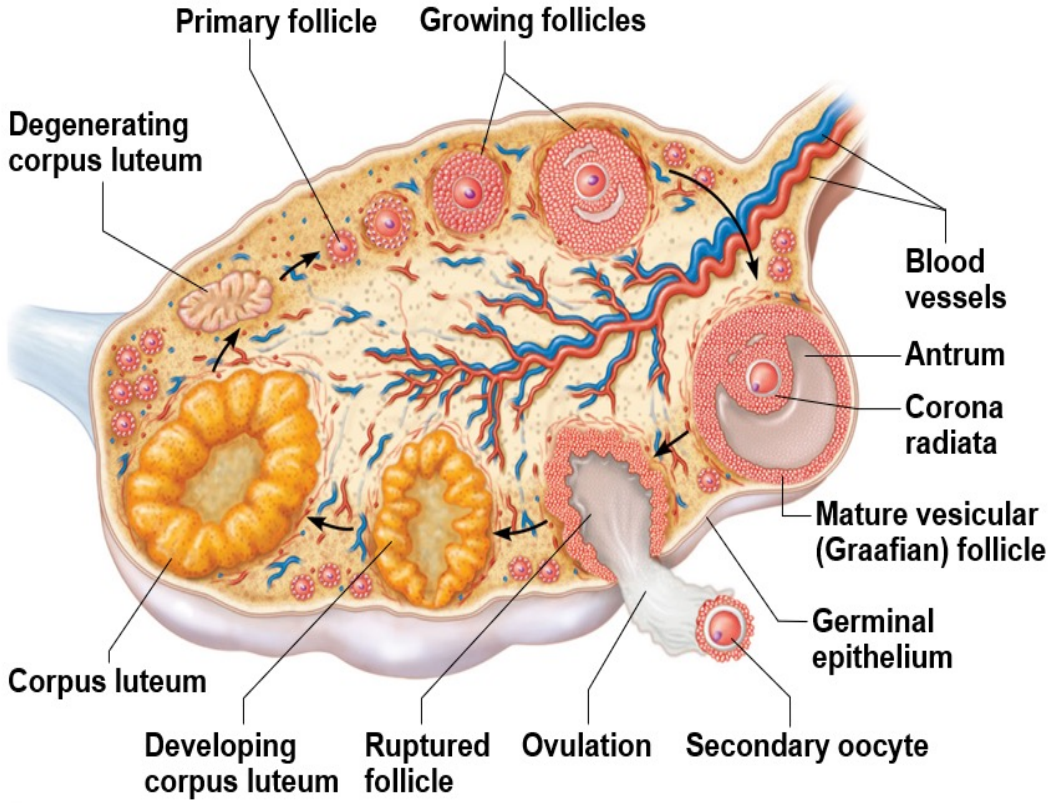

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# HORMONE PRODUCTION BY THE OVARIES

Progesterone is produced by the corpus luteum

- Production continues until LH diminishes in the blood
- Does *not* contribute to the appearance of secondary sex characteristics
- Helps maintain pregnancy
- Prepares the breasts for milk production

SAGITTAL SECTION OF A HUMAN OVARY SHOWING THE DEVELOPMENTAL STAGES OF AN OVARIAN FOLLICLE




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# UTERINE (MENSTRUAL) CYCLE

Cyclic changes of the endometrium, about 28 days in length

Regulated by cyclic production of estrogens and progesterone by the ovaries

FSH and LH, from the anterior pituitary, regulate the production of estrogens and progesterone by the ovaries

Ovulation typically occurs about midway through cycle, on day 14

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# UTERINE (MENSTRUAL) CYCLE

Stages of the menstrual cycle

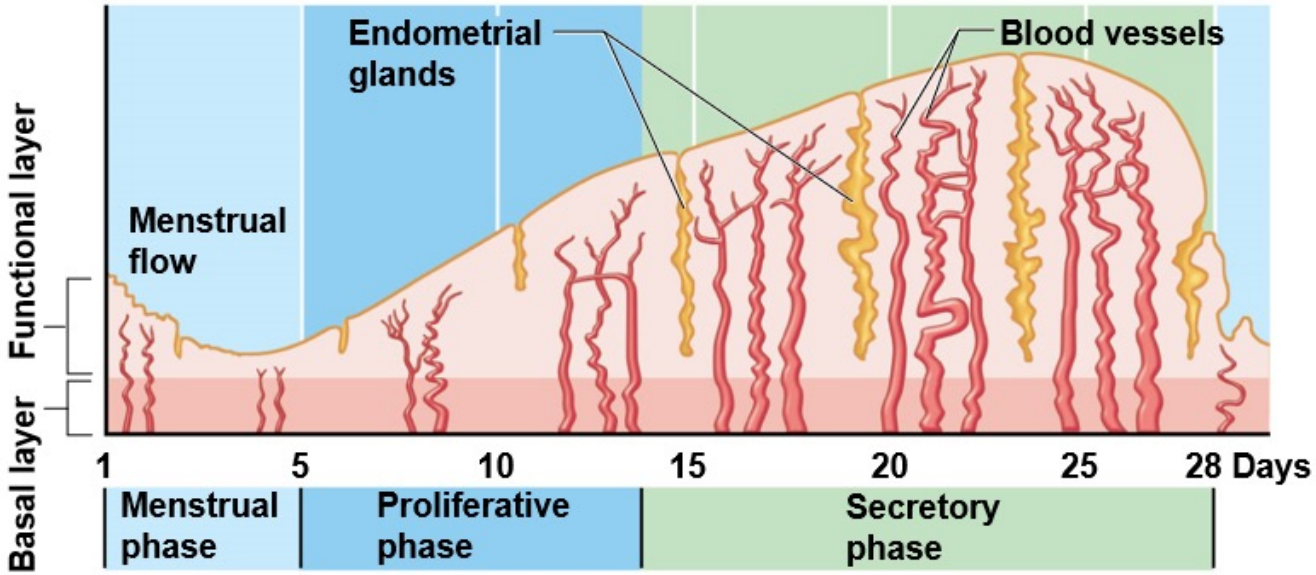
- Menstrual phase
- Proliferative stage
- Secretory stage

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# UTERINE (MENSTRUAL) CYCLE

Days 1–5: menstrual phase

- Functional layer of the endometrium is sloughed
- Bleeding occurs for 3 to 5 days
- Ovarian hormones are at their lowest levels
- By day 5, growing ovarian follicles are producing more estrogen




The menstrual and proliferative phases occur before ovulation and together correspond to the follicular phase of the ovarian cycle. The secretory phase corresponds in time to the luteal phase of the ovarian cycle.

**The three phases of the uterine cycle:**

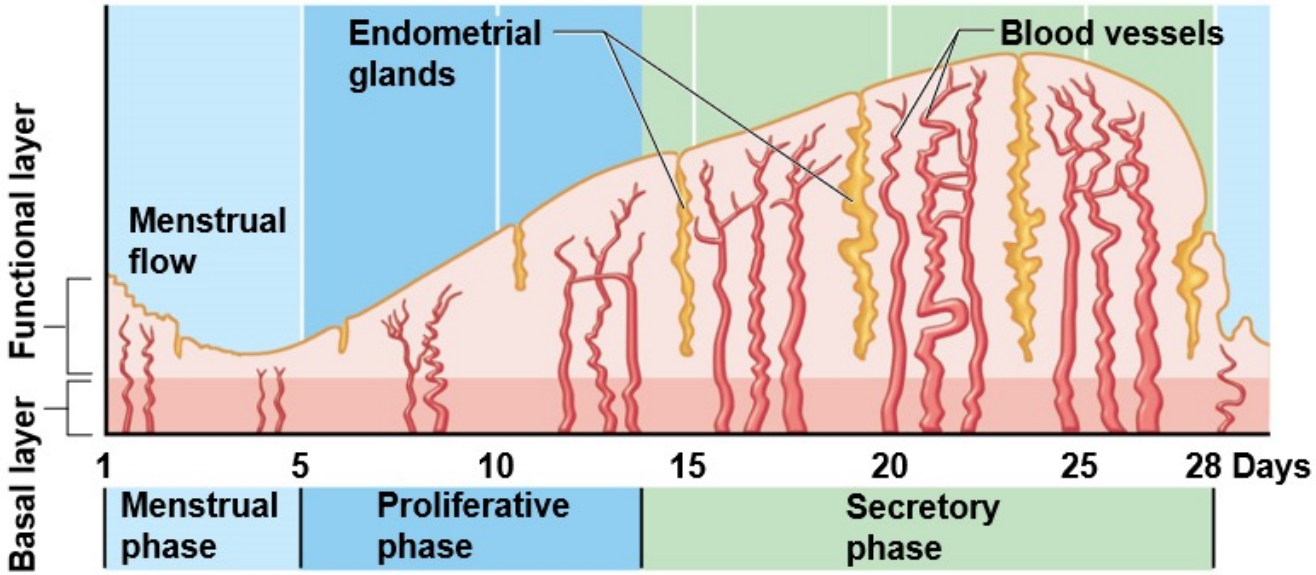
- Menstrual: Shedding of the functional layer of the endometrium.
- Proliferative: Rebuilding of the functional layer of the endometrium.
- Secretory: Begins immediately after ovulation. Enrichment of the blood supply and glandular secretion of nutrients prepare the endometrium to receive an embryo.

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# UTERINE (MENSTRUAL) CYCLE

Days 6–14: proliferative stage

- Regeneration of functional layer of the endometrium
  - Endometrium is repaired, thickens, and becomes well vascularized
- Estrogen levels rise
- Ovulation occurs in the ovary at the end of this stage




The menstrual and proliferative phases occur before ovulation and together correspond to the follicular phase of the ovarian cycle. The secretory phase corresponds in time to the luteal phase of the ovarian cycle.

**The three phases of the uterine cycle:**

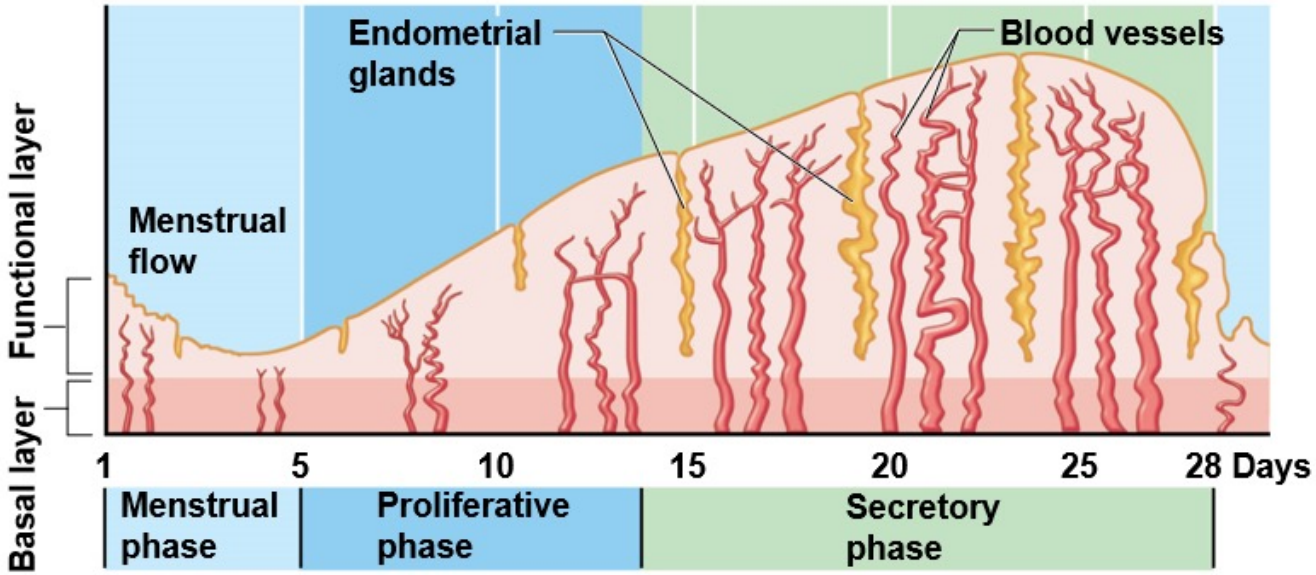
- Menstrual: Shedding of the functional layer of the endometrium.
- Proliferative: Rebuilding of the functional layer of the endometrium.
- Secretory: Begins immediately after ovulation. Enrichment of the blood supply and glandular secretion of nutrients prepare the endometrium to receive an embryo.

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# UTERINE (MENSTRUAL) CYCLE

Days 15–28: secretory phase

- Levels of progesterone rise and increase the blood supply to the endometrium, which becomes more vascular
- Endometrium increases in size and readies for implantation




The menstrual and proliferative phases occur before ovulation and together correspond to the follicular phase of the ovarian cycle. The secretory phase corresponds in time to the luteal phase of the ovarian cycle.

**The three phases of the uterine cycle:**

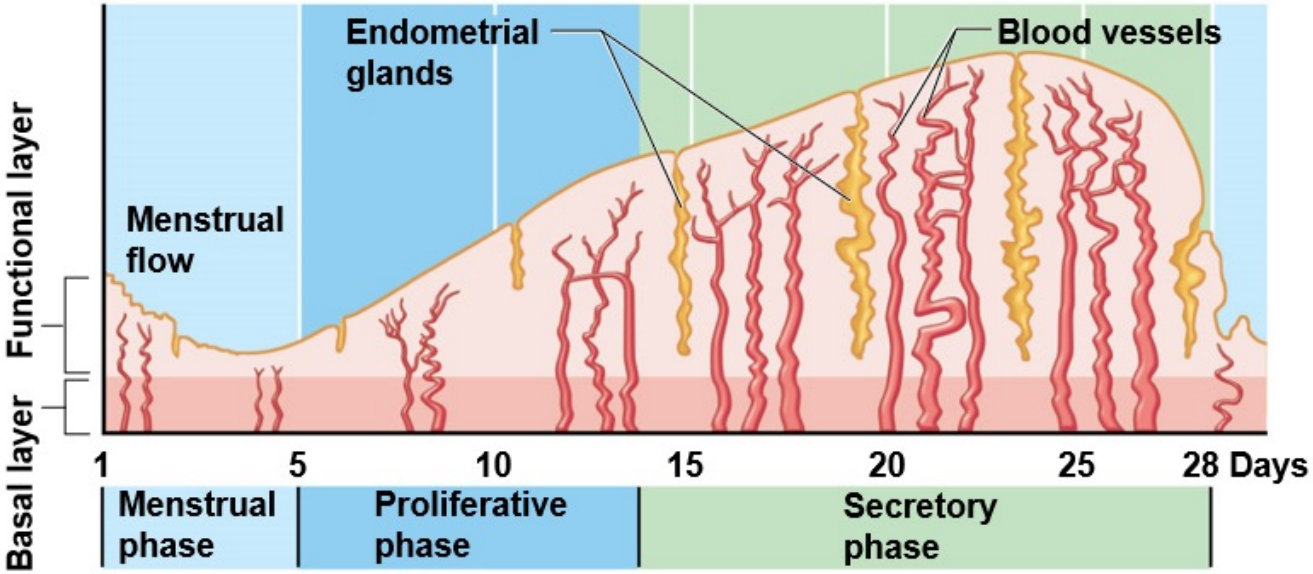
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# UTERINE (MENSTRUAL) CYCLE

Days 15–28: secretory phase (continued)

- If fertilization does occur:
  - Embryo produces a hormone that causes the corpus luteum to continue producing its hormones
- If fertilization does NOT occur:
  - Corpus luteum degenerates as LH blood levels decline
  - The phases are repeated about every 28 days

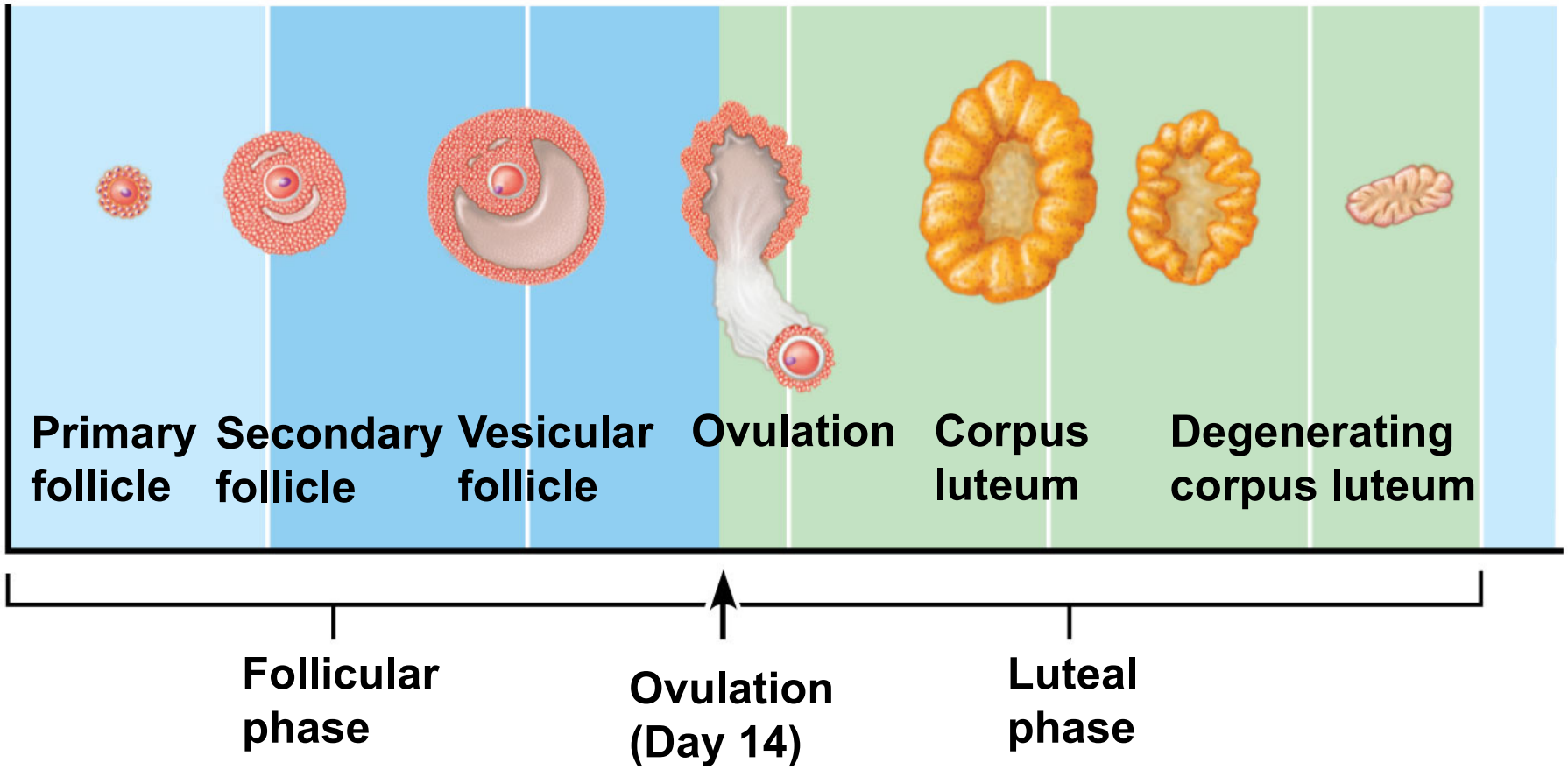



The menstrual and proliferative phases occur before ovulation and together correspond to the follicular phase of the ovarian cycle. The secretory phase corresponds in time to the luteal phase of the ovarian cycle.

**The three phases of the uterine cycle:**

- Menstrual: Shedding of the functional layer of the endometrium.
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**(b) Ovarian cycle:** Structural changes in the ovarian follicles during the ovarian cycle are correlated with (d) changes in the endometrium of the uterus during the uterine cycle.

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# MAMMARY GLANDS

- Present in both sexes, but function only in females
  - Modified sweat glands
- Function is to produce milk to nourish a newborn
- Stimulated by sex hormones (mostly estrogens) to increase in size

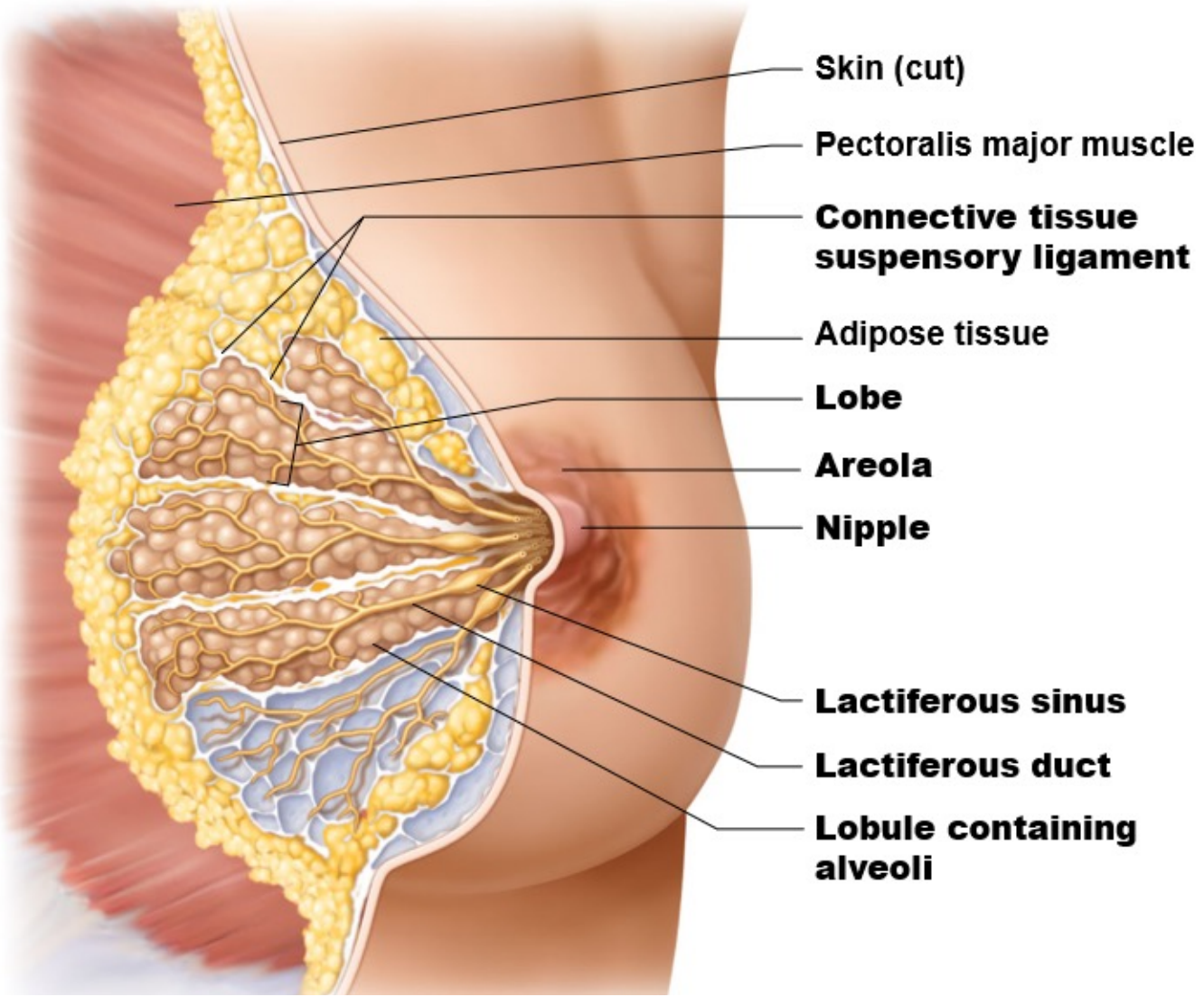
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# MAMMARY GLANDS

## ANATOMY

Parts of the mammary gland

- Areola—central pigmented area
- Nipple—protruding central area of areola
- Lobes—internal structures that radiate around nipple

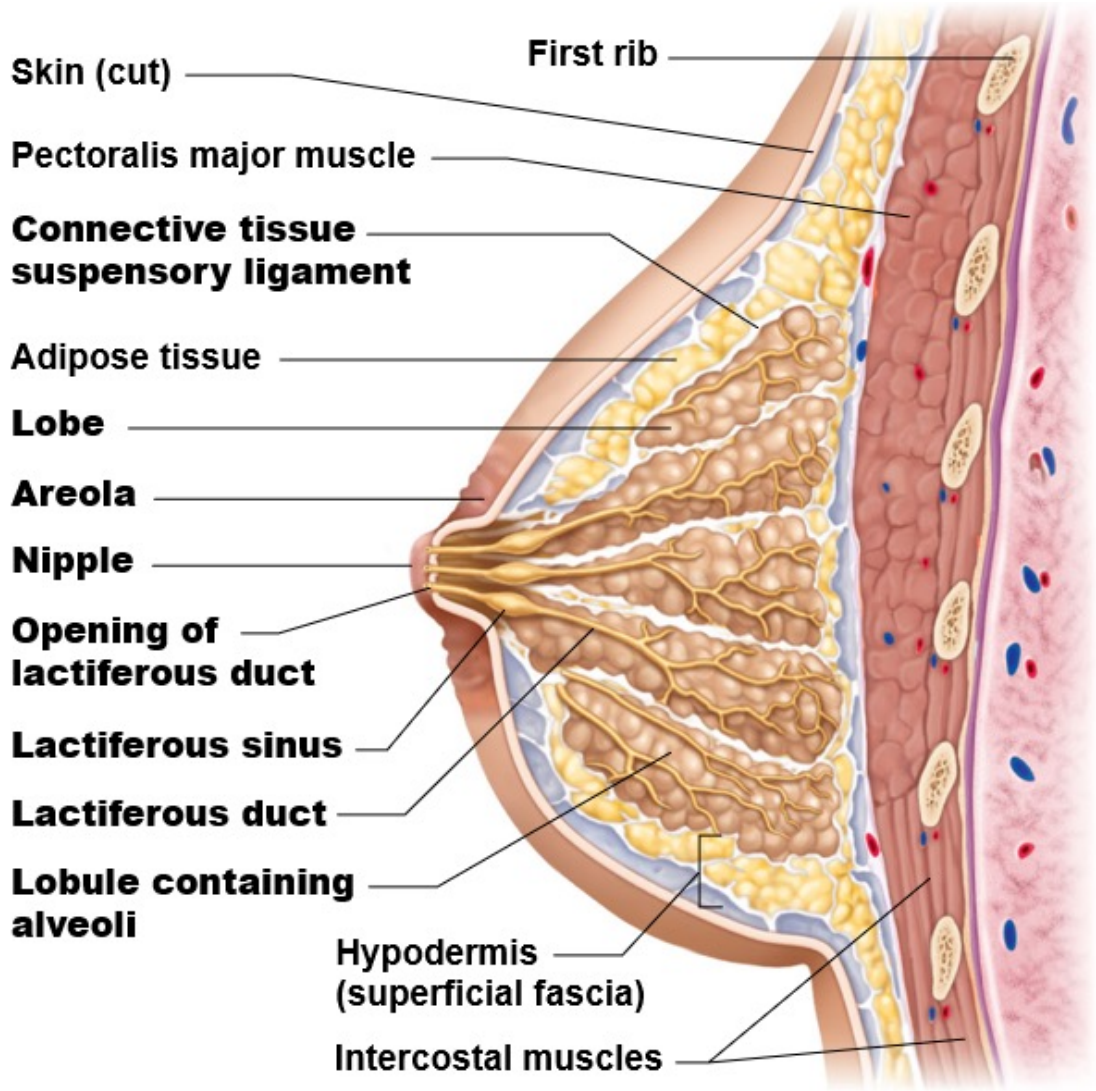



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# MAMMARY GLANDS

## ANATOMY

- Lobules—located within each lobe and contain clusters of alveolar glands
- Alveolar glands—produce milk when a woman is lactating (producing milk)
- Lactiferous ducts—connect alveolar glands to nipple
- Lactiferous sinus—dilated portion where milk accumulates



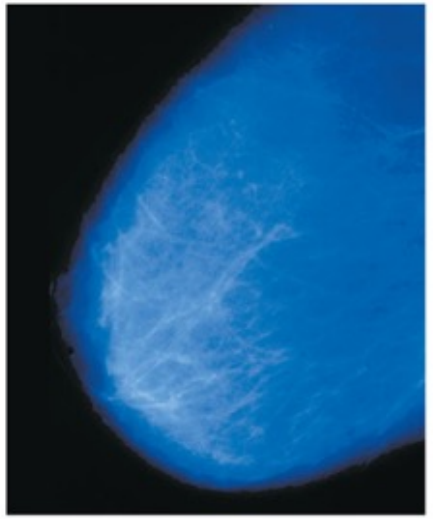

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# MAMMOGRAPHY

- X-ray examination that detects breast cancers too small to feel
- American Cancer Society recommends mammography annually for women between 45 and 54 years old and every 2 years thereafter if the results are normal
- Breast cancer is often signaled by a change in skin texture, puckering, or leakage from the nipple



(a) Mammogram procedure



(b) Film of normal breast




(c) Film of breast with tumor

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